23rd Annual Meeting of the Complex Trait Community

Workshop: Tools For Variant Visualization and Analysis at the Rat Genome Database (RGD) and the Alliance of Genome Resources (the Alliance)

Jennifer R. Smith

Rat Genome Database

https://rgd.mcw.edu

Alliance of Genome Resources

https://www.alliancegenome.org/

https://rgd.mcw.edu

MEDICAL COLLEGE OF WISCONSIN

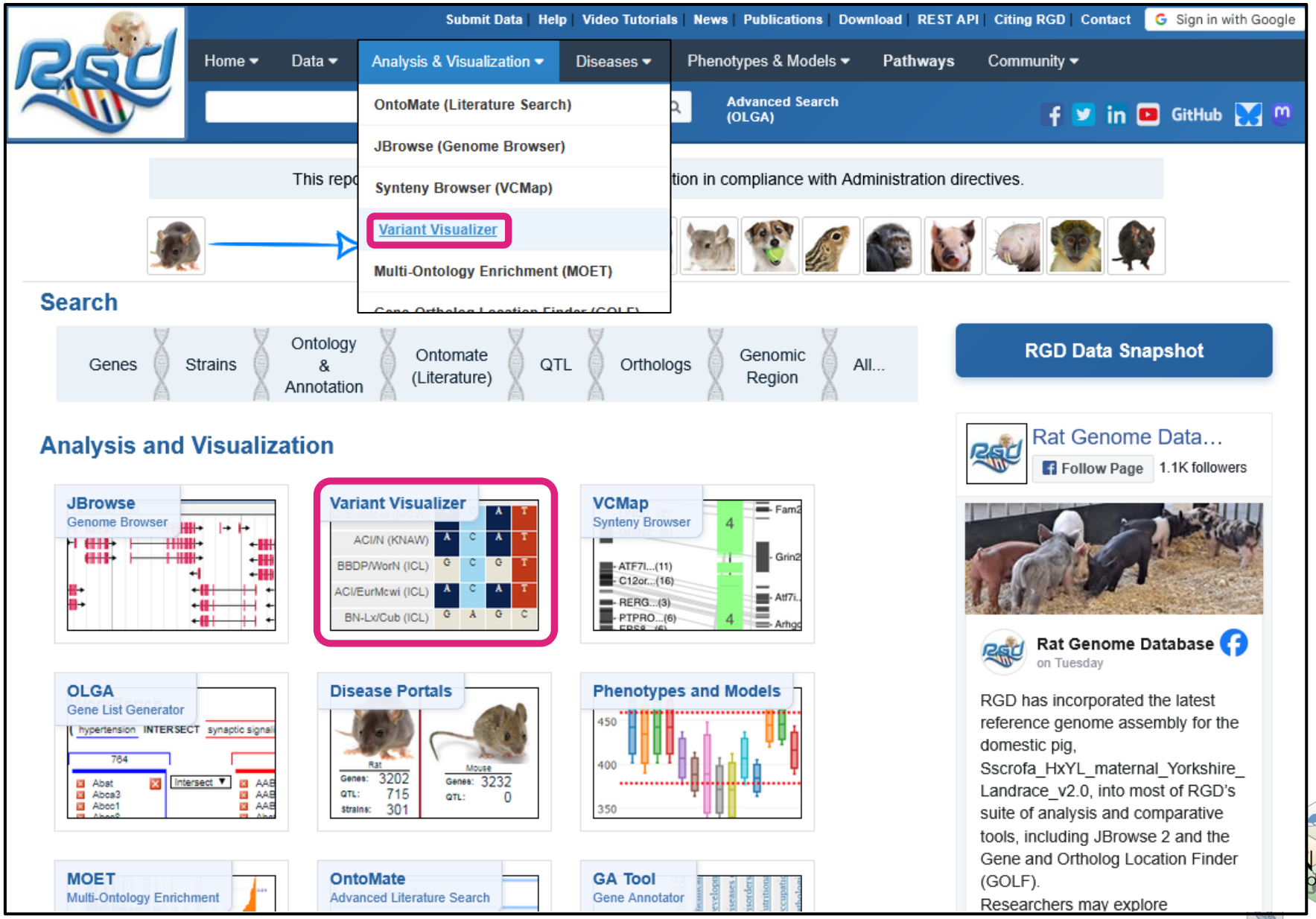
physiology

GBC GLOBAL CORE BIODATA RESOURCE

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RGD: Accessing Variant Visualizer

Access Variant Visualizer from the dropdown "Analysis & Visualization" menu in the main menu bar at the top of any RGD page or use the "Variant Visualizer" button on the RGD home page.

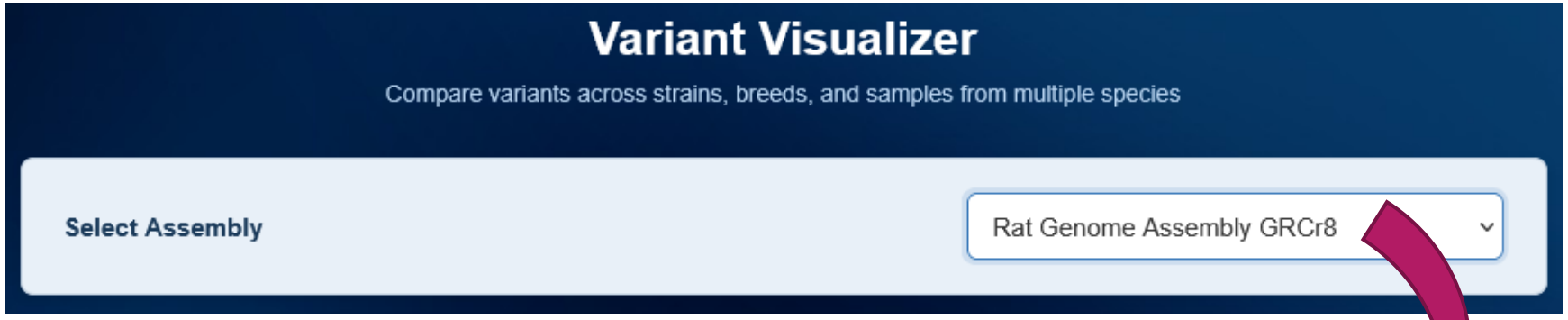


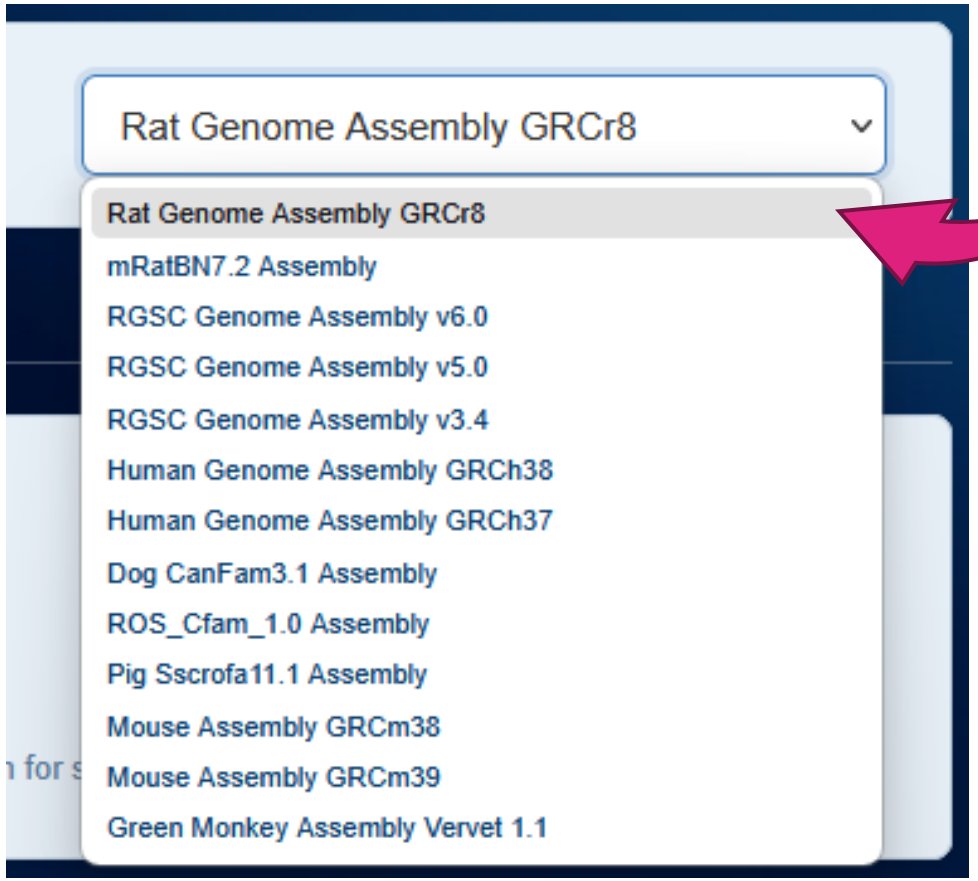
https://rgd.mcw.edu

RGD: The Variant Visualizer Landing Page

Start a search in Variant Visualizer by selecting a species and an assembly.

For this example we will use GRCr8.





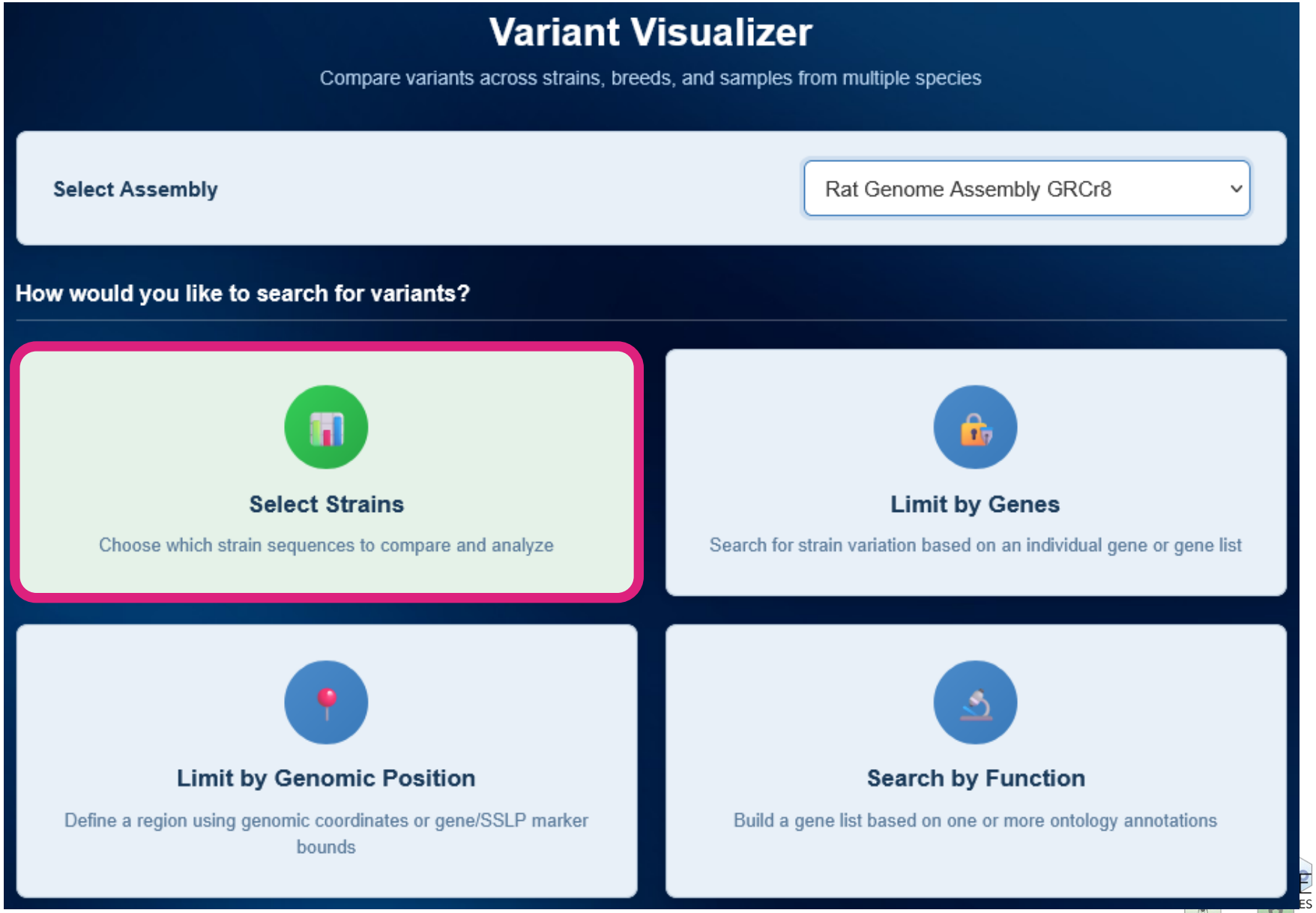
Variant Visualizer

https://rgd.mcw.edu

RGD: The Variant Visualizer Landing Page

Once the species and assembly are selected, you have several options for searching for variants, any of which can be selected first.

For this example, we will begin by selecting strains. The other options will be available again later.



https://rgd.mcw.edu

Variant Visualizer

RGD: The Variant Visualizer Strain/Sample Selection

- The strain/sample selection page has been completely redesigned.
- Sequence groups are listed at the top.
- Below that the samples are grouped by strain.
- These are closed by default.

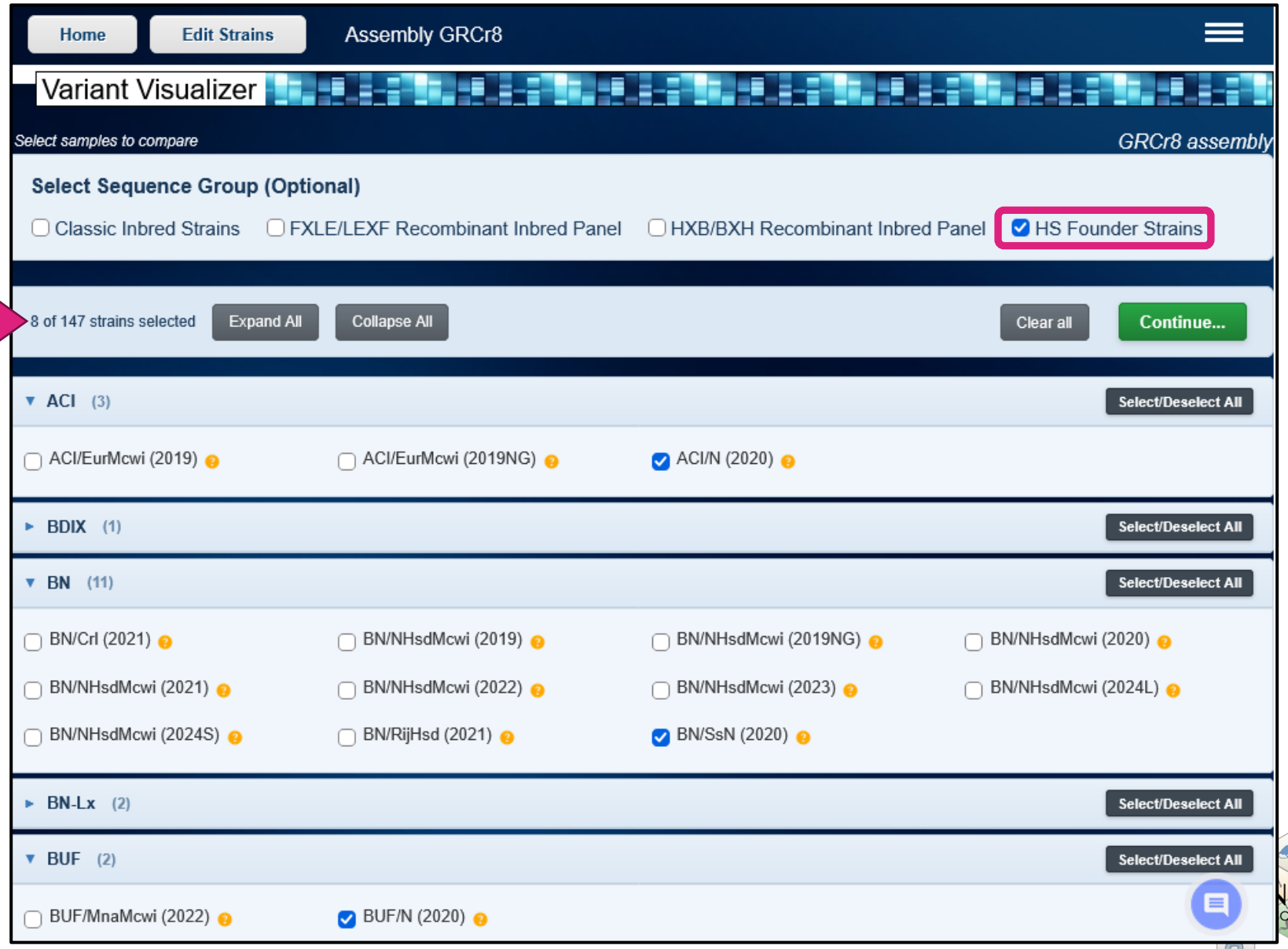


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RGD: The Variant Visualizer Strain/Sample Selection

Click a collapsed row to open it or select a Sequence Group in the top list to open the rows containing samples in that group.

Selecting the HS Founder Strains group automatically selects 8 of the 147 available samples and opens the corresponding rows

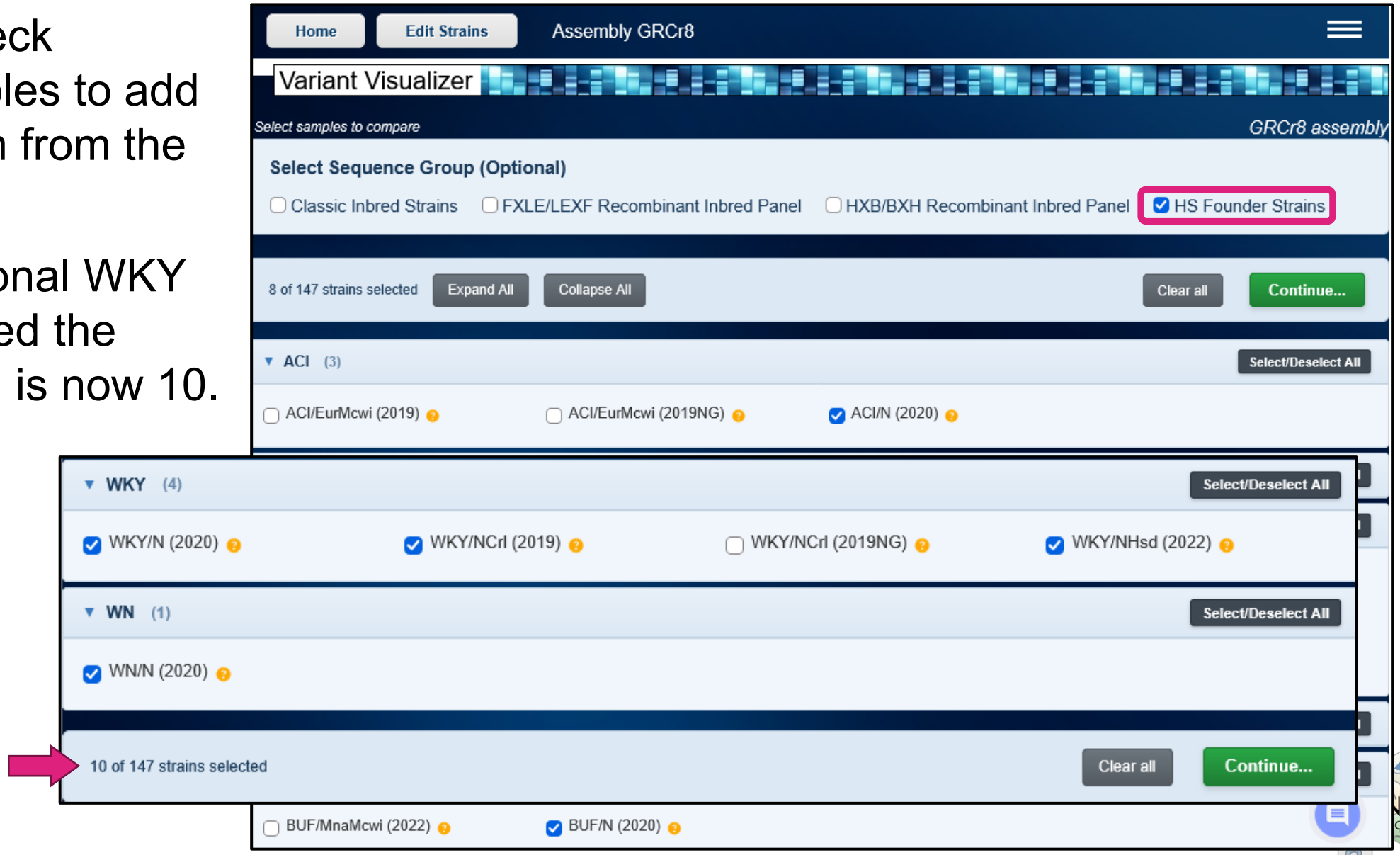


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RGD: The Variant Visualizer Strain/Sample Selection

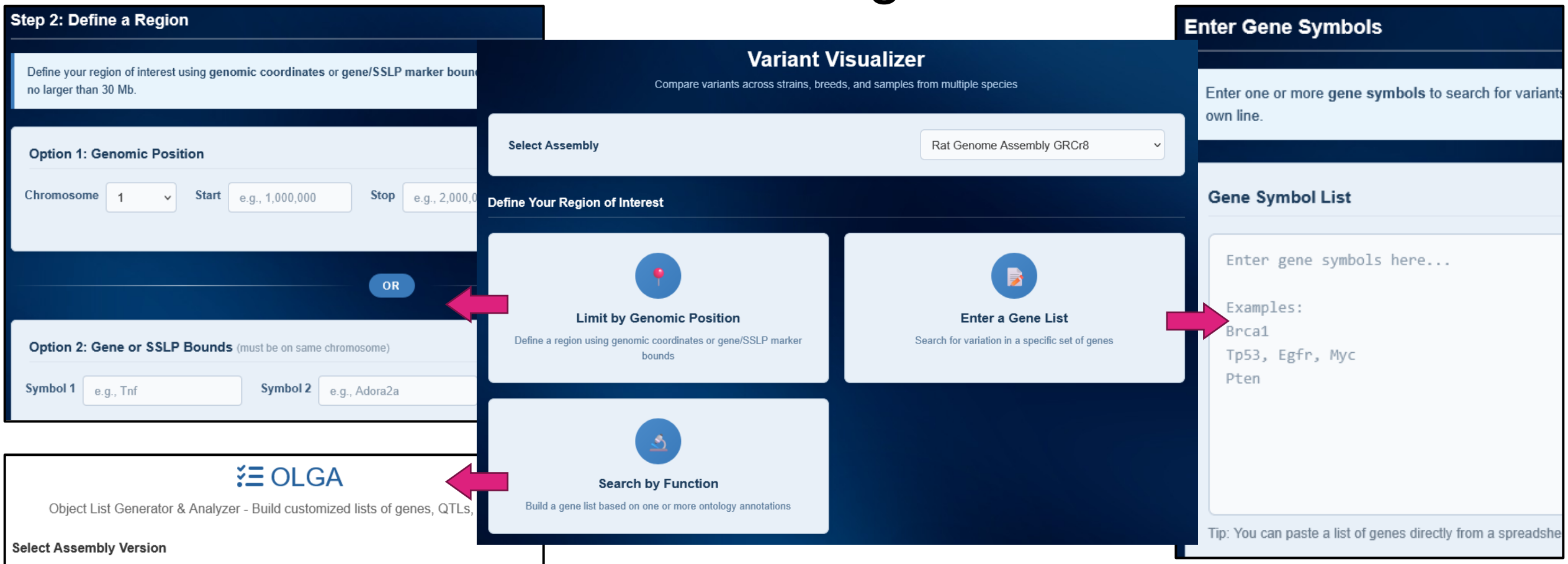
Check or uncheck individual samples to add or remove them from the selected list.

With two additional WKY samples selected the number chosen is now 10.



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RGD: The Variant Visualizer Region Selection

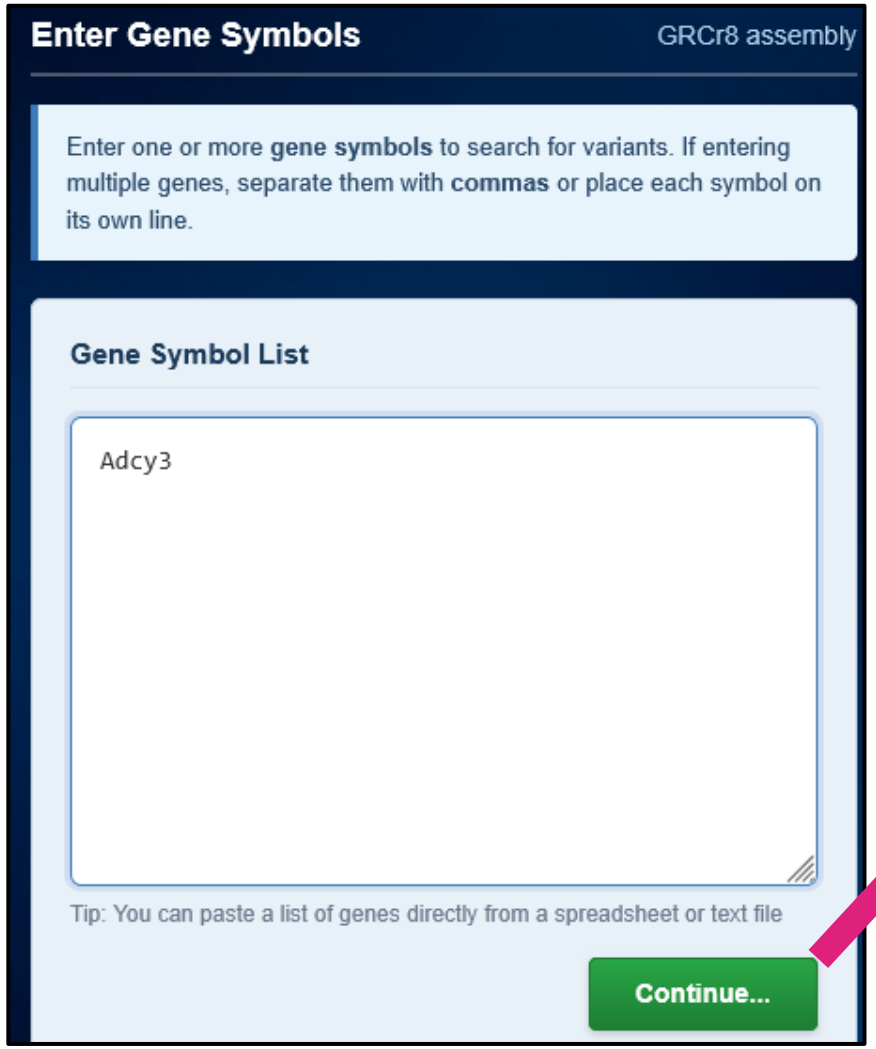


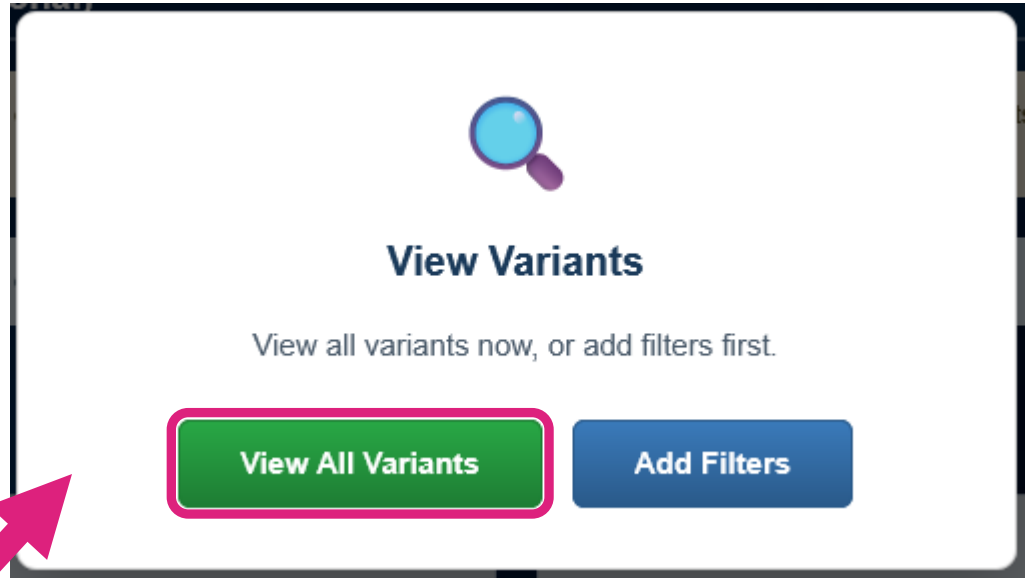
Three options are provided for defining one or more regions in which to search for variants

For this example, choose "Enter a Gene List" and type **Adcy3** in the Gene Symbol List box.

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RGD: The Variant Visualizer "View Variants"





After clicking "Continue", choose whether to view all of the variants in the selected region or gene(s) or filter the selection.

For the moment, we will choose "View All Variants" and will come back to the filter options later.

https://rgd.mcw.edu

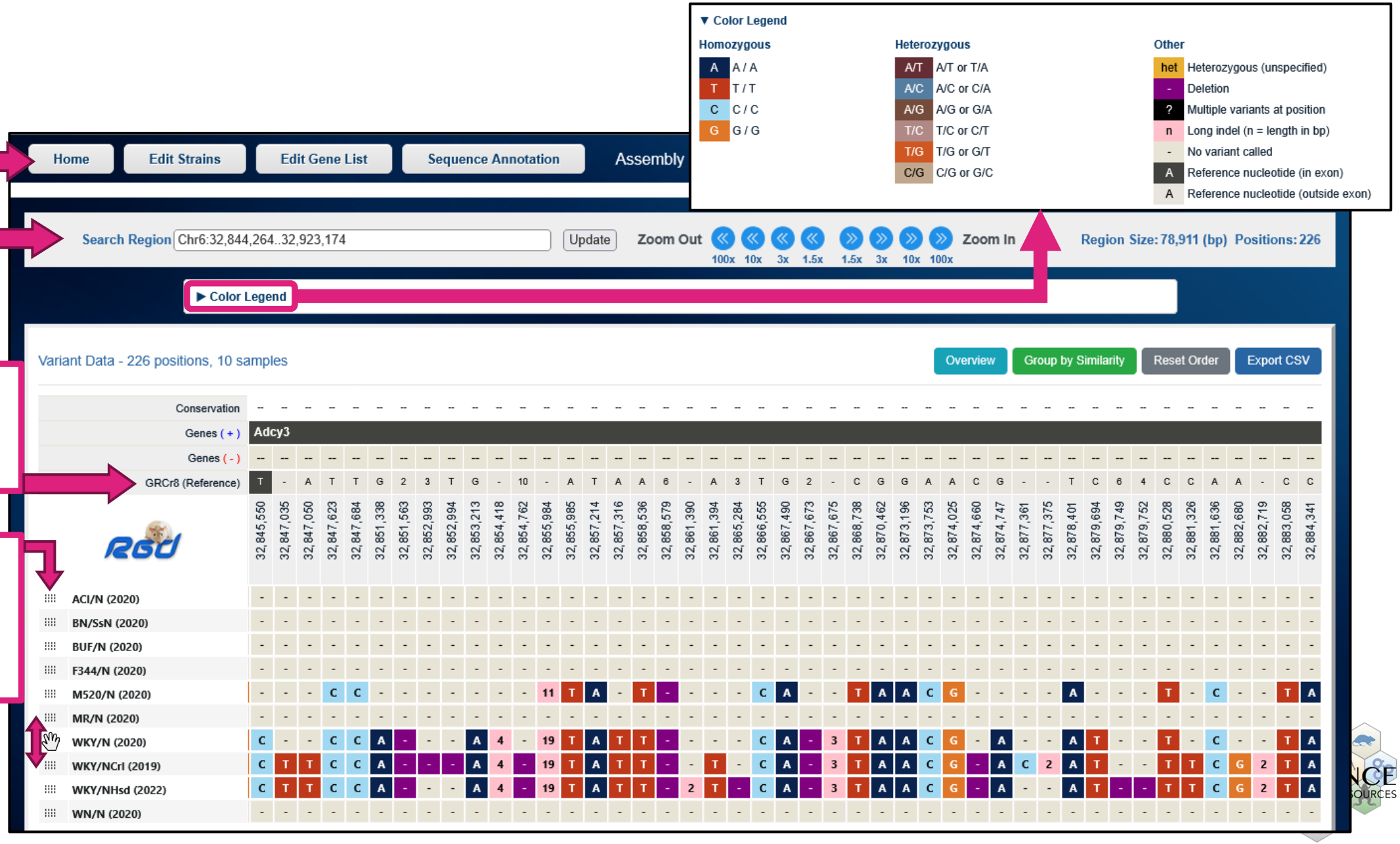
RGD: The Variant Visualizer Viewer (All Variants)

Go back to previous selection screens

Adjust the region

Shaded boxes denote exonic positions

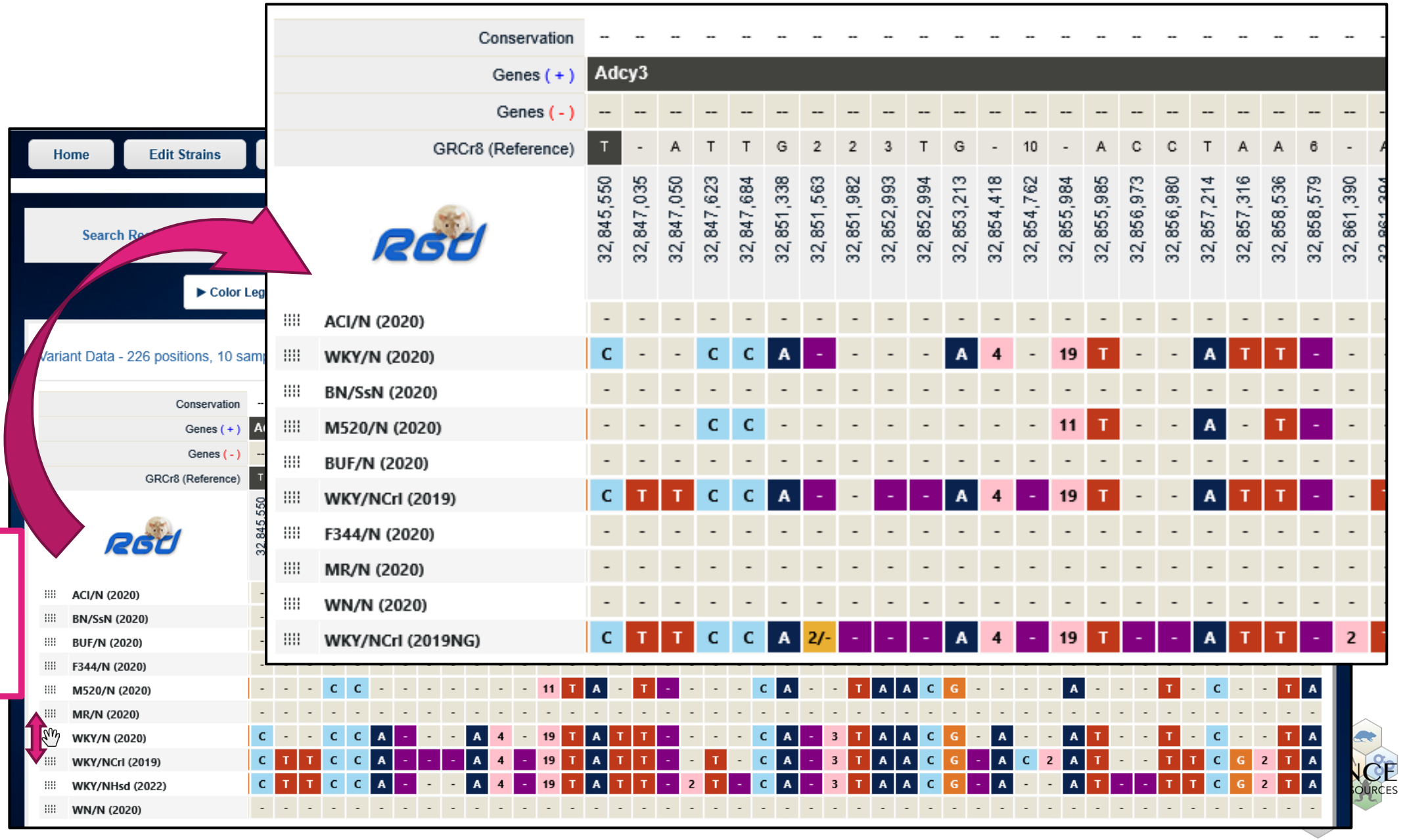
Use "handles" to drag and drop rows to reorder strains



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RGD: The Variant Visualizer Viewer (All Variants)

Here the WKY samples have been dispersed throughout the list.

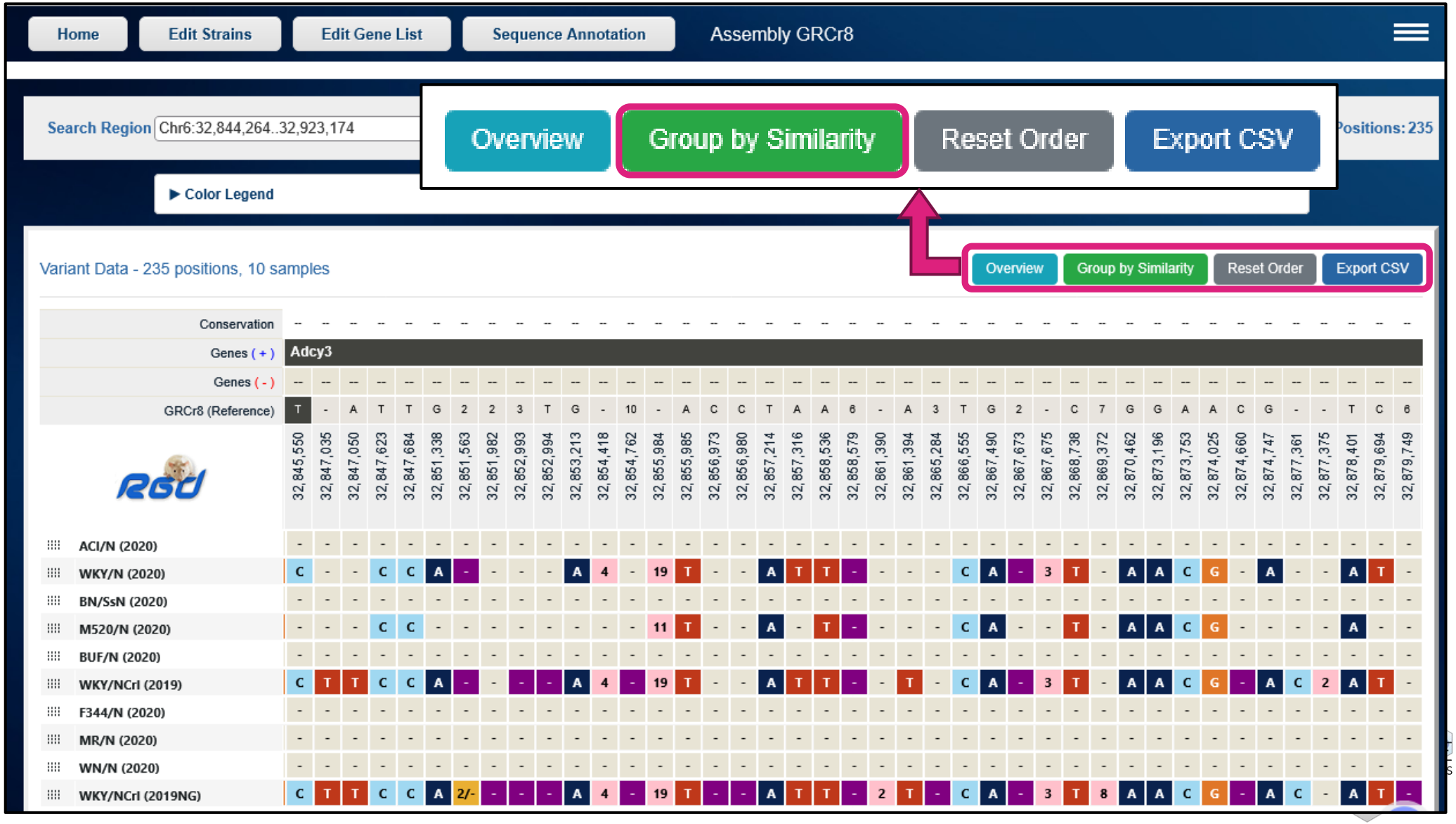


Use "handles" to drag and drop rows to reorder strains

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RGD: The Variant Visualizer Viewer (All Variants)

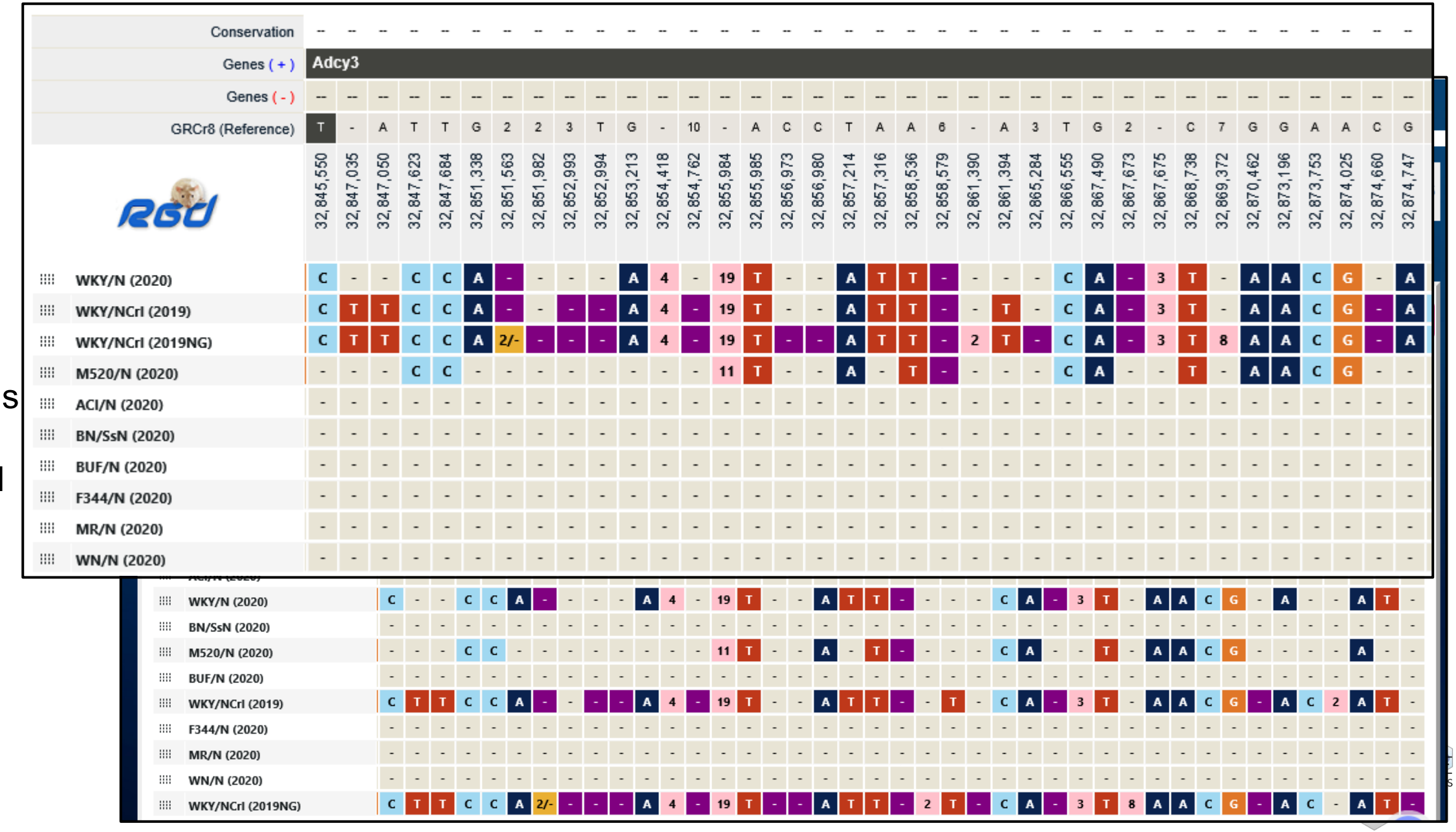
Buttons on the right above the viewer provide additional functionality, for example "Group by Similarity".



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RGD: The Variant Visualizer Viewer (All Variants)

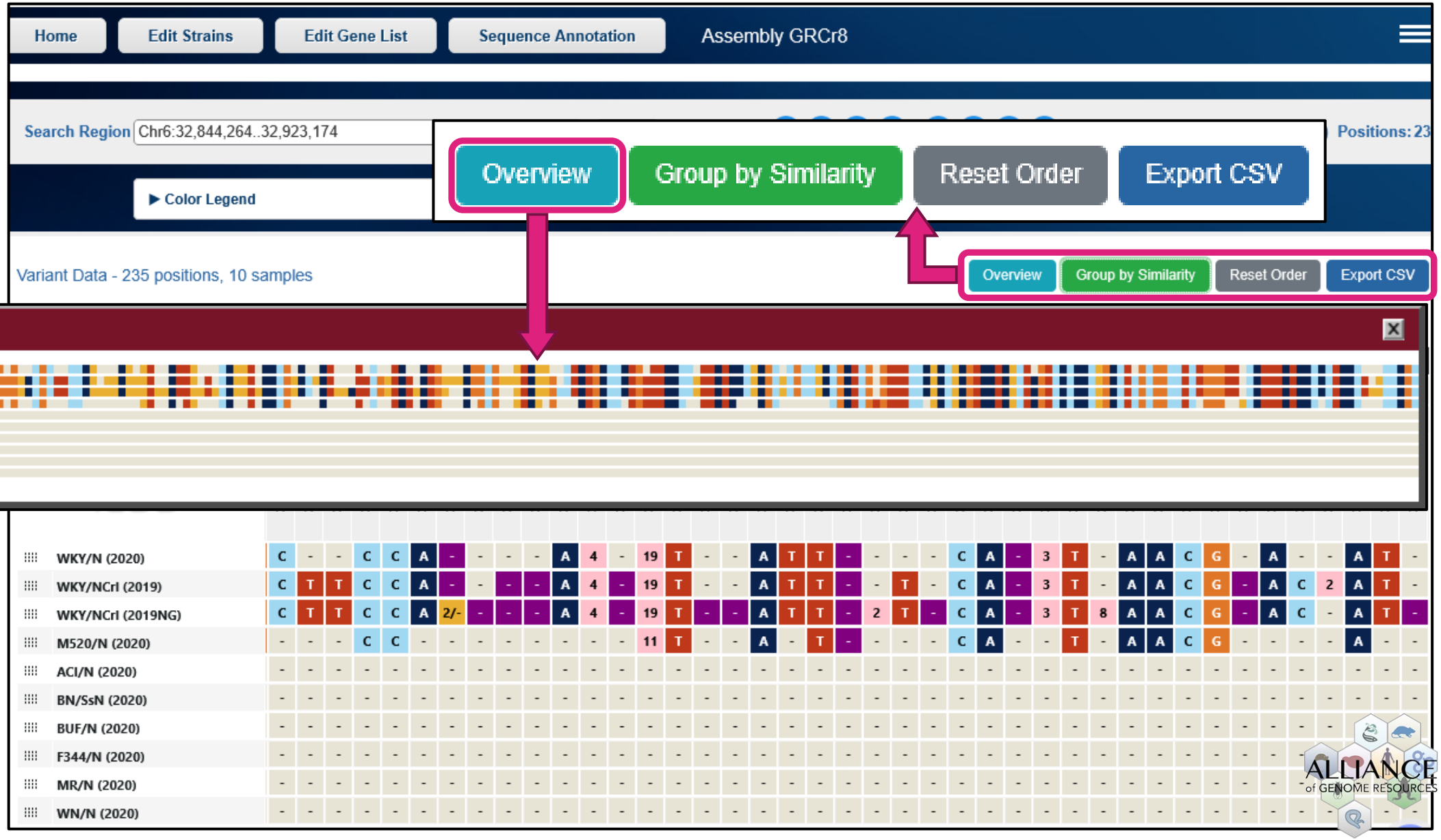
Group by Similarity has been used to group the rows with variant distributions similar to the WKY/N (2020) sample.



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RGD: The Variant Visualizer Viewer (All Variants)

"Overview" gives a bird's eye view of all of the variants in the selected region,

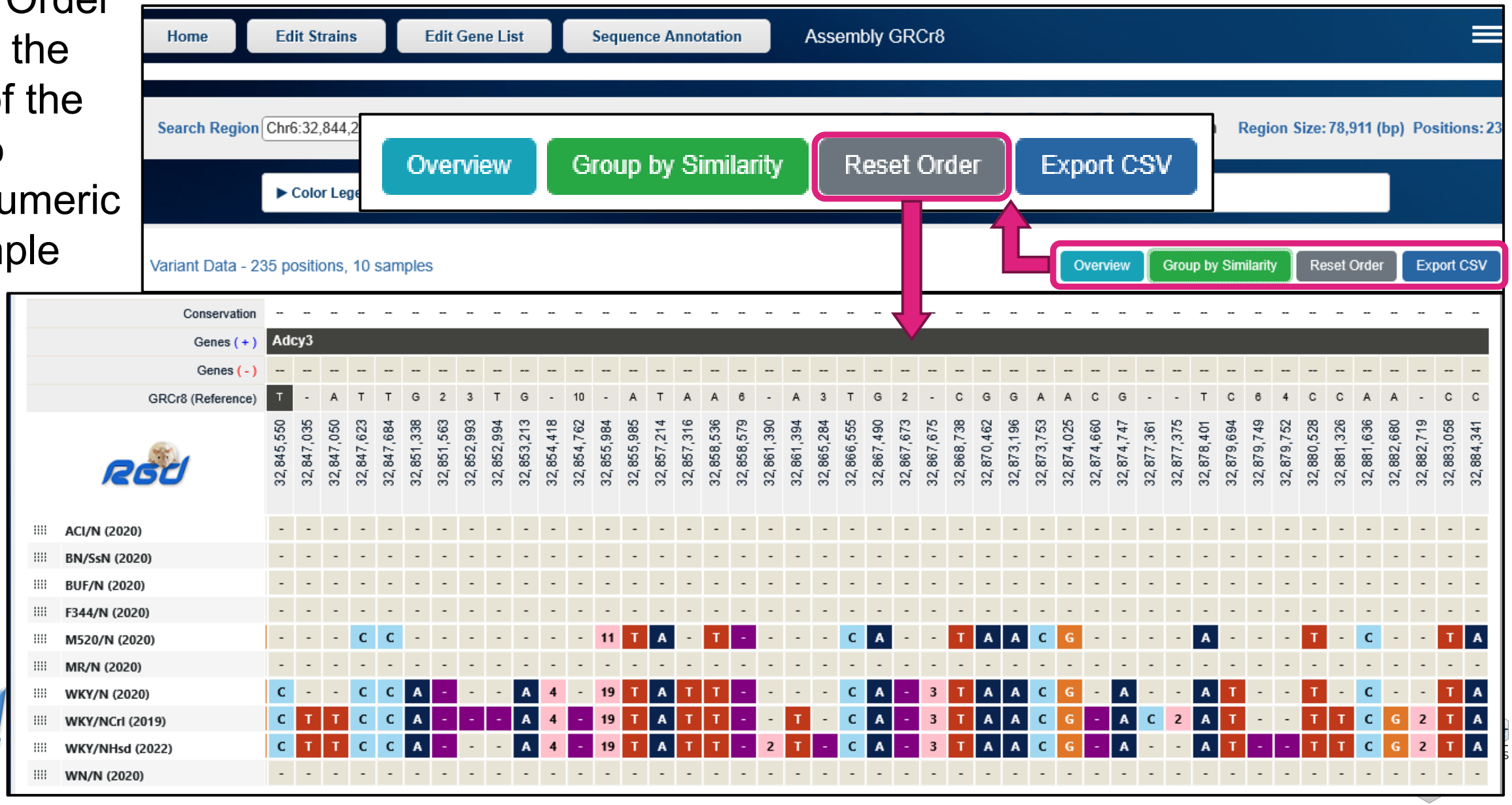


including those that are "off screen" in the main viewer.

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RGD: The Variant Visualizer Viewer (All Variants)

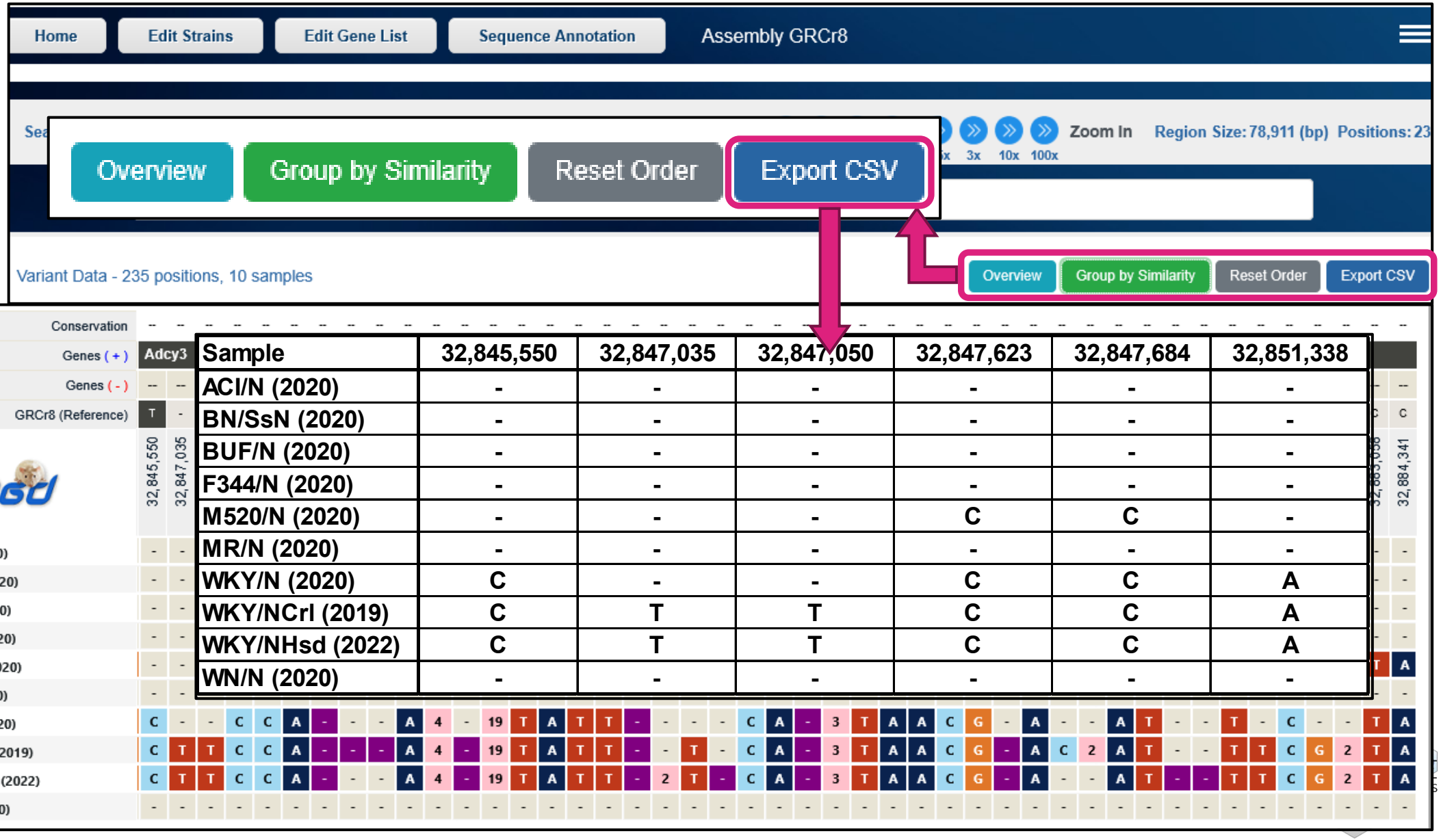
"Reset Order" returns the order of the rows to alphanumeric by sample name.



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RGD: The Variant Visualizer Viewer (All Variants)

"Export CSV" downloads a CSV file that recapitulates the view of the variants. All variants in the region are included.

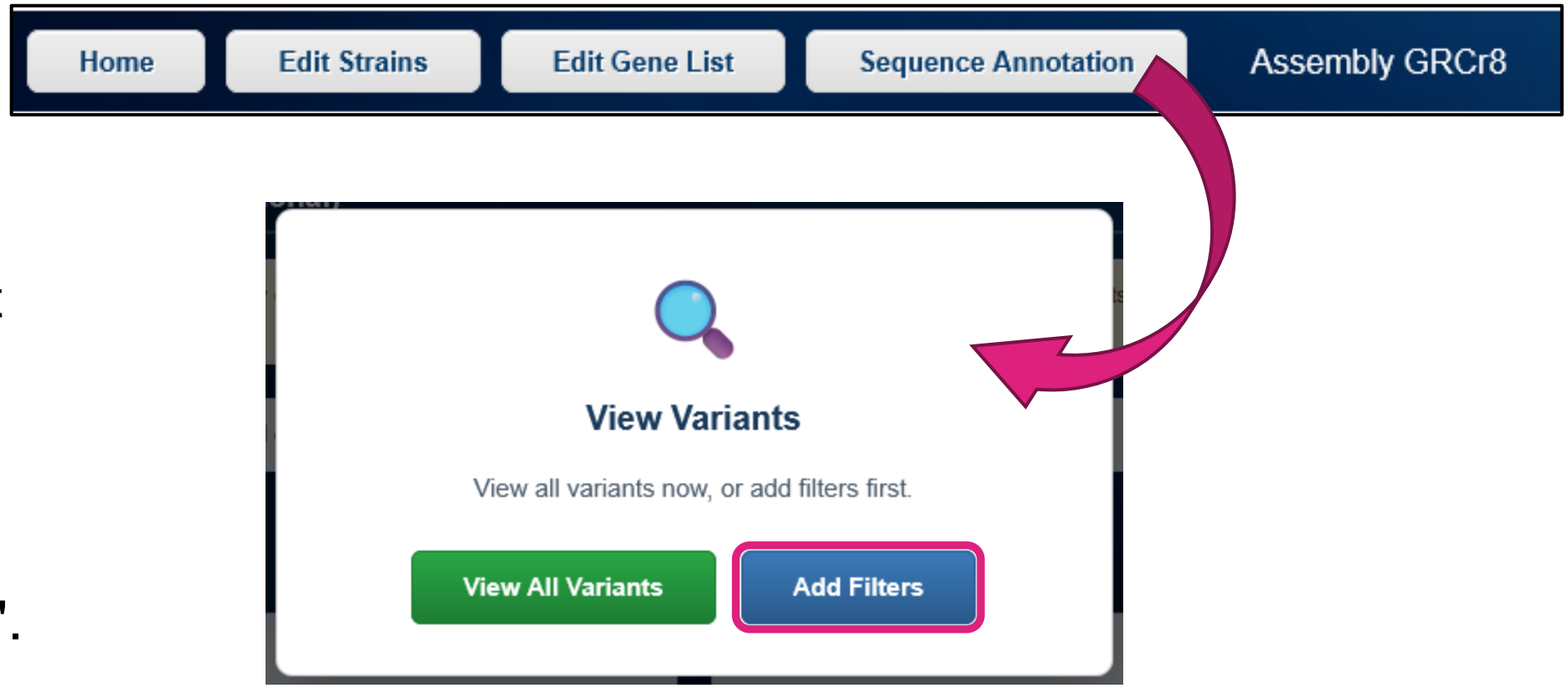


Sample	32,845,550	32,847,035	32,847,050	32,847,623	32,847,684	32,851,338
ACI/N (2020)	-	-	-	-	-	-
BN/SsN (2020)	-	-	-	-	-	-
BUF/N (2020)	-	-	-	-	-	-
F344/N (2020)	-	-	-	-	-	-
M520/N (2020)	-	-	-	C	C	-
MR/N (2020)	-	-	-	-	-	-
WKY/N (2020)	C	-	-	C	C	A
WKY/NCrl (2019)	C	T	T	C	C	A
WKY/NHsd (2022)	C	T	T	C	C	A
WN/N (2020)	-	-	-	-	-	-

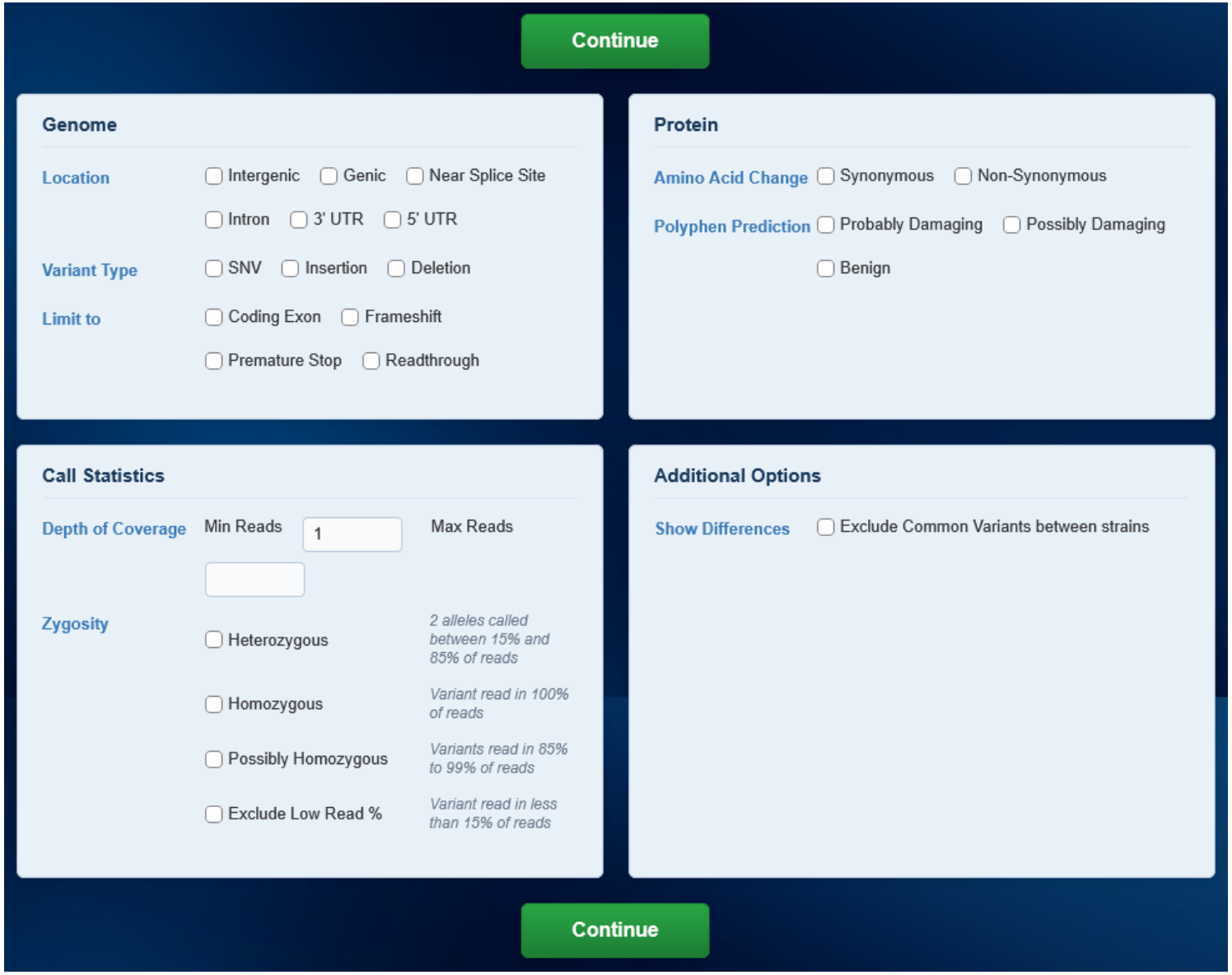
https://rgd.mcw.edu

RGD: The Variant Visualizer Viewer (Filtering Variants)

In the Variant Visualizer Viewer, click on the "Sequence Annotation" button at the top of the viewer window. This will once again bring up the selector to "View All…" or "Add Filters". This time, click "Add Filters".



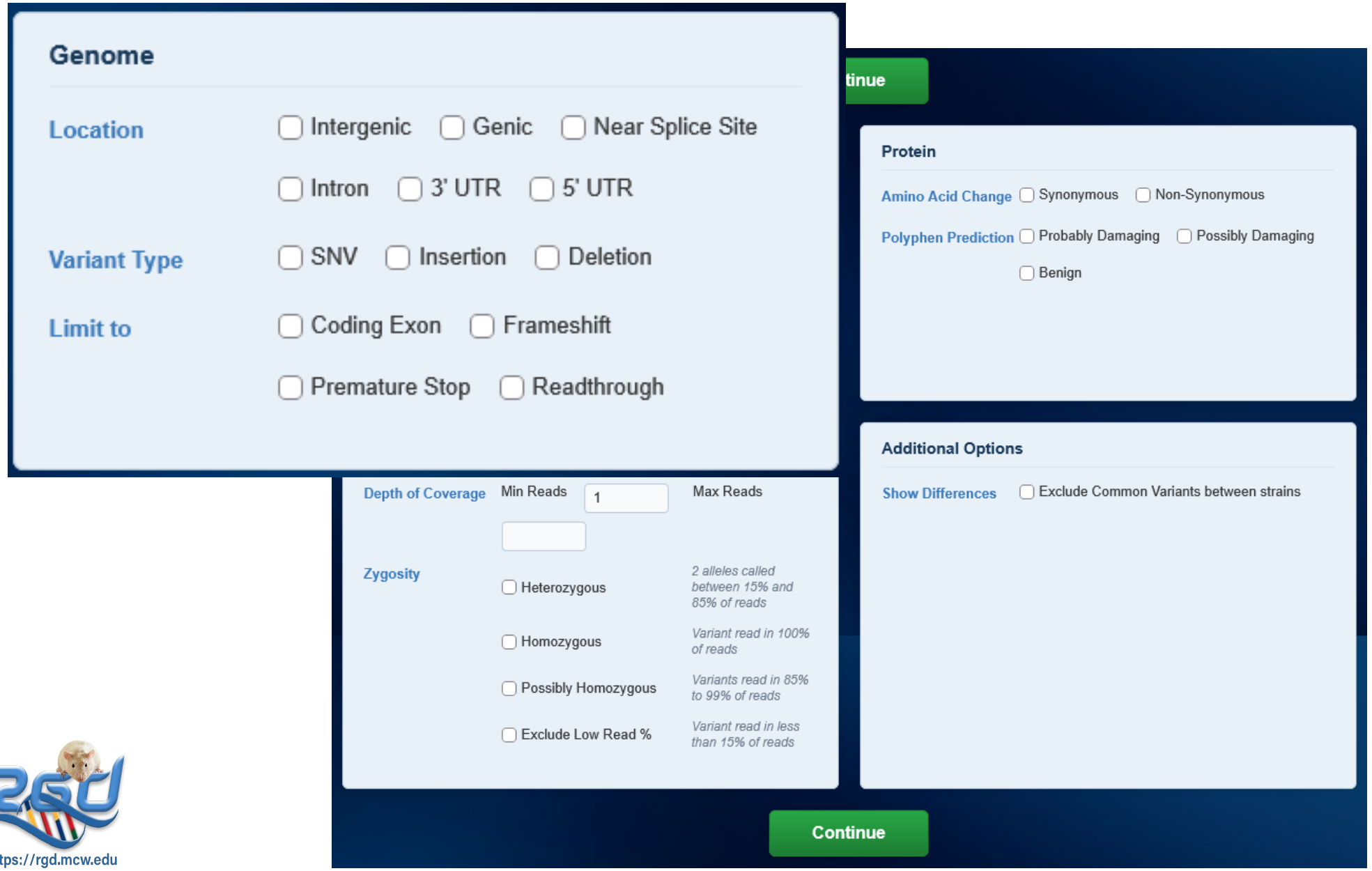
RGD: The Variant Visualizer Viewer (Filtering Variants)



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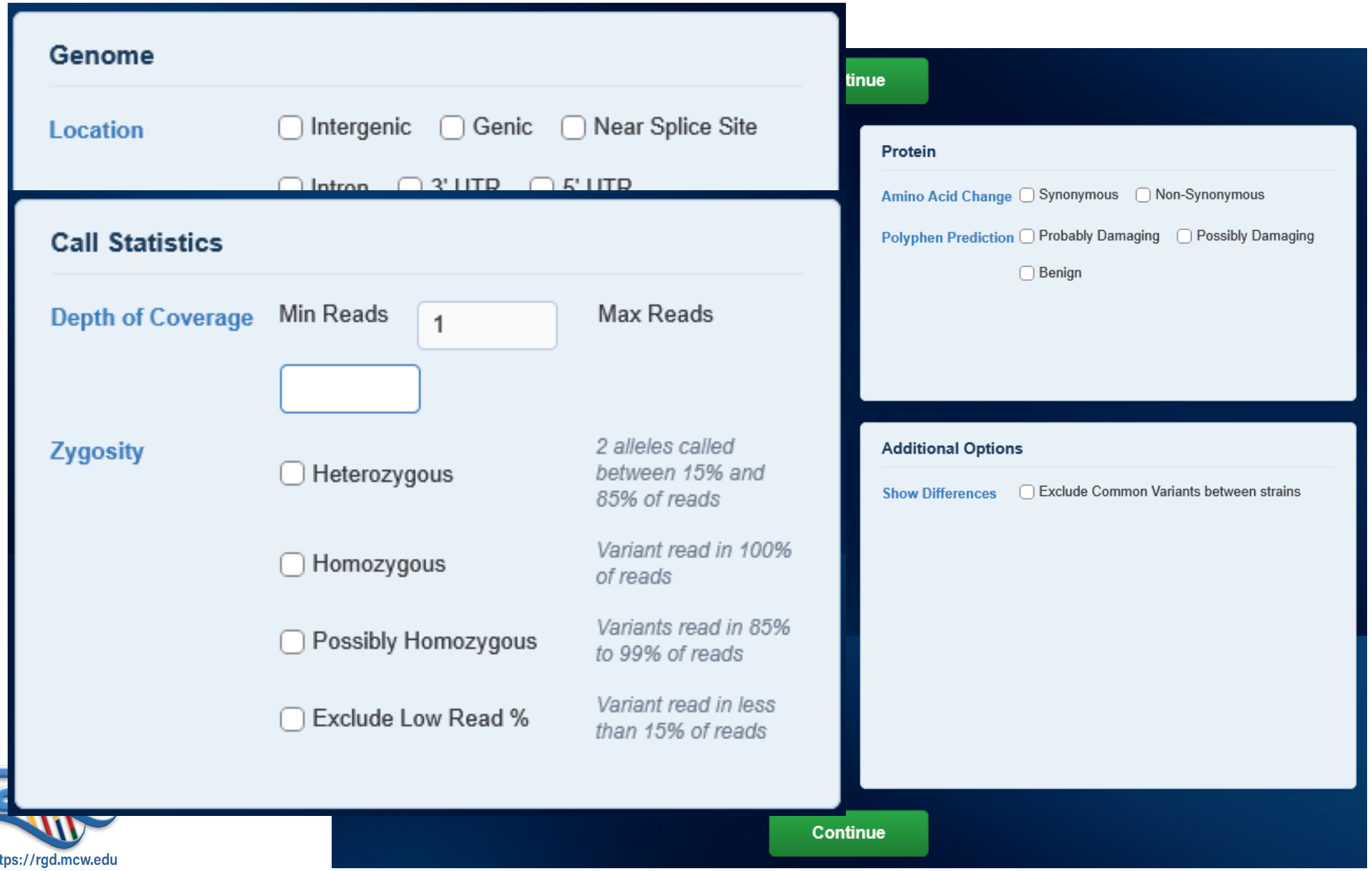
RGD: The Variant Visualizer Viewer (Filtering Variants)



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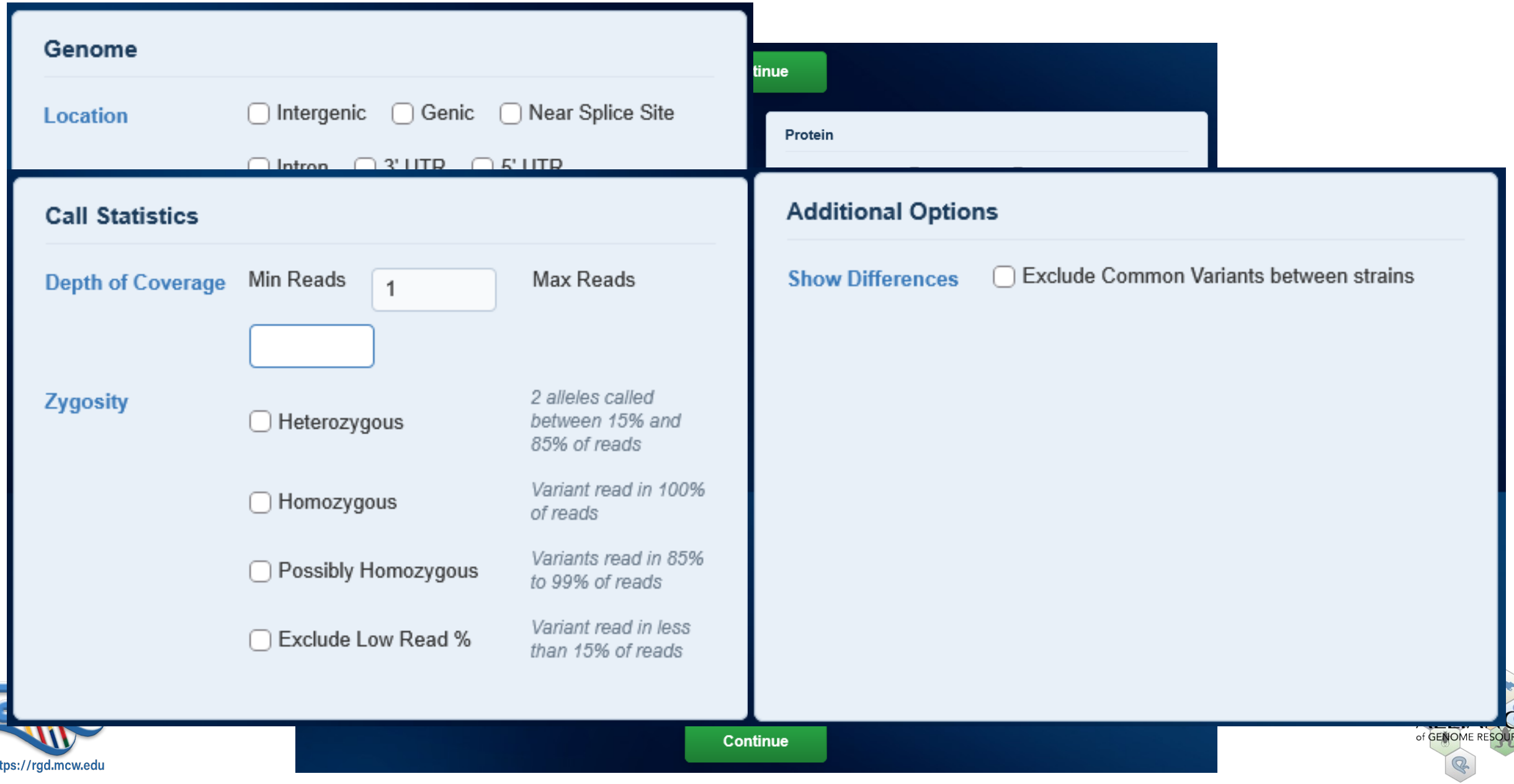
RGD: The Variant Visualizer Viewer (Filtering Variants)



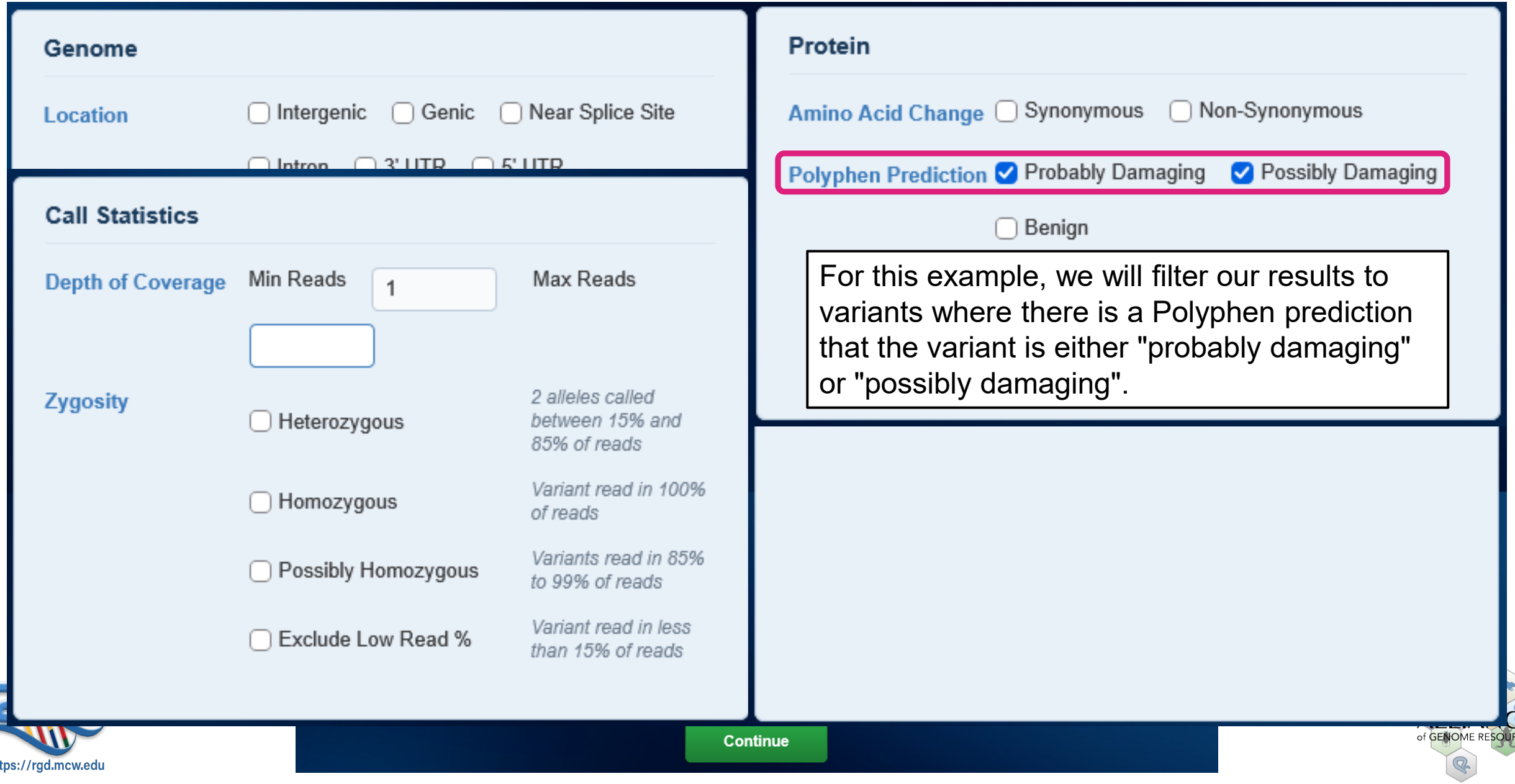
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RGD: The Variant Visualizer Viewer (Filtering Variants)

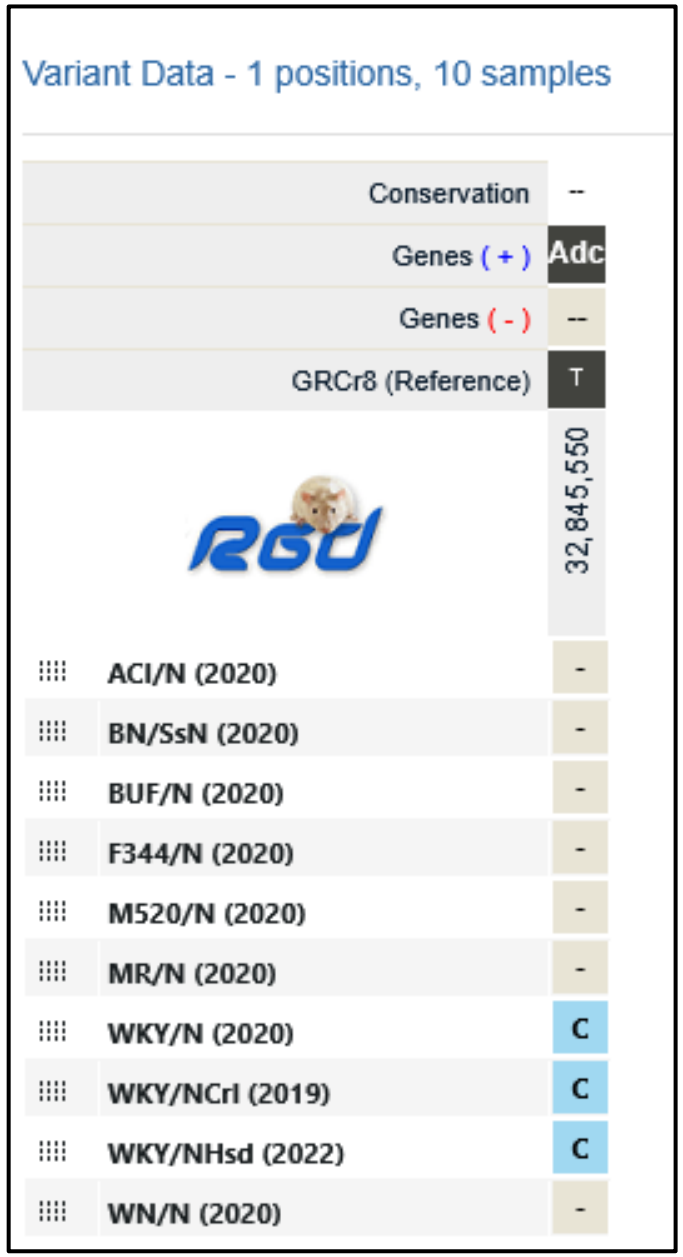


RGD: The Variant Visualizer Viewer (Filtering Variants)



RGD: The Variant Visualizer Viewer (Filtering Variants)

After filtering for only possibly and probably damaging variants, a single T-to-C variant found in the three WKY samples remains.

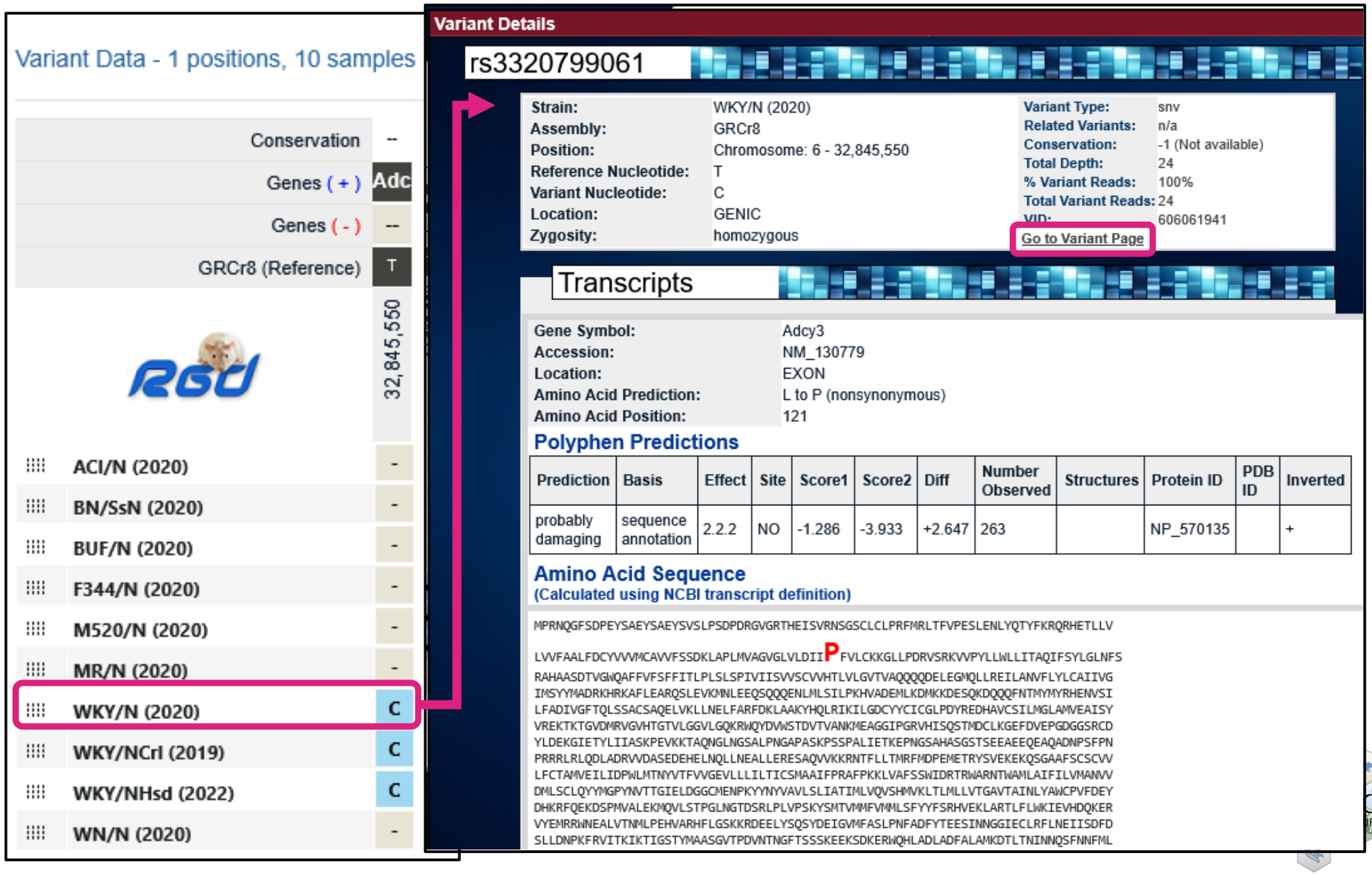


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RGD: The Variant Visualizer Viewer (Filtering Variants)

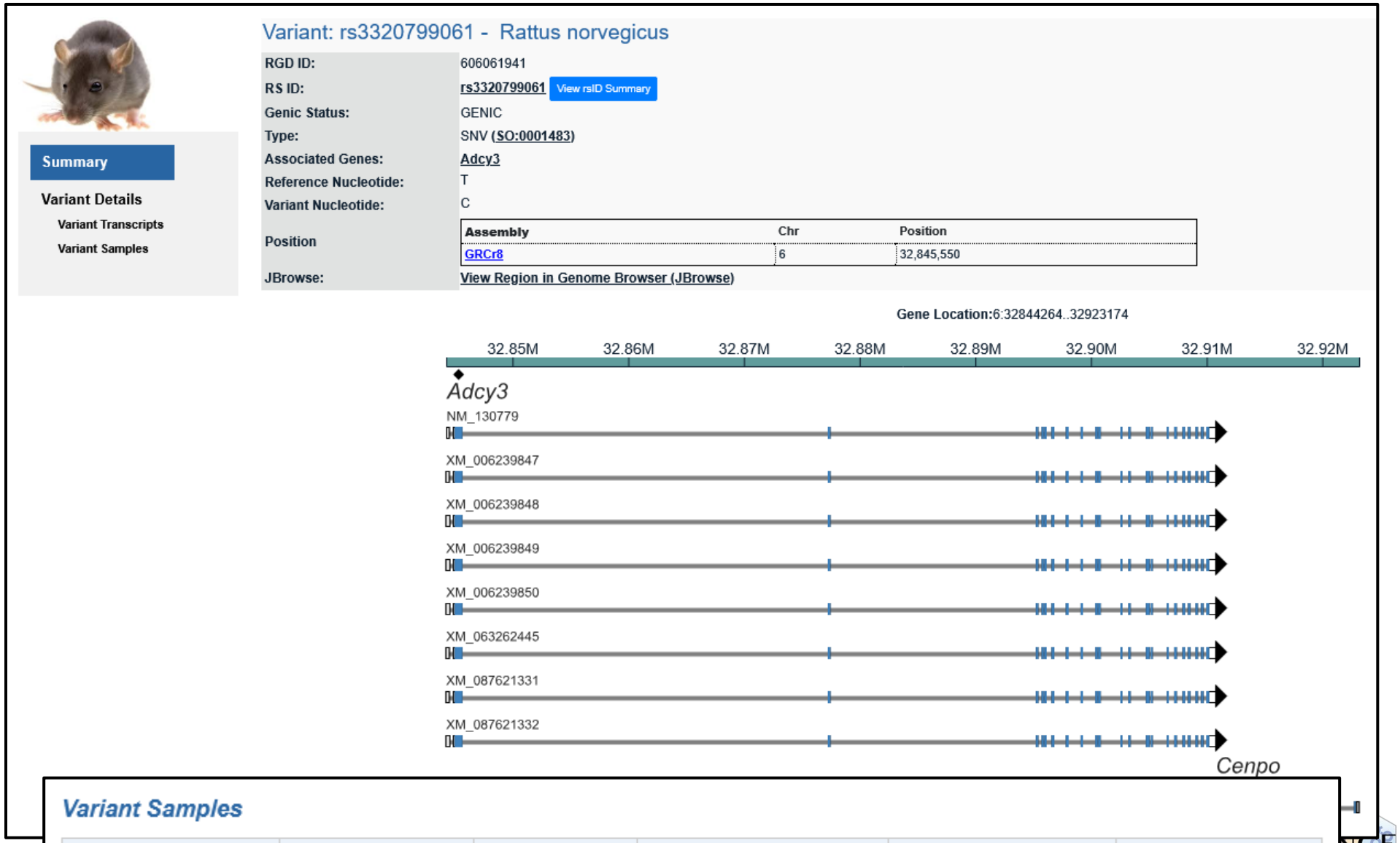
Click on a square in the viewer to bring up a Variant Details popup with information about the variant and its effects.



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RGD: The Variant Report Page

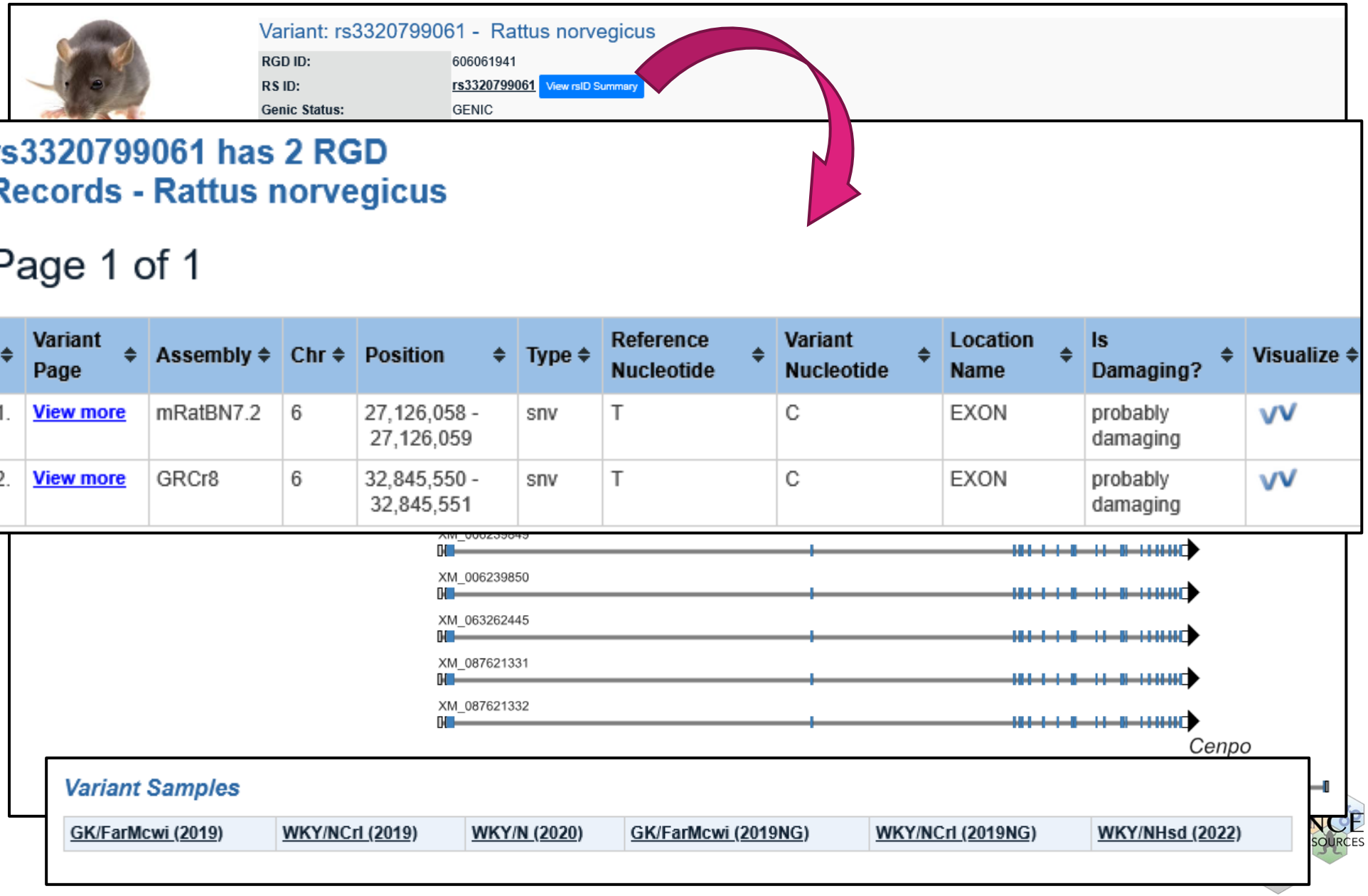
The Variant Details popup links to the RGD variant report page for that variant as shown here.



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RGD: The Variant Report Page

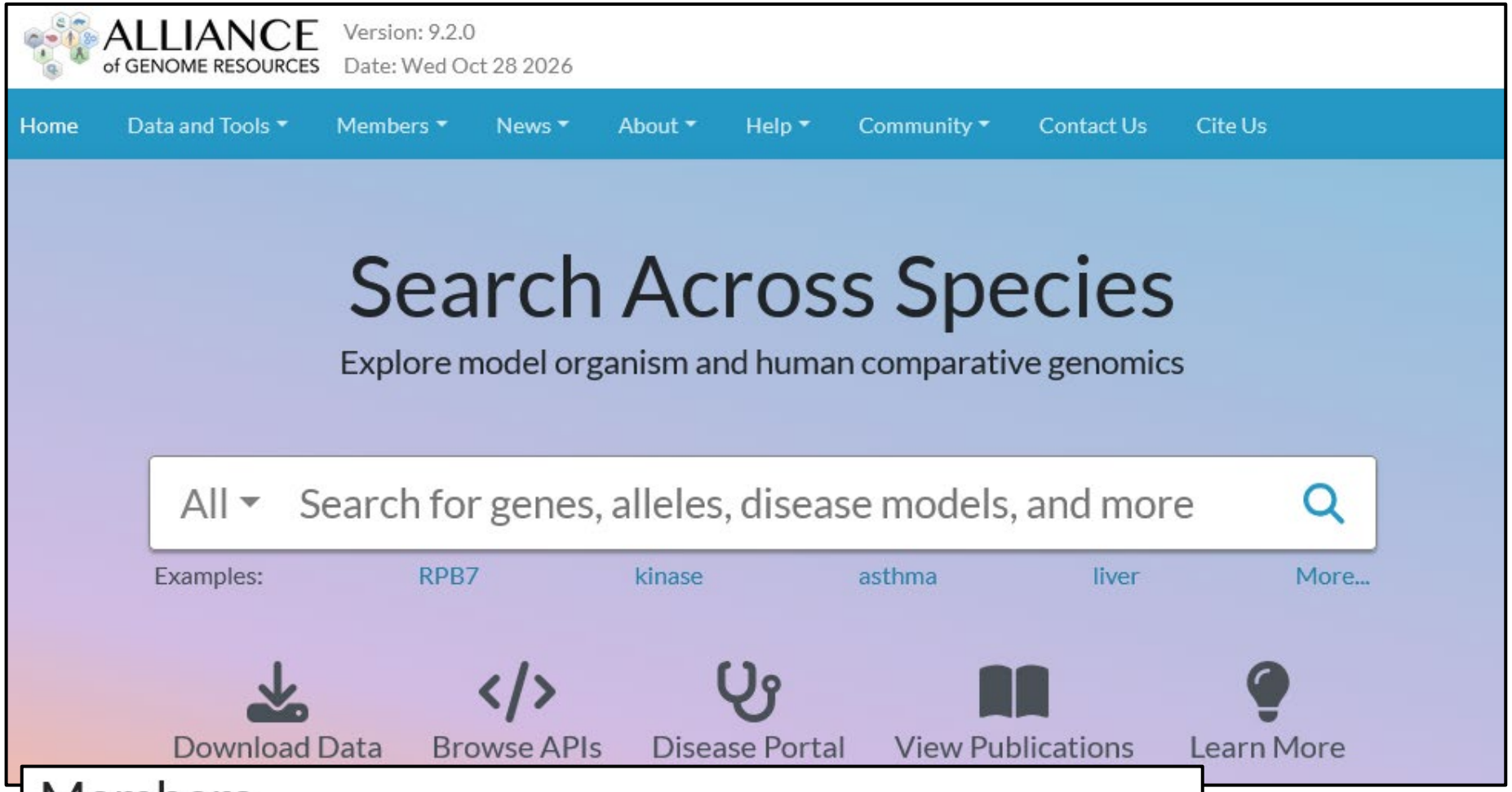
RGD has submitted the strain-specific variants from our sequence analyses to the European Variation Archive (EVA) and imported the assigned rs IDs from them. Using the rs ID, variants can be tracked across assemblies. For example, this variant was seen in both the mRatBN7.2 assembly and GRCr8.

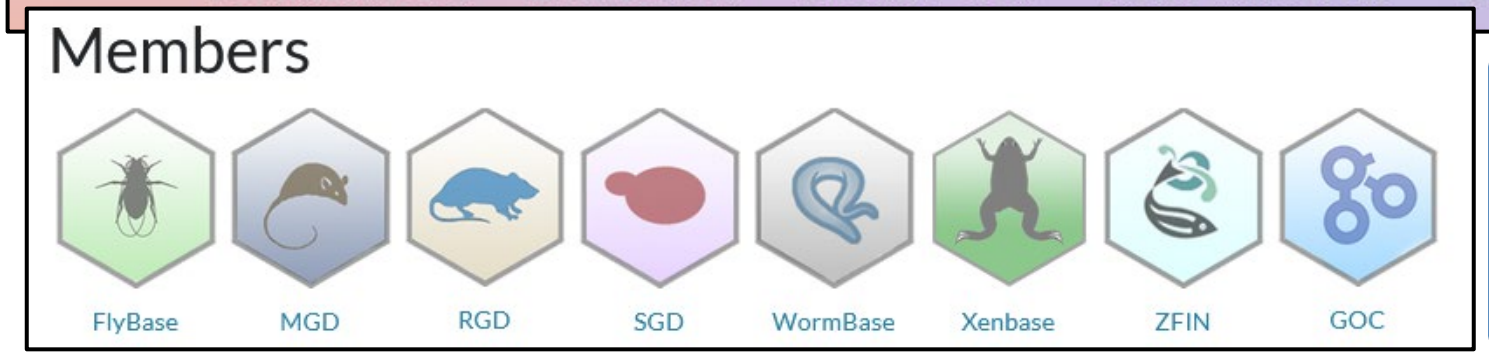


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The Alliance of Genome Resources

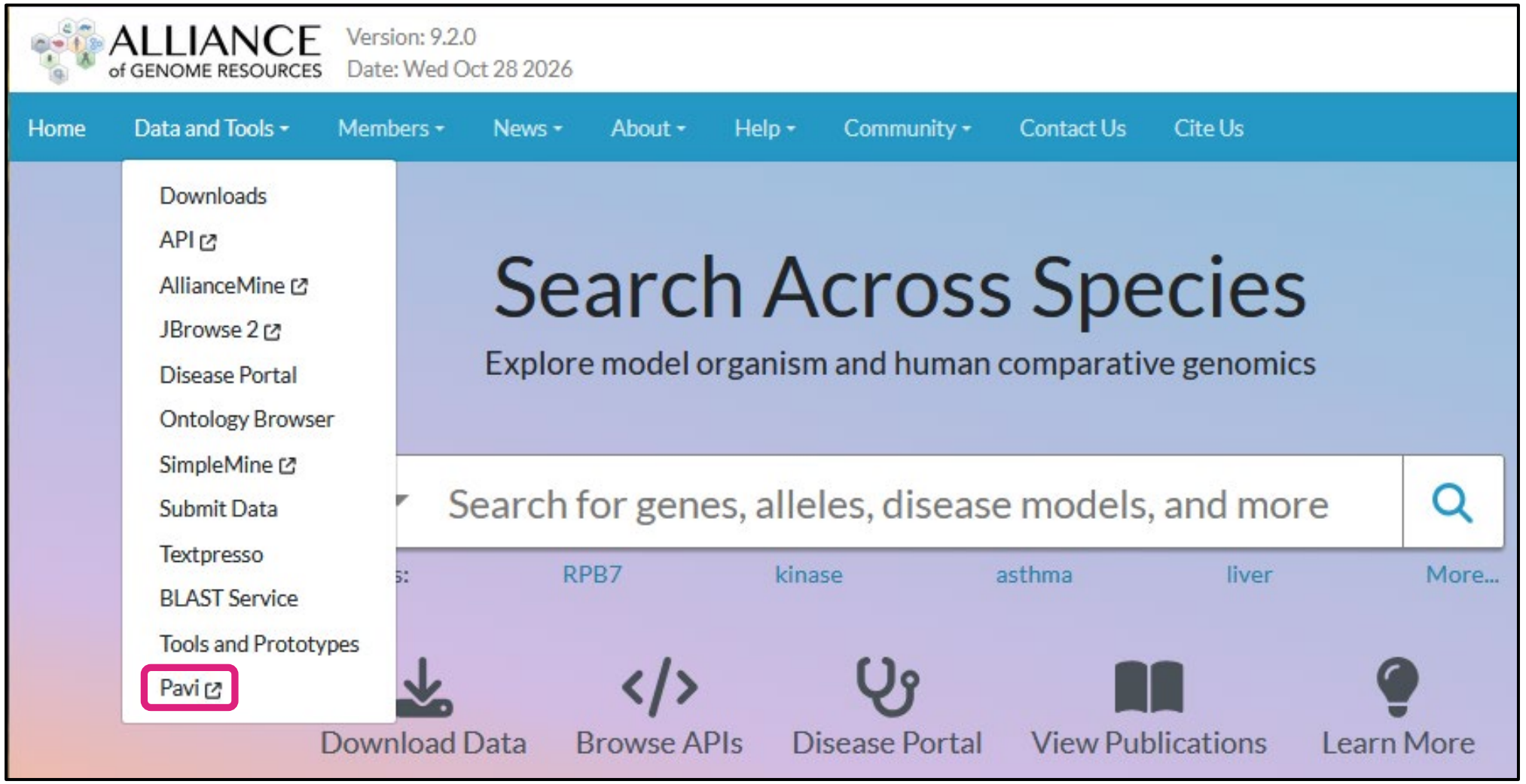
The Alliance of Genome Resources (the Alliance) is "a consortium of seven model organism databases and the Gene Ontology (GO) Consortium whose goal is to provide an integrated view of their data".





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The Alliance: the Protein Annotation and Variant Inspector



Access PAVI, the Protein Annotation and Variant Inspector tool using the "Data and Tools" menu on any Alliance page.

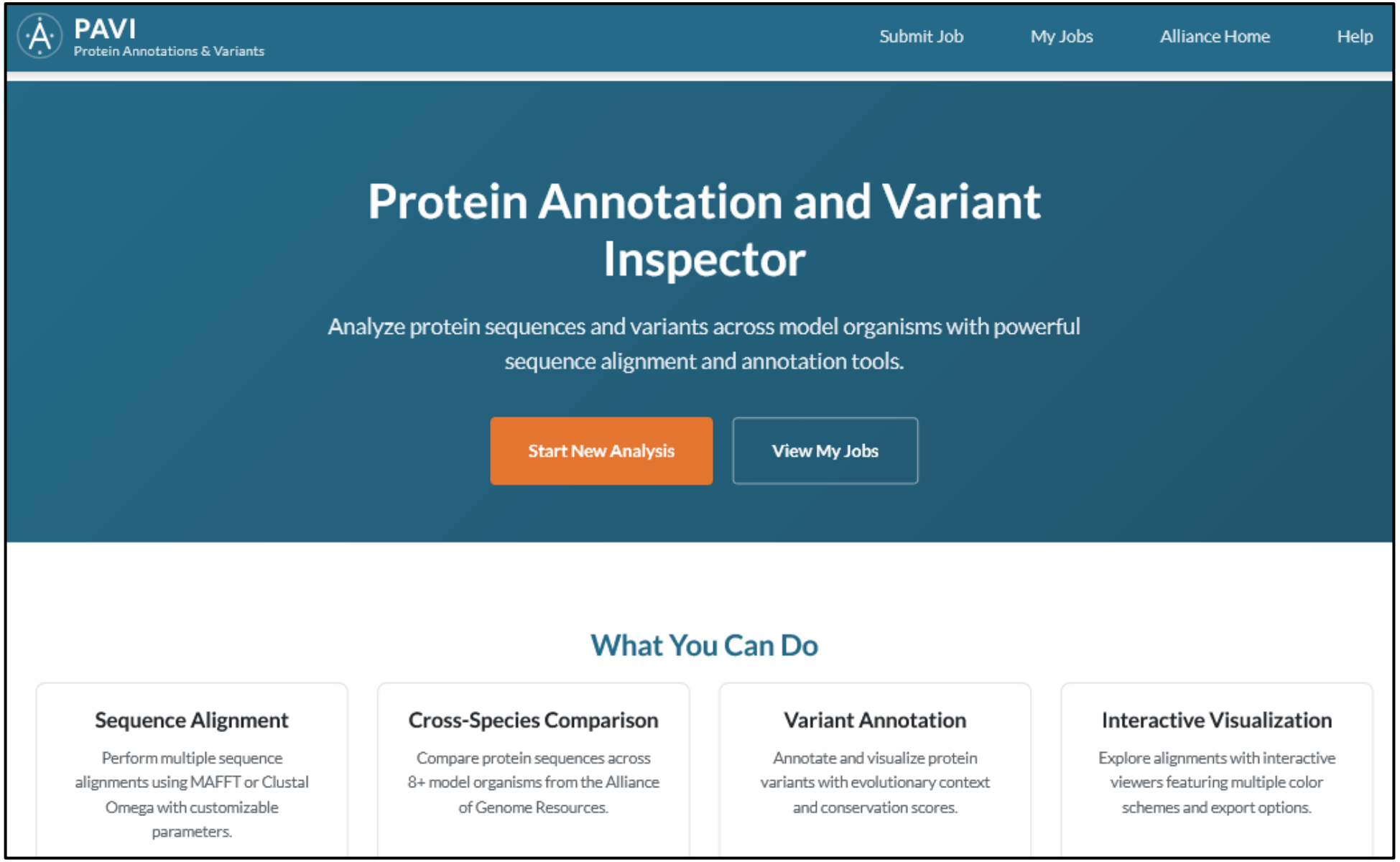
https://rgd.mcw.edu

The Alliance: the Protein Annotation and Variant Inspector

Access PAVI using this QR code or the URL below.

PAVI

https://rgd.mcw.edu



https://www.alliancegenome.org/pavi/

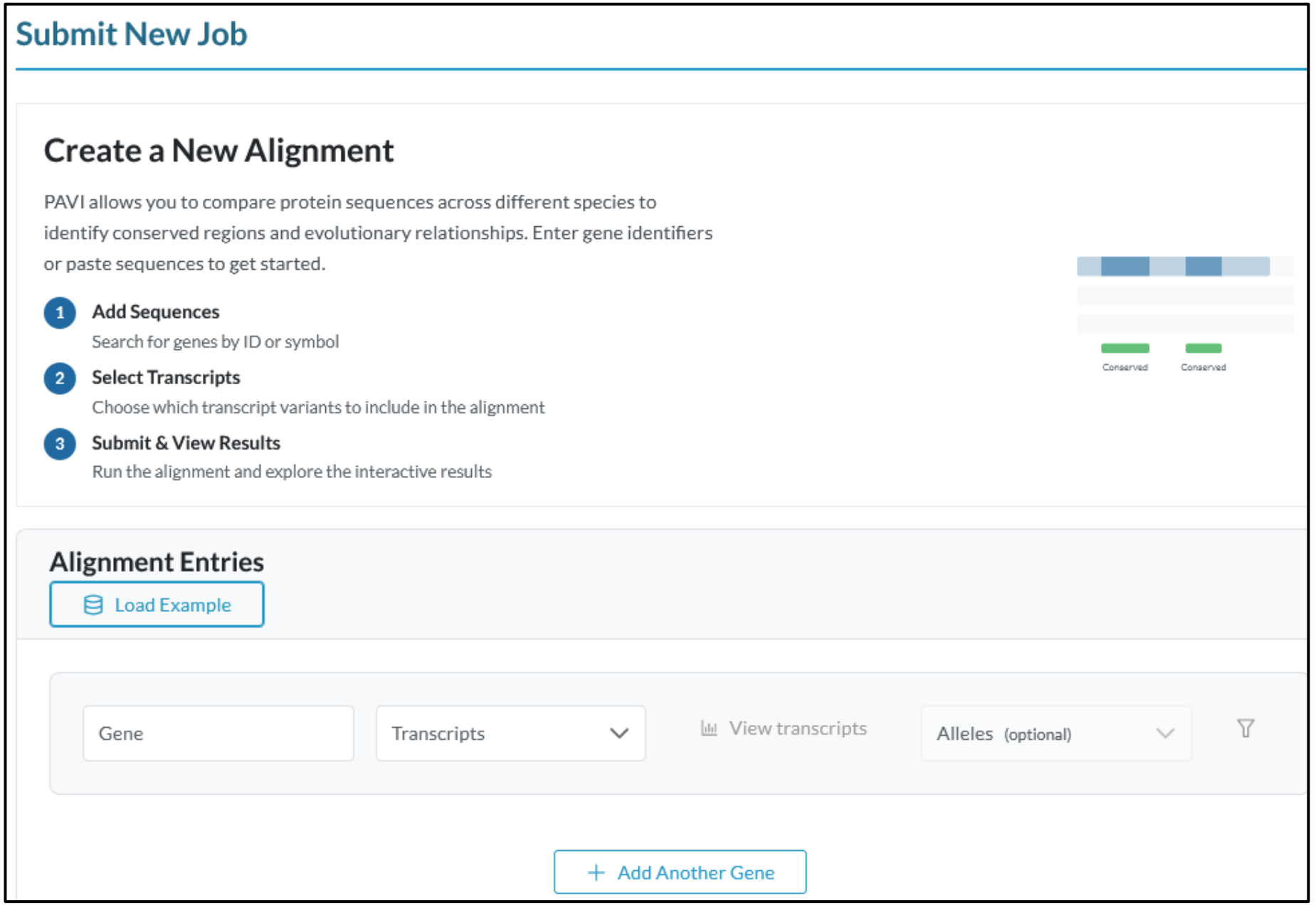
Access the recent webinar about PAVI using this QR code

About PAVI

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The Alliance: the Protein Annotation and Variant Inspector

Clicking "Start New Analysis" on the PAVI home page opens the "Create a New Alignment" page.

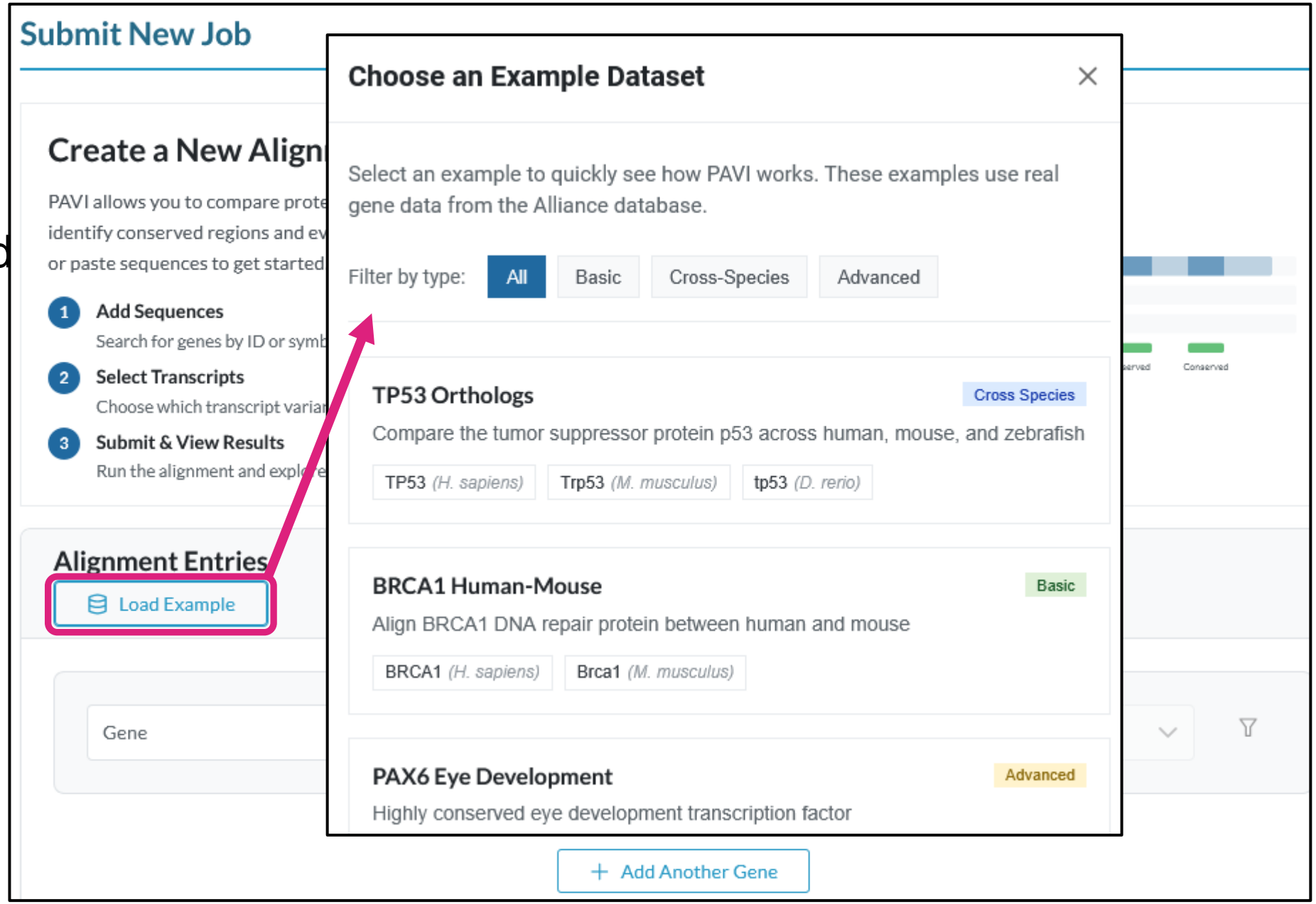


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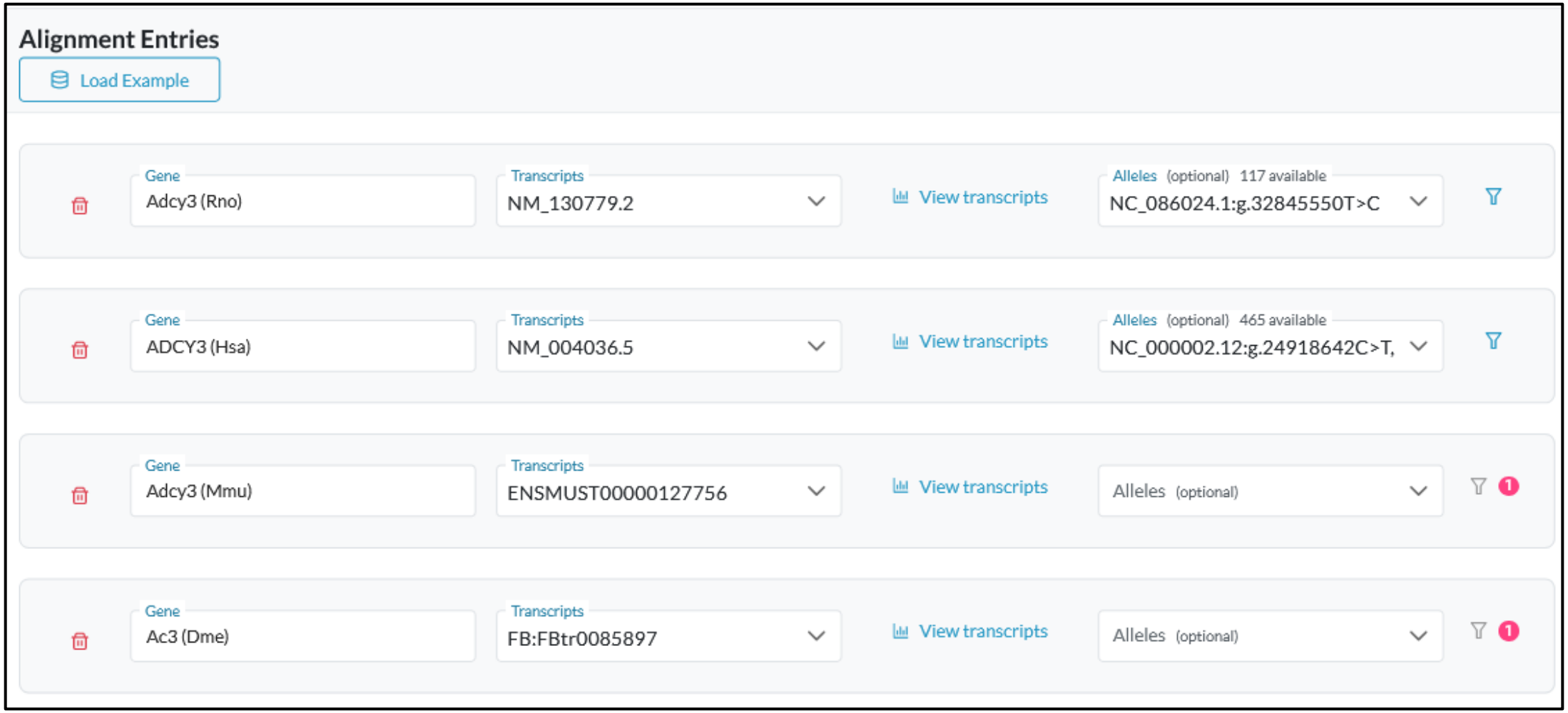
The Alliance: the Protein Annotation and Variant Inspector

To easily see how PAVI works, choose a preconfigured example dataset from the list provided.



https://rgd.mcw.edu

The Alliance: PAVI Alignment Entries



This is the entry page for the example that will be shown in later slides. The values, presented in a format that can be copied and pasted into the tool, are on the next slide.

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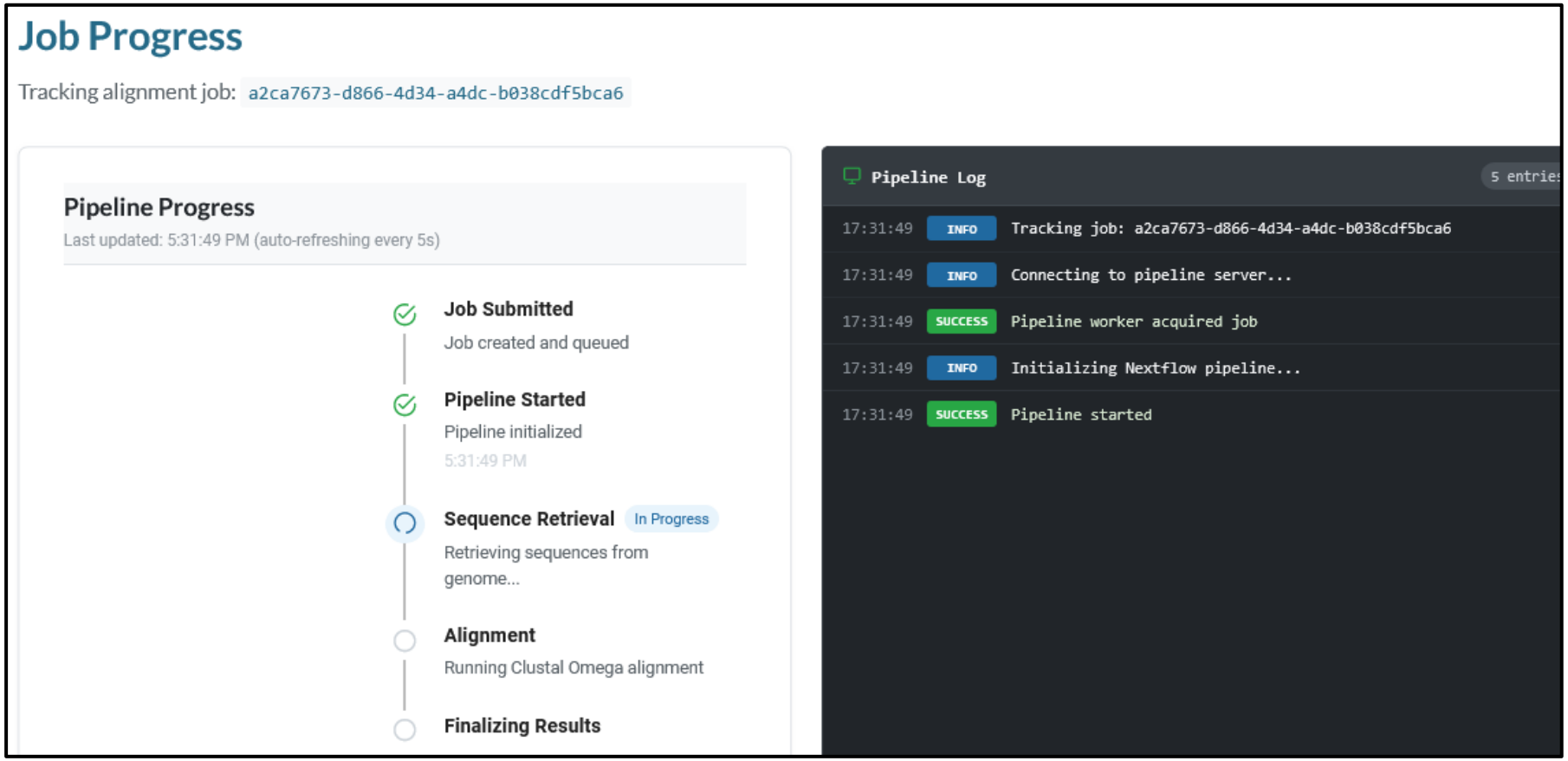
The Alliance: PAVI Alignment Entries

The input values to create the same view as the example are below. Use any or all of these to try the tool yourself. For reference and as will be shown in the coming slides, the variant for rat and the third variant for human are at the same location in the alignment.

Gene (species)	**Transcript**	**Variant(s)**
Adcy3 (Rno)	NM_130779.2	NC_086024.1:g.32845550T>C
ADCY3 (Hsa)	NM_004036.5	NC_000002.12:g.24918642A>T NC_000002.12:g.24918630T>C NC_000002.12:g.24918626A>T
Adcy3 (Mmu)	ENSMUST00000127756	no variant
Ac3 (Dme)	FB:FBtr0085897	no variant

The Alliance: PAVI Alignment Progress

The progress page follows the movement of the process through these steps in real time.



The tool uses the start and stop positions of the exons and coding sequence for each protein-coding transcript to create the transcript sequence from the genome assembly sequence, then translates it into the corresponding protein. If variants have been selected, the transcript sequence is altered based on the variant position and sequence and translated into an altered protein sequence. The reference proteins and any variant proteins are then aligned.

https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

Alignment details

Variant details

Adjust color scheme

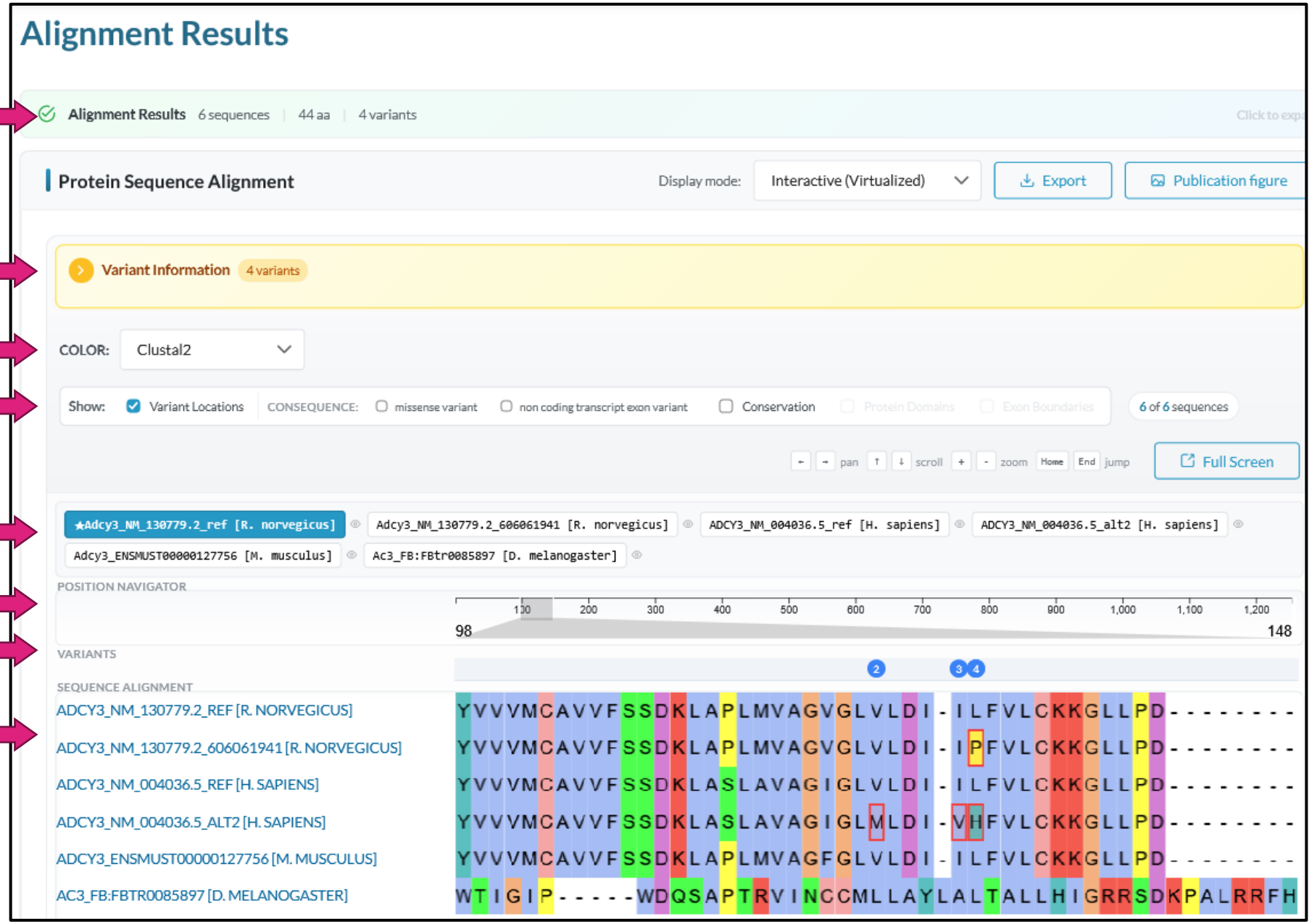
Display options

Aligned sequences

Alignment position

Variant positions

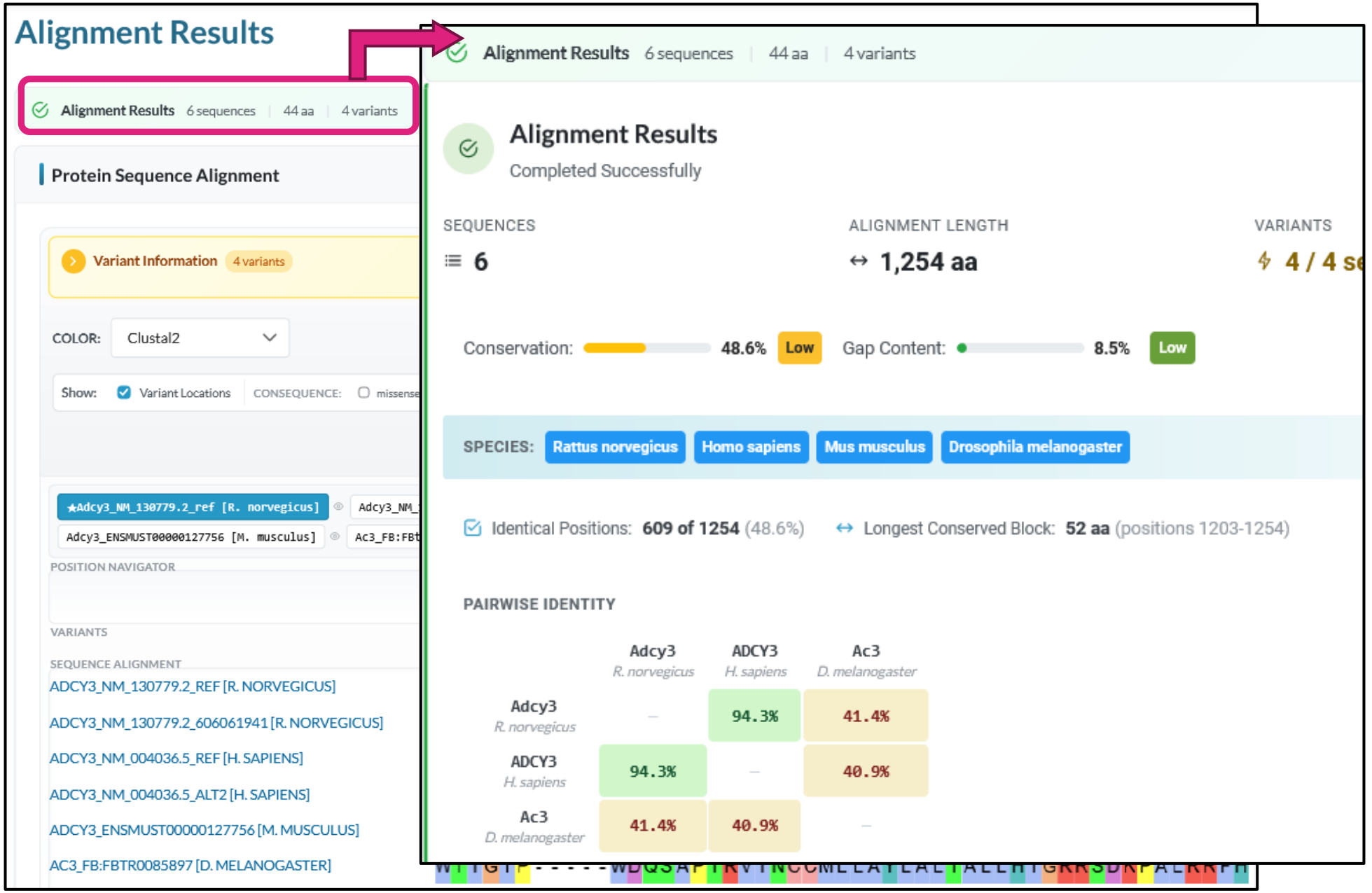
Protein sequence alignment



https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

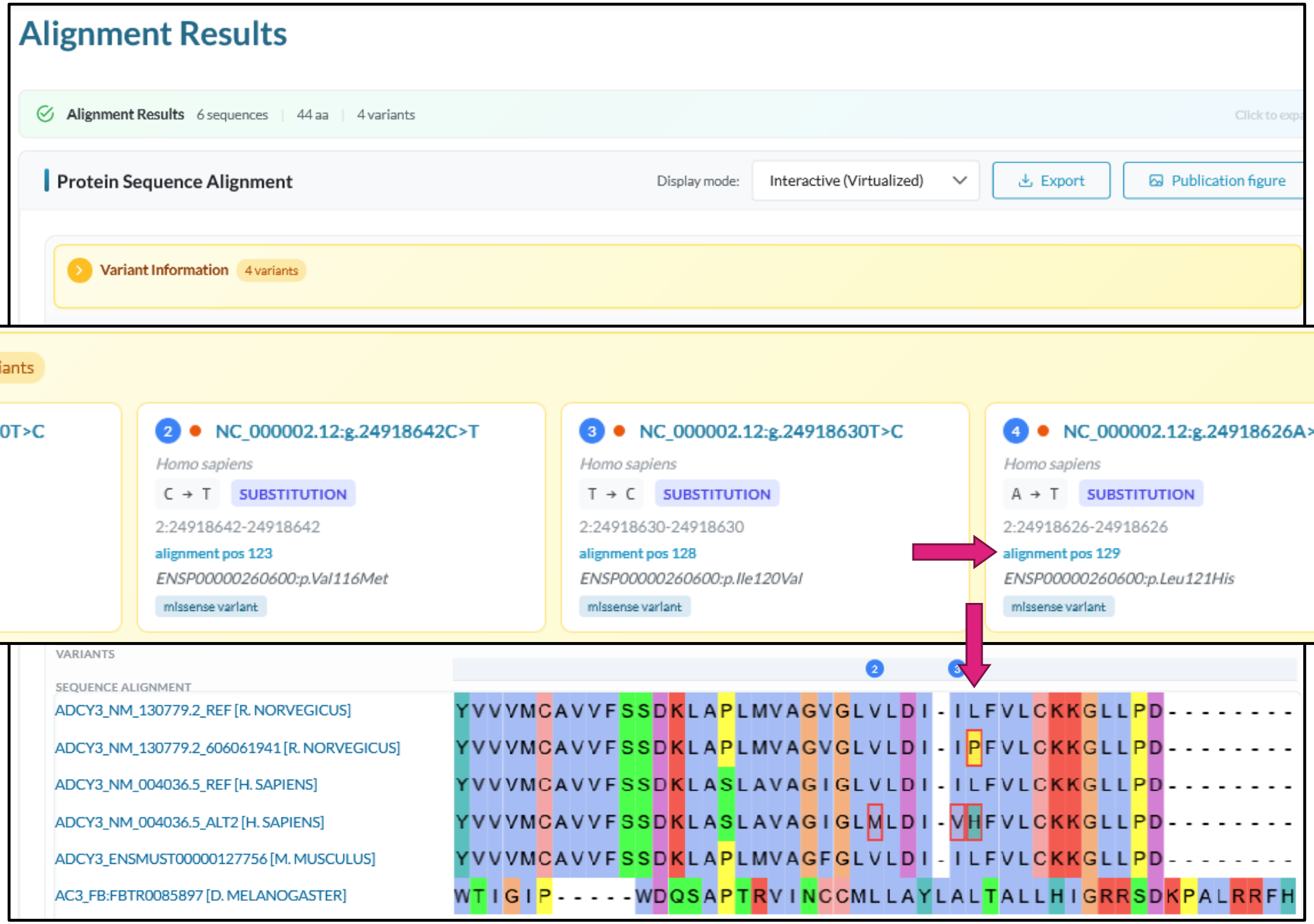
Details about the alignment are available but are not shown by default.



https://rgd.mcw.edu

The Alliance: PAVI Alignment Results

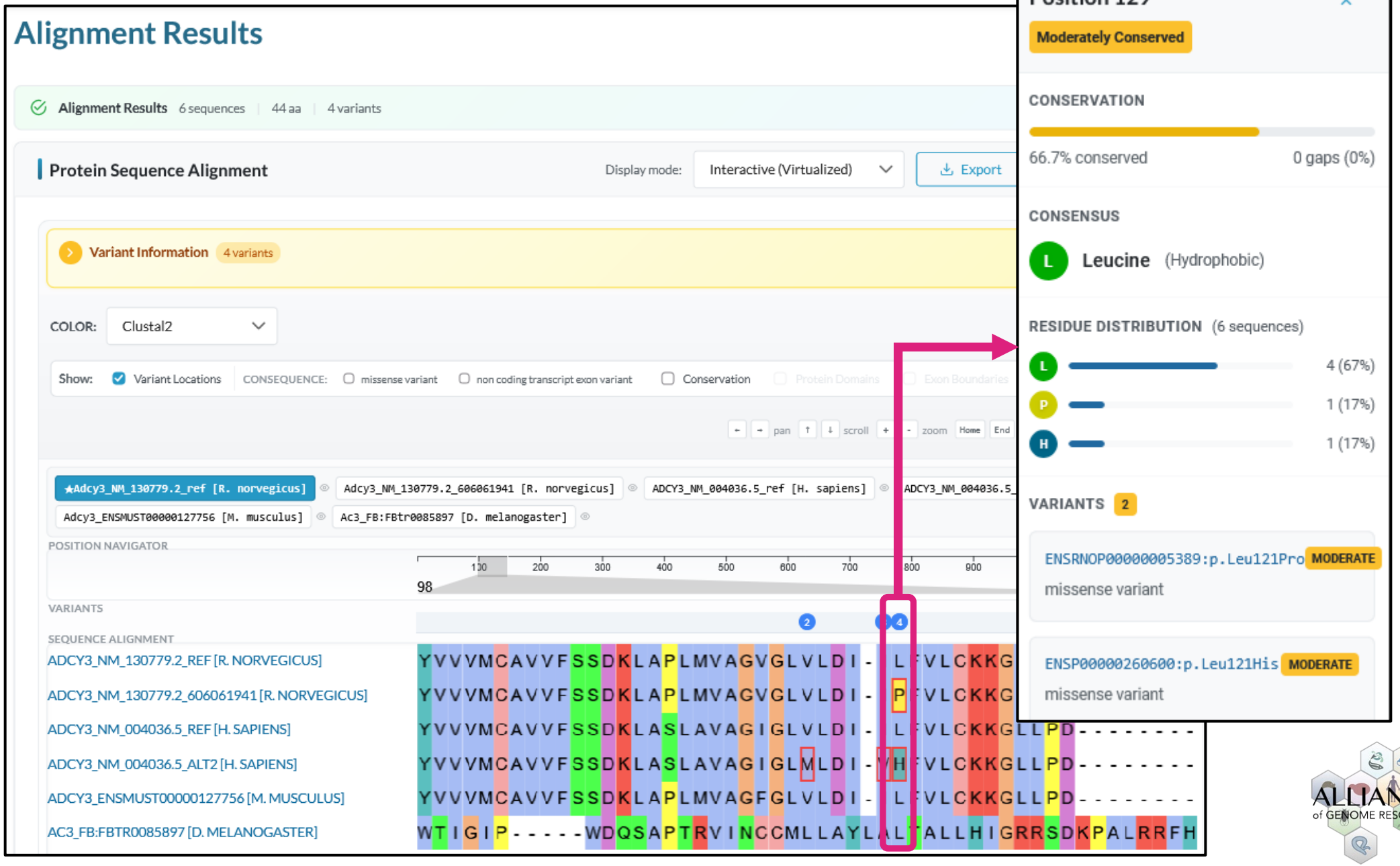
Information about the selected variants and their positions in the alignment are also available but not shown by default.



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The Alliance: PAVI Alignment Results

Click on a column in the alignment to to view a detail popup about that position.



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Thank you!

jrsmith@mcw.edu
https://rgd.mcw.edu



We acknowledge the RGD Team

(Principal Investigator: Anne Kwitek, PhD and Co-Investigator, Mindy Dwinell, PhD)

And the Alliance of Genome Resources Variants Working Group, particularly Paulo Nuin, primary developer of PAVI.

We gratefully acknowledge our funders:

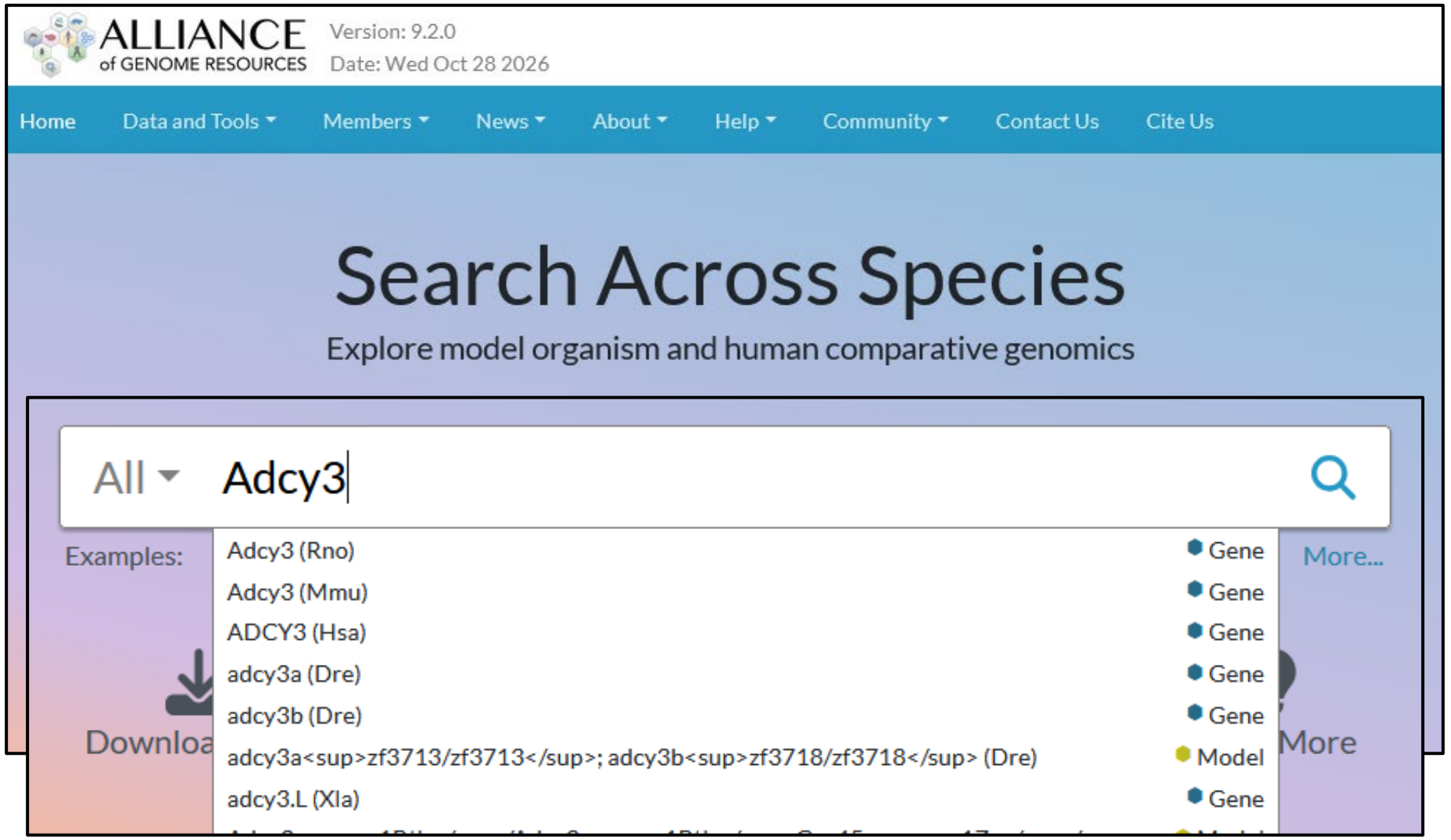
RGD is funded by the National Heart, Lung, and Blood Institute (NHLBI, R01HL064541); the Alliance of Genome Resources, and RGD as a founding member, are funded by the National Human Genome Research Institute (NHGRI, U24HG010859); the HRDP is funded by the NIH Office of the Director (R24OD024617 and R24OD039886).

We also acknowledge our collaborators and the researchers who contribute data and who faithfully use our website and data!

https://rgd.mcw.edu

The following slides give information about the Alliance gene and variant pages which may or may not be covered in the workshop, depending on time.

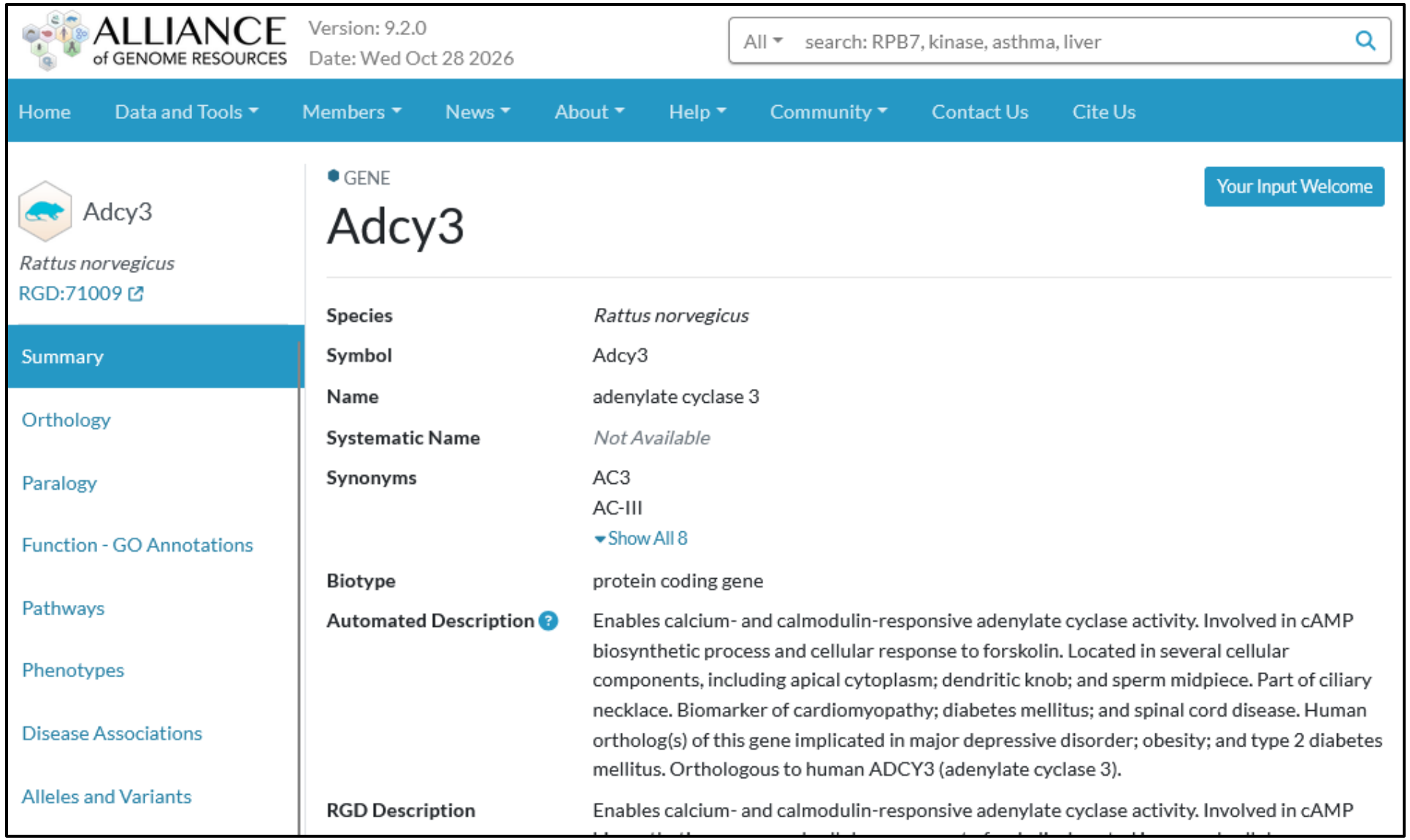
The Alliance of Genome Resources: Genes and Variants



The Alliance gene search; for this example, we will select the rat gene.

https://rgd.mcw.edu

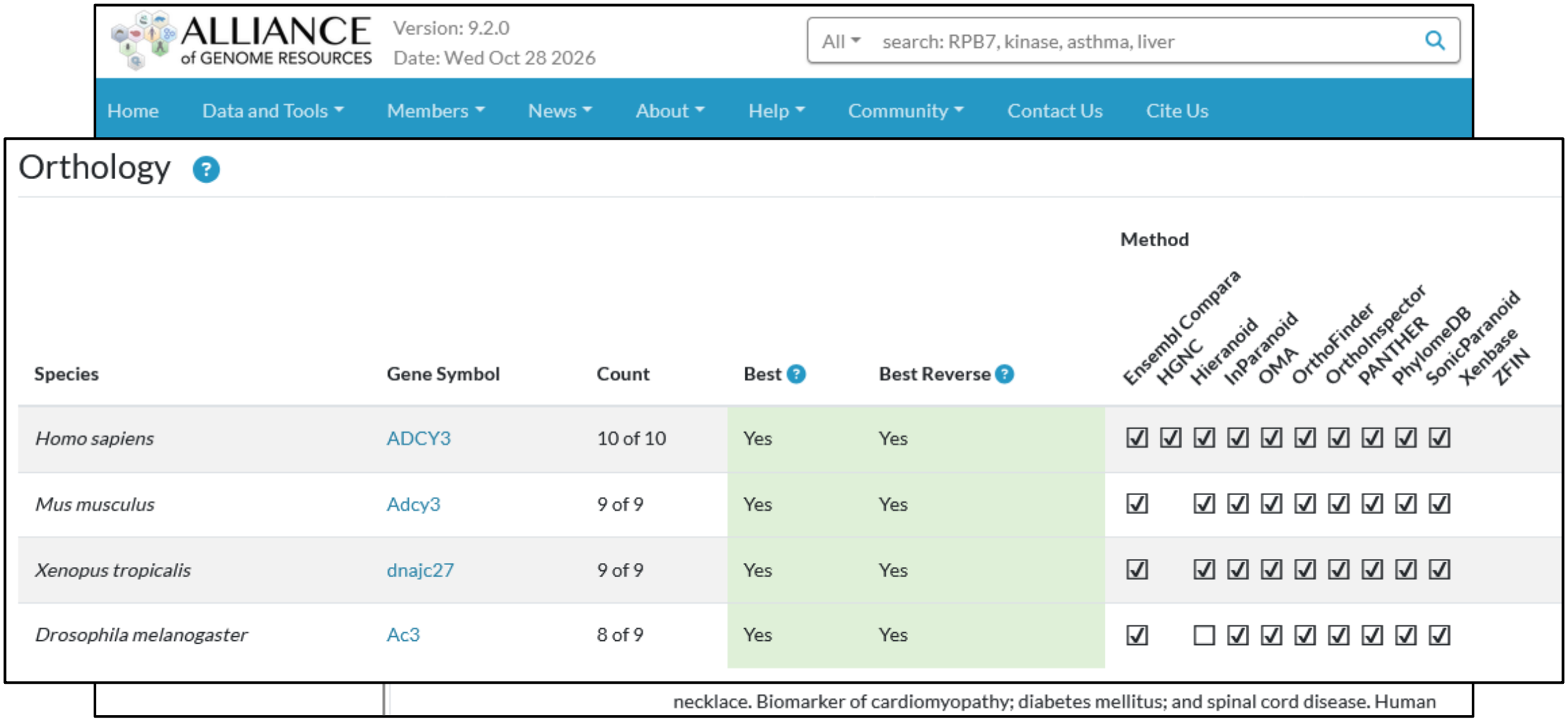
The Alliance of Genome Resources: Genes and Variants



https://rgd.mcw.edu

The Alliance gene page; the summary section is shown here. Use links in the left sidebar to easily access other sections of the page.

The Alliance of Genome Resources: Genes and Variants

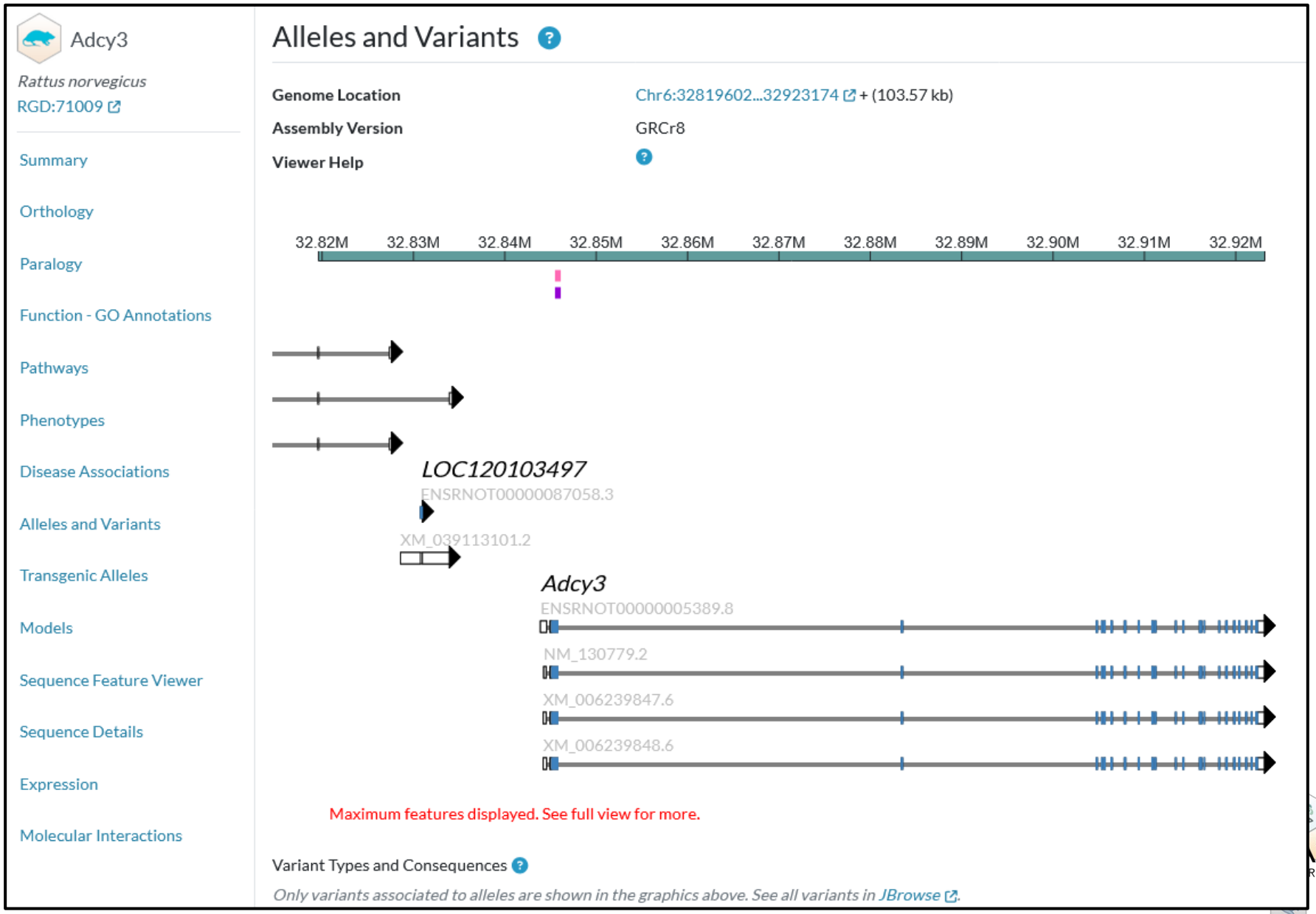


The orthology section of the rat Adcy3 gene page.

https://rgd.mcw.edu

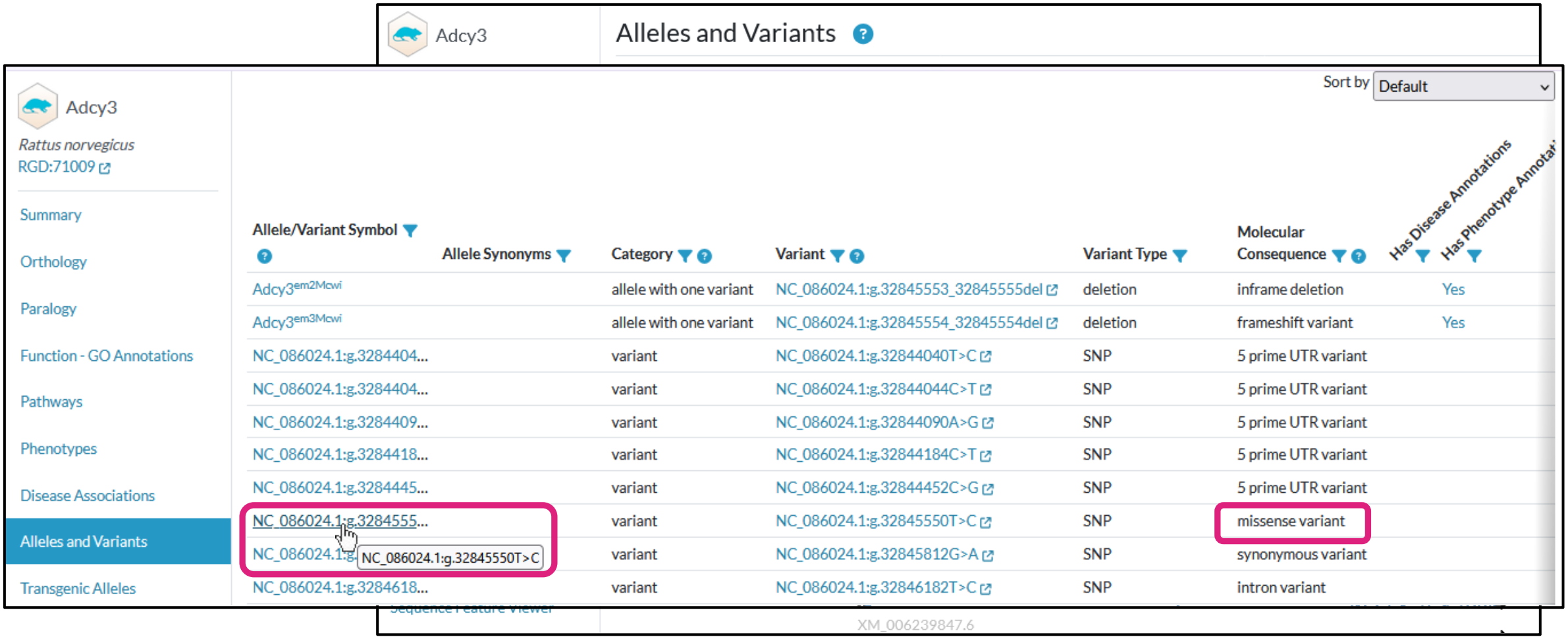
The Alliance of Genome Resources: Genes and Variants

The Alleles and Variants section of the gene page, in this case, the rat Adcy3 gene page. Shown here is the sequence viewer.



https://rgd.mcw.edu

The Alliance of Genome Resources: Genes and Variants



The alleles and variants table on the rat Adcy3 gene page. The circled variant is the rat one used in the PAVI example. The name in the first column links to the allele/variant page.

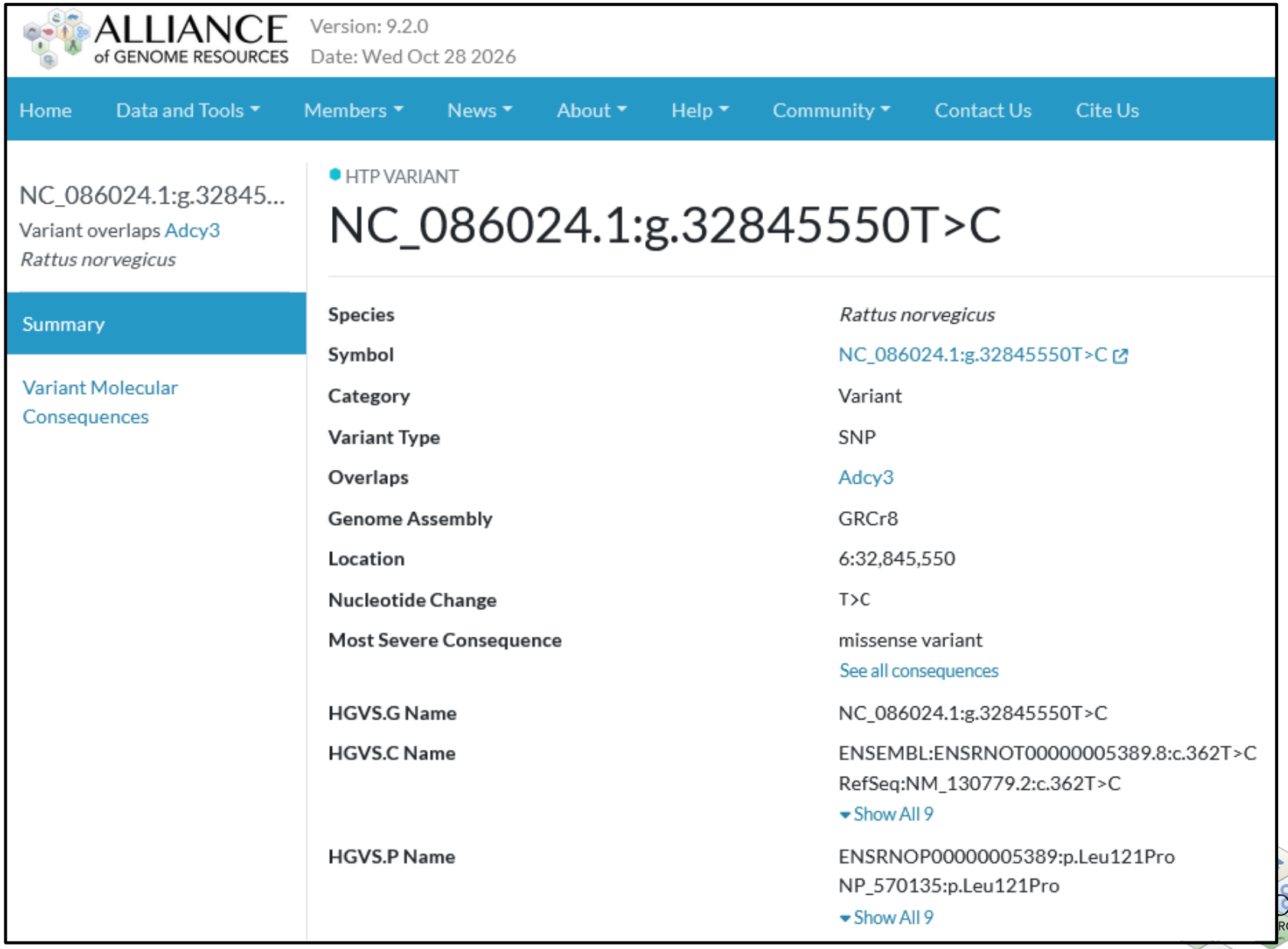
https://rgd.mcw.edu

The Alliance of Genome Resources: Genes and Variants

The Allele/Variant page gives detailed information about an allele, including its variants

In the case of high throughput (HTP) variants, as shown here, details about the variant are provided and HGVS nomenclature is assigned.

HGVS nomenclature is used as the unique identifier for the variant. This variant, NC_086024.1:g.32845550T>C, is the one used in the PAVI example.



https://rgd.mcw.edu

The Alliance of Genome Resources: Genes and Variants

The Variant Molecular Consequences section shows transcript-specific effects of the variant including SIFT and Polyphen predictions from Ensembl's Variant Effect Predictor (VEP).

Variant Molecular Consequences

Hide all details	Sequence Feature	Sequence Feature Type	Associated Gene	Location	Molecular Consequence	Codon and amino acid change
Show details	ENSEMBL:ENSRNOT00000005389.8	protein_coding	Adcy3	Exon 2/22	- missense variant	361 cTt 363 [121] L [121] => cCt P
Hide details	RefSeq:NM_130779.2	protein_coding	Adcy3	Exon 2/22	- missense variant	361 cTt 363 [121] L [121] => cCt P

Predicted effect of NC_086024.1:g.32845550T>C on **RefSeq:NM_130779.2**

Variant Type	SNP
HGVS Names	NP_570135:p.Leu121Pro ▾ Show All 3
Affected Sequence Feature Name	RefSeq:NM_130779.2
Affected Sequence Feature Type	protein_coding
Affected Sequence Feature Accession	RefSeq:NM_130779.2
Variant Location In Feature	Exon 2/22
Molecular Consequence	missense variant
VEP Impact	moderate
SIFT Prediction (Score)	deleterious (0)
PolyPhen Prediction (Score)	probably damaging (1)
Position of Variant in Transcript	803
Position of Variant in Coding Sequence	362
Position of Variant in Protein	121
Protein Change	

VEP Impact	moderate
SIFT Prediction (Score)	deleterious (0)
PolyPhen Prediction (Score)	probably damaging (1)

Reference				Variant
361	cTt	363	=>	cCt
[121]	L	[121]		P

https://rgd.mcw.edu

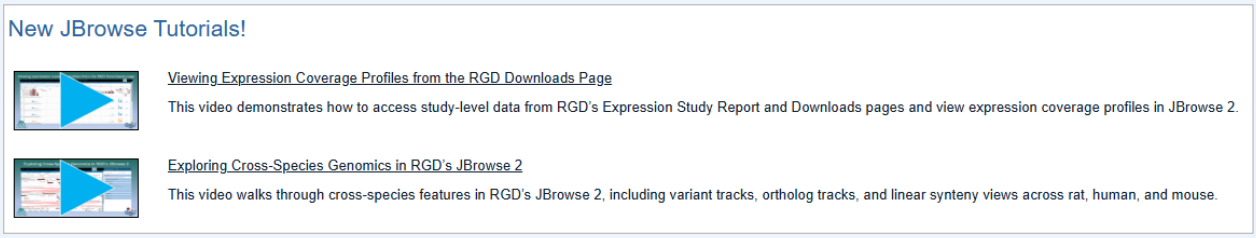
23rd Annual Meeting of the Complex Trait Community

Workshop: Comparative Genomics in JBrowse 2 at the Rat Genome Database (RGD)

November 11, 2026

RGD website: rgd.mcw.edu

RGD Video Tutorials: https://rgd.mcw.edu/wg/home/rgd_rat_community_videos/rgd-tutorials/



JBrowse website: https://jbrowse.org/jb2/

Paper resource: PMID: 37069644

Outline:

- Access JBrowse 2 through RGD
- Navigation
- Variant Tracks
- Ortholog Tracks
- Linear Genome Views
- Human Variant Track
- Synteny Track
- Linear Synteny View
- Download and Open Expression Tracks

> *Note: there was a recent JBrowse 2 update, so your view and what is shown in the workshop may slightly differ, but the overall process and content will be the same.*

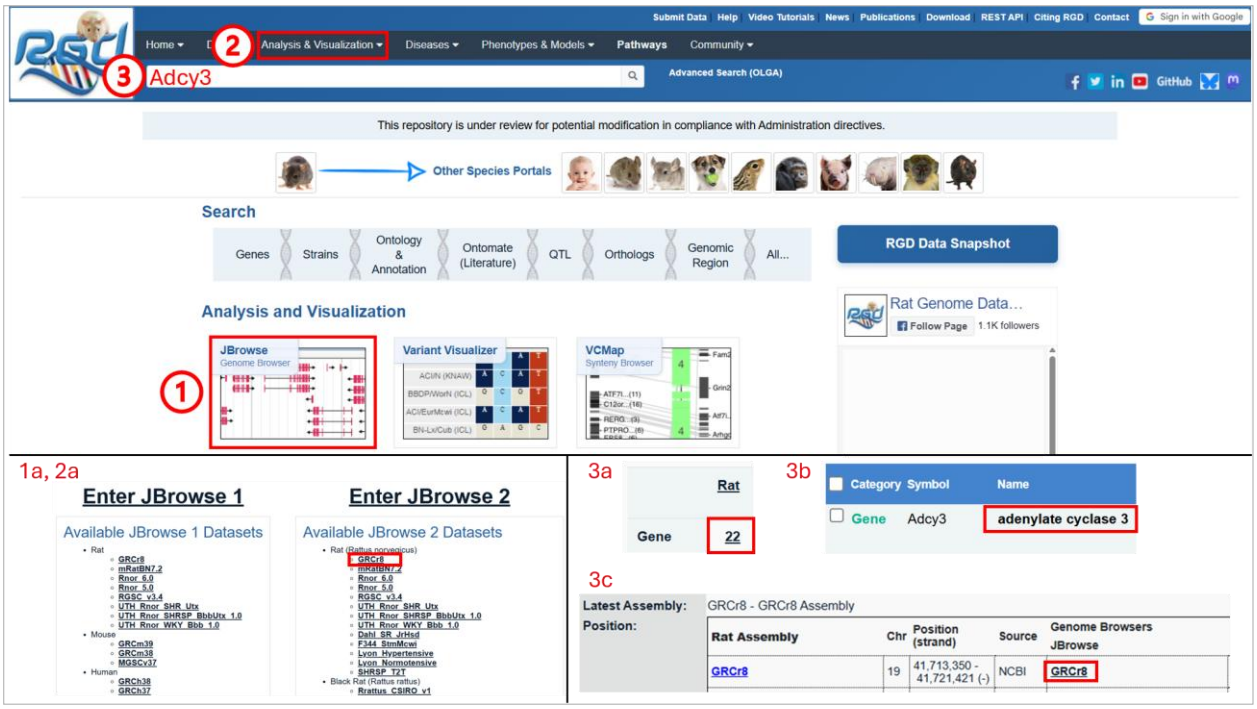
Access JBrowse 2 through RGD (rgd.mcw.edu)

There are multiple ways to access JBrowse from RGD. Here are a few:

1) Homepage → "Analysis and Visualization" section → JBrowse button
2) Homepage → top menu "Analysis & Visualization" tab → "JBrowse (Genome Browser)"
   a) These options will bring you to a new page with options to enter JBrowse 1 or JBrowse 2 by selecting the species assembly of interest. This will open a default view in JBrowse.

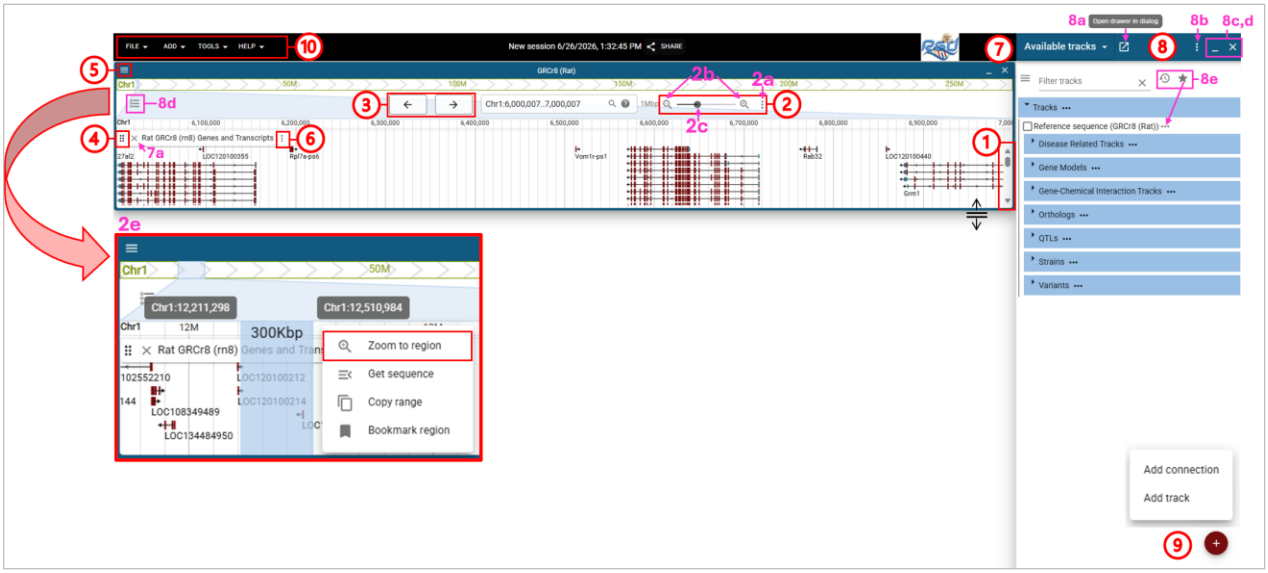
> *Note: **GRCr8 (rn8)** is the most recent assembly for rat, and there are new rat genome annotations now available from NCBI, including annotations for GRCr8 (strain: BN/NHsdMcwi, RRID:RGD_61498) and SHRSP_T2T (strain: SHRSP/BbbUtx, RRID:RGD_8142383).*

3) Homepage → search for a gene (e.g., *Adcy3*)
   a) Select linked number in the gene row for the species of interest (e.g., rat)
   b) Select the gene from the list (e.g., adenylate cyclase 3)
   c) Gene page → "Position" section → "Genome Browsers" → "JBrowse" → select the linked assembly (e.g., GRCr8). This will automatically open to the position of the *Adcy3* gene in JBrowse 2



> *Note: Updates are coming to the RGD Report pages (e.g., Gene page) soon. The layout and content will remain the same, but there will be a new look and feel!*

Navigation



https://rgd.mcw.edu/jbrowse2/?session=share-50A4mlXTRb&password=JxBRU

Adjust track/window size: click and drag when this icon appears ⇳

1) **Scroll** in track up/down with mouse wheel or by clicking the arrows/dragging the bar in the vertical scroll bar
2) **Zoom options:**
   a) 3 vertical dots → zoom in/out options (e.g., Zoom in 100x)
   b) +/- magnifying glass icons
   c) Sliding bar
   d) Ctrl + mouse scroll wheel in a track (not shown)
   e) Click and drag in the detail position bar to select the region → "Zoom to region"
3) **Pan options:**
   a) In a track, click and drag left/right
   b) Left/right arrow buttons scroll left/right

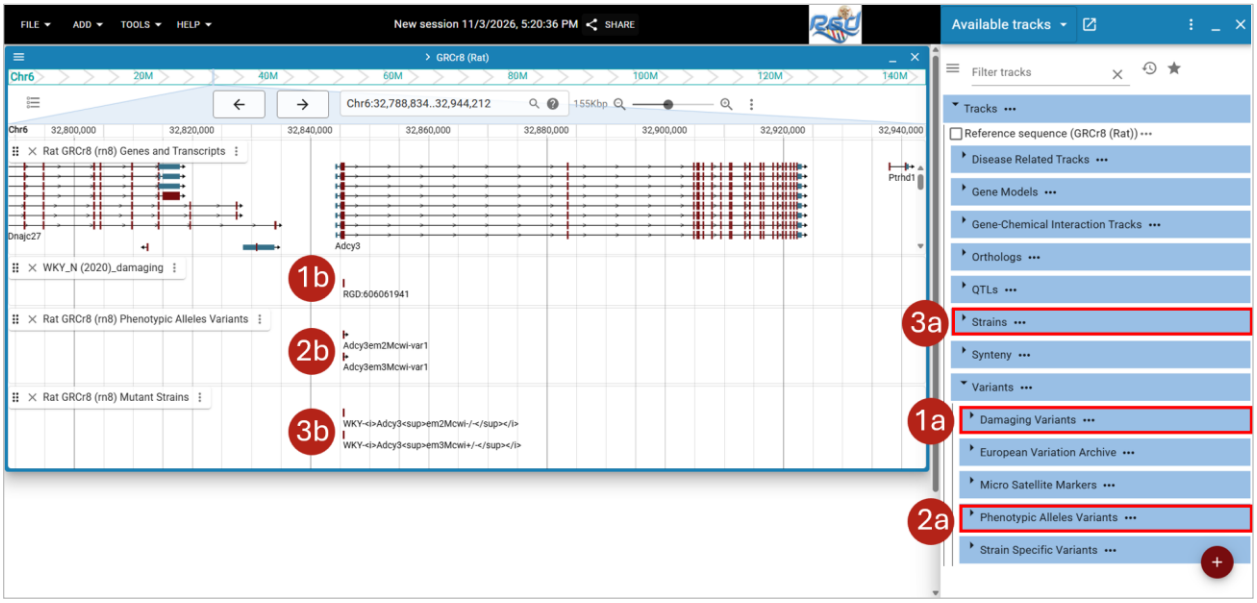
Bonus navigation tips:

4) Re-order tracks: click and drag up/down on the drag handle on the track label (six vertical dots)
5) Re-ordering views: unlike tracks, views cannot be reordered by drag-and-drop. Instead, use the view menu (hamburger icon) and select "Move up"/"Move down"
6) Track menu: About track info, track actions, display settings, and more
7) Each window can be minimized or X'ed out
   a) You can also exit out of individual tracks within the window (see X next to the 4 callout in the track name)

8) Available tracks:
   a) Next to the “Available tracks” title is an icon in which you can open this menu as a dialog box
   b) The vertical 3 dots (right of the 8 label) allow you to toggle the “Available tracks” menu to the left or right of the screen/track window
   c) On the far right of the header, you can minimize the track menu. To re-open, there will be a button on the bottom right of the screen to re-open the track menu
   d) Use the X to completely close the available tracks menu. To re-open, select the hamburger menu (3 horizontal lines beginning with a circle)
   e) Recently used tracks (clock with arrow). Favorite tracks (star): you can mark your favorite tracks from the Track menu (in “Available tracks” menu, find the track and select the 3 horizontal dots next to the track name and select “Add to favorites”). Click on the clock or star icons to access recently used or favorited tracks
9) Red circular plus (+) icon in the “Available tracks” menu → add connection or track
   a) Also see top left menu in the black bar → “FILE” → Open track/connection options
   b) Also see 3-line hamburger menu at the top of the “Available tracks” menu, next to “Filter tracks” → open faceted track selector, Add track, Connections, Sort, Collapse
10) Menu options (File, Add, Tools, Help)

> TIP: Undo and redo: you can undo/redo any action via “TOOLS” → “Undo”/“Redo”, or use keyboard shortcuts Ctrl+Z/Cmd+Z (undo) and Ctrl+Y/Cmd+Shift+Z (redo)

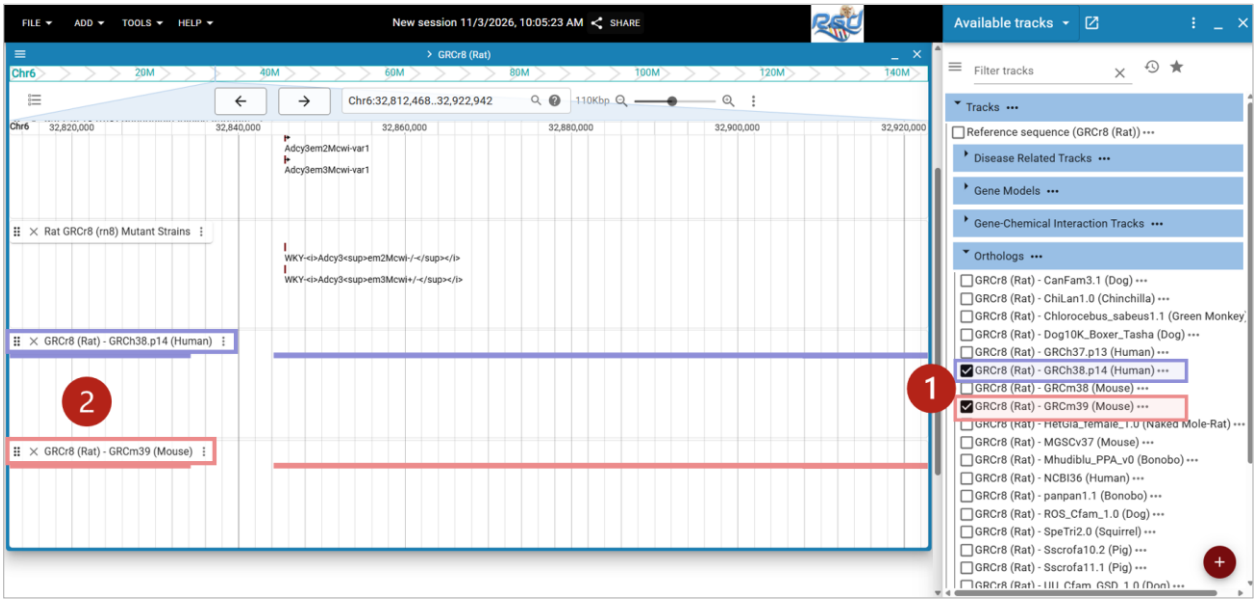
Variant Tracks



https://rgd.mcw.edu/jbrowse2/?session=share-dc_9a-x7yx&password=T9Bu3

1) **Damaging Variants**
   a) In the "Available tracks" menu, open the "Variants" → "Damaging Variants" subcategories. Select the "WKY_N (2020)_damaging" track
   b) Damaging variant in WKY/N strain in *Adcy3* (RGD:606061941)
      i) Clicking on the variant will provide "Feature details" in the right sidebar. You can also search the RGD ID number on the RGD site for more information
2) **Phenotypic Alleles Variants**
   a) In the "Available tracks" menu, open the "Variants" → "Phenotypic Alleles Variants" subcategories. Select the "Rat GRCr8 (rn8) Phenotypic Alleles Variants" track
   b) Here we see 2 variants, including Adcy3em2Mcwi-var1 associated with the Adcy3^{em2Mcwi} allele carried by the WKY-*Adcy3*$^{em2Mcwi-/-}$ strain
      i) Clicking on the variant will provide "Feature details" in the right sidebar
3) **Strains**
   a) In the "Available tracks" menu, open the "Strains" subcategory. Select the "Rat GRCr8 (rn8) Mutant Strains" track
   b) Here we see 2 mutant rat strains: WKY-*Adcy3*$^{em2Mcwi-/-}$ and WKY-*Adcy3*$^{em2Mcwi+/-}$
      i) Clicking on the strain will provide "Feature details" in the right sidebar, including phenotype terms (MP) annotated to the strain

Ortholog Tracks

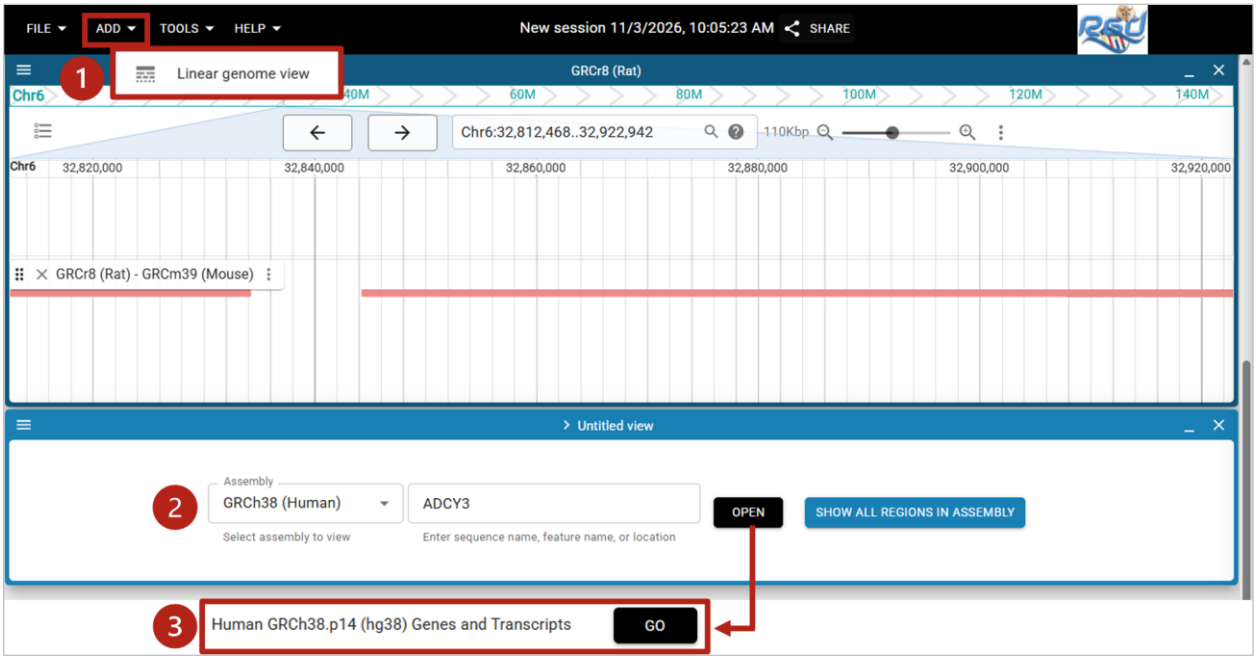


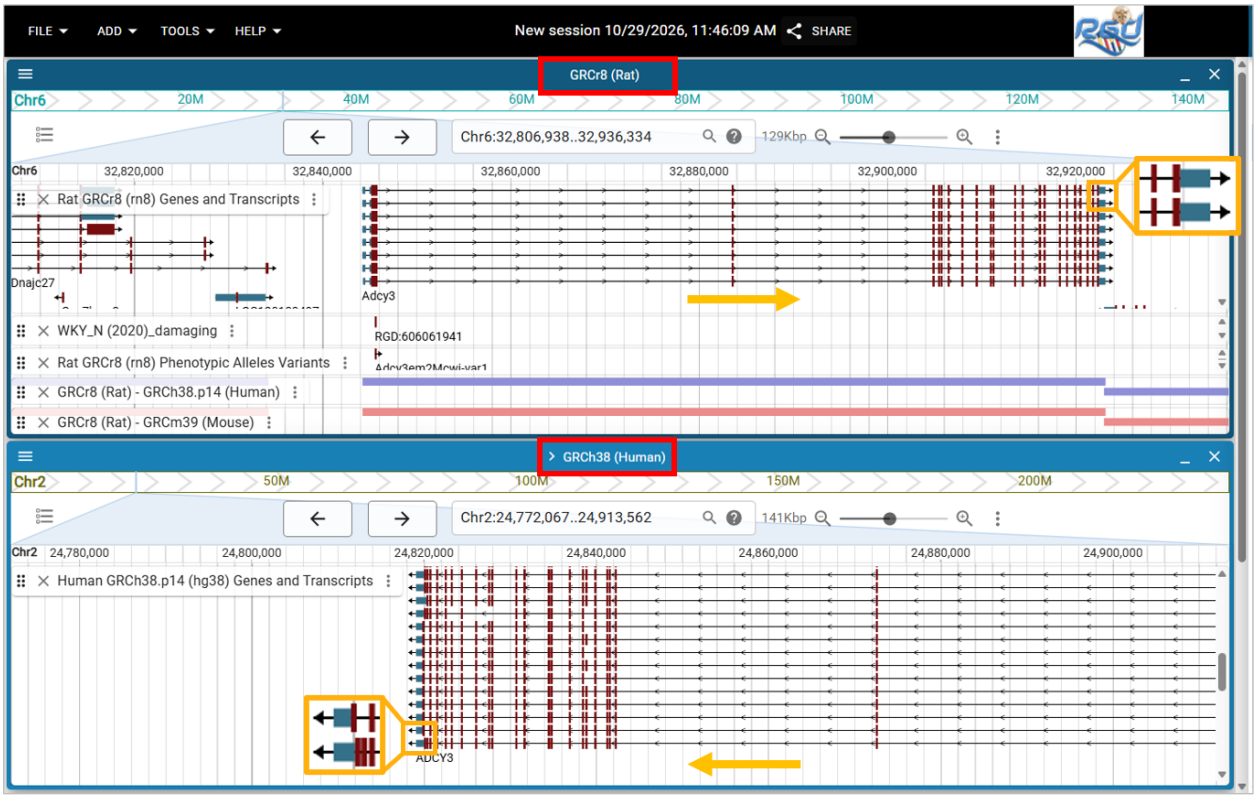
https://rgd.mcw.edu/jbrowse2/?session=share-lhsxlCUeM7&password=BQQpJ

In the “Available tracks” menu, open the “Orthologs” subcategory.

1) Select the “GRCr8 (Rat) – GRCh38.p14 (Human)” and “GRCr8 (Rat) – GRCm39 (Mouse)” tracks
2) The purple ortholog glyphs for human indicate that those genes are reversed compared to the genes in rat. The orange glyphs for mouse indicate that mouse genes are in the same orientation as rat.
    a) Click on the block glyph to bring up “Synteny feature details” in the right sidebar. This display provides the positions and lengths of the corresponding genes in the two assemblies

Linear Genome Views





https://rgd.mcw.edu/jbrowse2/?session=share-oZBXSIG2I9&password=BQJpx

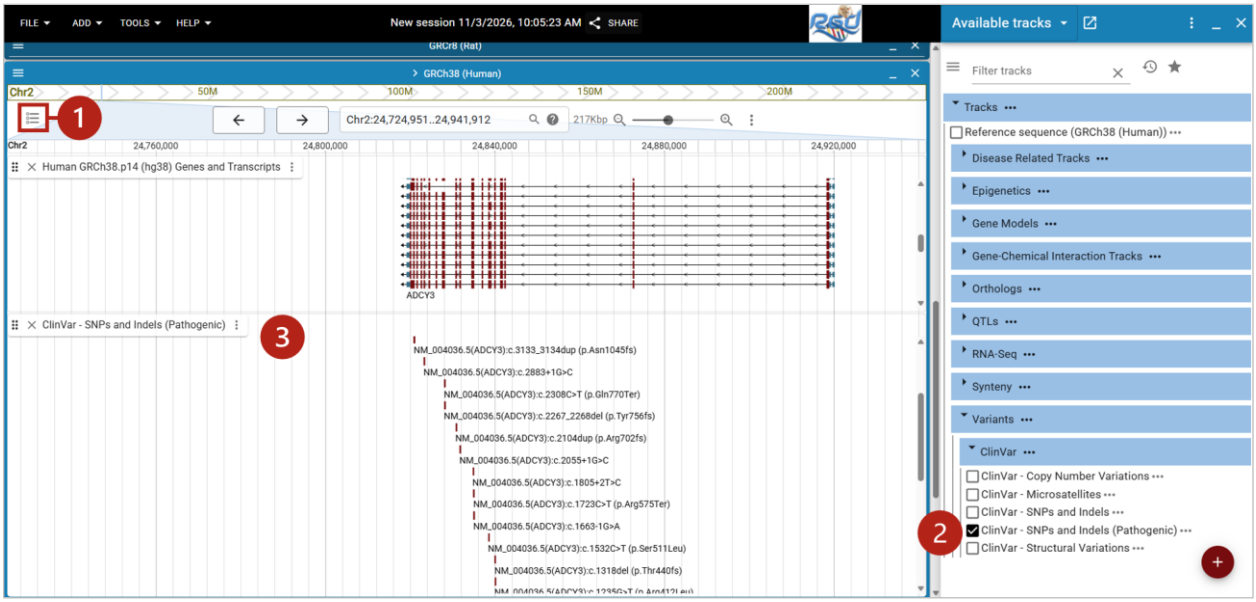
To add the human linear genome view in the same session:

1) Top left menu select “ADD” → “Linear genome view”
2) Select the assembly (e.g., human) and location (e.g., ADCY3) → click “OPEN”
3) Select “Human GRCh38.p14 (hg38) Genes and Transcripts” track from the resulting list by clicking “GO”

Now there are 2 sub-windows with the GRCr8 (Rat) assembly open at the top and the GRCh38 (Human) assembly open at the bottom. The arrows on the transcript icons indicate human genes are reversed compared to rat (yellow callout boxes and arrows).

> *Note: The gene symbol for each gene is displayed below the transcript glyphs. Hovering over a gene region will highlight the gene in gray and a popup will display the gene symbol.*

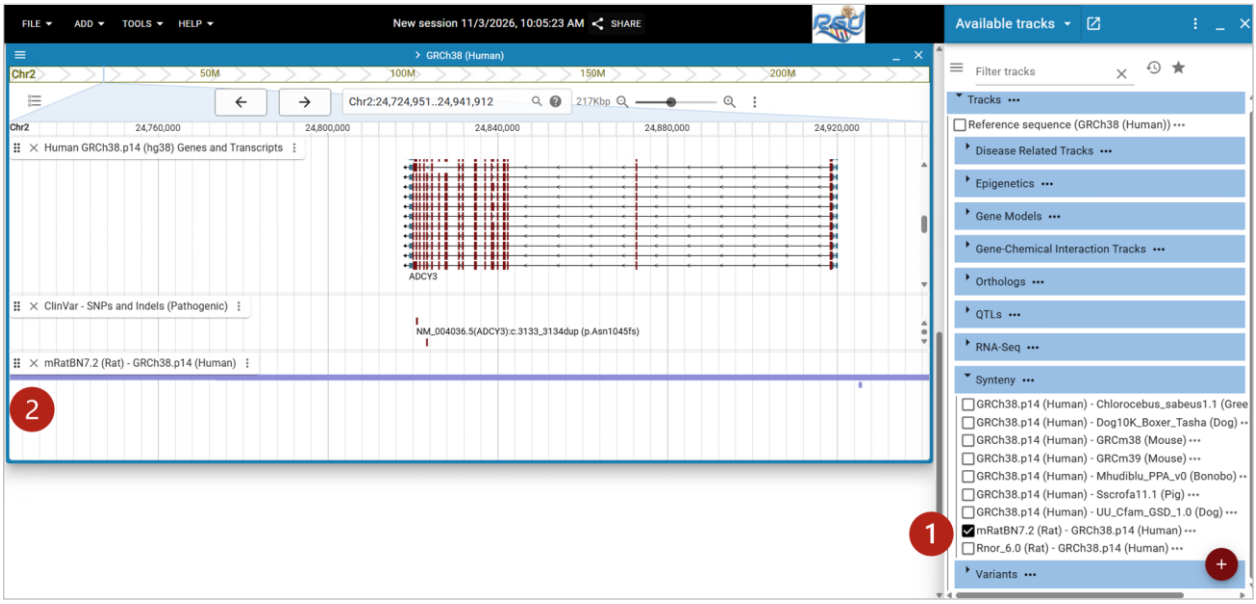
Human Variant Track



https://rgd.mcw.edu/jbrowse2/?session=share-hzRSx-8gLx&password=apjFp

1) Click the hamburger menu (3 horizontal lines beginning with a circle) in the top left of the human panel to open the “Available tracks” menu for human
2) In the “Available tracks” menu, open the “Variants” subcategory → “ClinVar” → select “ClinVar – SNPs and Indels (Pathogenic)” track
3) Scroll down (if needed) to see the *ADCY3* ClinVar variants designated as pathogenic or likely pathogenic

Rat – Human Synteny Track

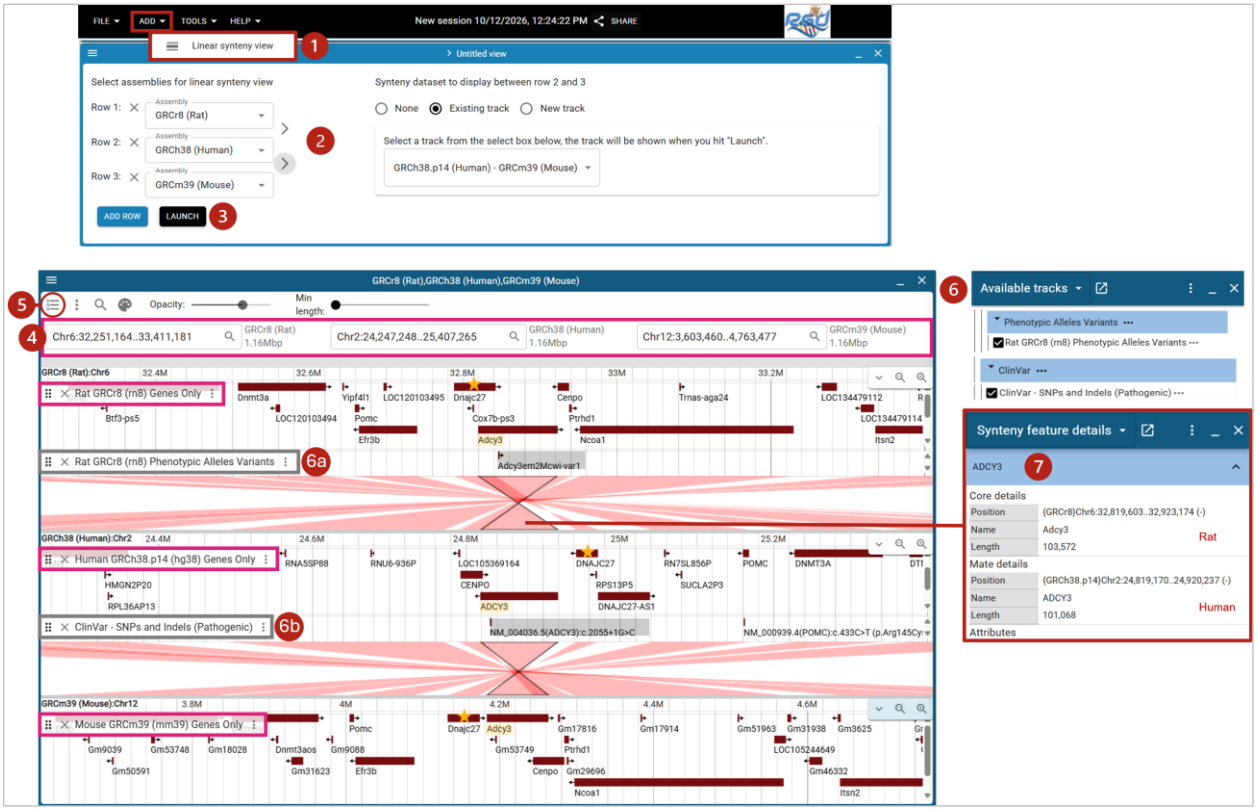


https://rgd.mcw.edu/jbrowse2/?session=share-XAxpmjJSEE&password=nB8ik

1) In the “Available tracks” menu, open the “Synteny” subcategory → select “mRatBN7.2 (Rat) – GRCh38.p14 (Human)” track
2) These blocks are generally large for closely related species, as seen here (33-35 Mb), reflecting the large conserved syntenic regions typical between rat and human

> *Note: The rat – human synteny track currently uses the previous assembly, mRatBN7.2.*

Linear Synteny View



https://rgd.mcw.edu/jbrowse2/?session=share-yfOx29AcPu&password=qgyVH

To explore synteny across multiple species:

1) Top left menu select "ADD" → "Linear synteny view"

2) Choose a species for each row (e.g., rat, human, mouse)

a) Default is 2 rows → click "ADD ROW" to select more assemblies

b) You have to click the bottom arrow (>) for the "LAUNCH" button to become activated

> *Note: On the right side, we will leave "Existing track" selected. You can also choose to add a new track – select that option and follow the prompts. Currently, it is unnecessary to select the dropdown option, as there is only one option.*

3) "LAUNCH" displays genome-wide synteny blocks connected with red lines (initial display not shown in this figure)

> *Note: Chromosomes are listed across the top of each section of the display with the width of each column determined by the relative length of the corresponding chromosome.*

4) Search "*Adcy3*" in each location box (top pink outline); select the NCBI "genes only" track for each species (tracks outlined in pink; *Adcy3* highlighted in yellow). This view shows a single bar glyph for each gene, consolidating these tracks

> *Note: The magnifying glass icon (🔍, in the same line as the (5) callout) allows you to modify the orientation of the search boxes (side-by-side, as shown, or stacked vertically).*
>
> *Zoom in/out and pan will move each section separately. "Link views" setting (3 vertical dots menu option next to the (5) callout) will zoom/pan sections together.*

5) Additional tracks can be added to any or all of the syntenic regions shown. To bring up the "Available tracks" menu in the right sidebar, select the menu icon in the top left (≡, red circle) → "Row track selectors" → select row

6) Select tracks from the "Available tracks" menu

a) Select "Rat GRCr8 (rn8) Phenotypic Alleles Variants" track

b) Select "ClinVar – SNPs and Indels (Pathogenic)" track

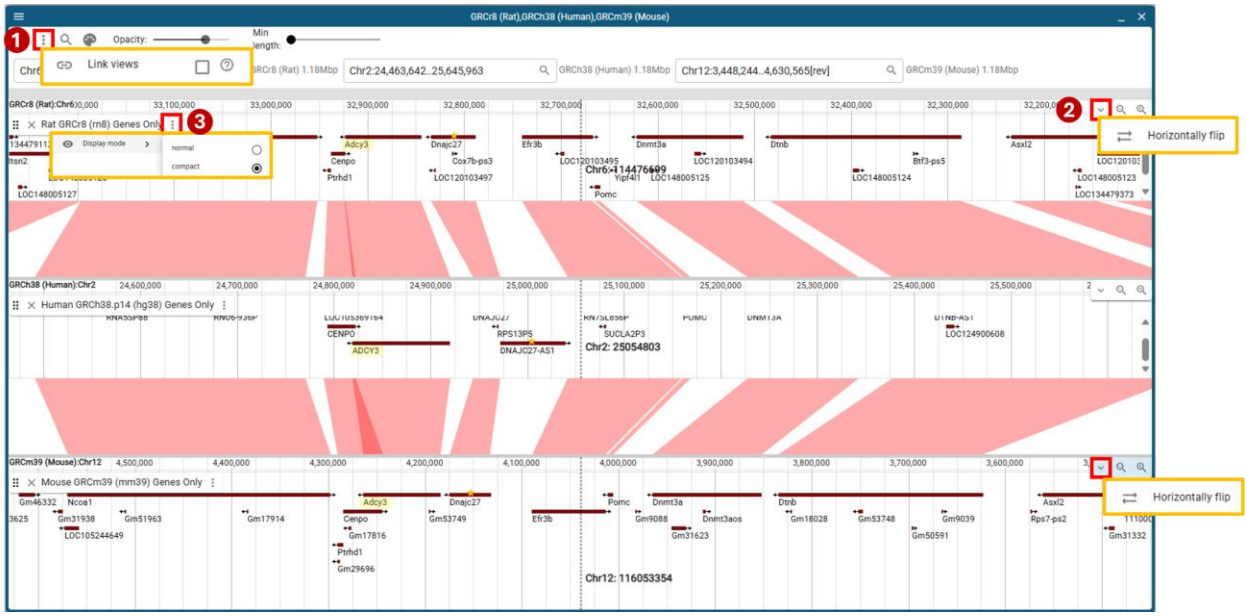
- Track names outlined in gray and variants highlighted in gray. These are the same tracks added in the previous figures

> *Note: The opacity of the red synteny tracks can be modified by the sliding bar at the top of the panel, in the same line as the (5) callout. The color of the synteny tracks can be changed by selecting the paint palette icon to the left of the opacity slider.*

7) Clicking a synteny link opens "Synteny feature details" in the right sidebar (red callout)

The human block is reversed relative to rat and mouse, as shown by gene order (*Dnajc27* starred for reference) and by the distinctive (black) X pattern of the lines in the synteny tracks that join the orthologs between the genome assembly tracks.

Bonus: uncross synteny tracks by horizontally flipping the tracks



https://rgd.mcw.edu/jbrowse2/?session=share-DMLoU9t_03&password=l5gxz

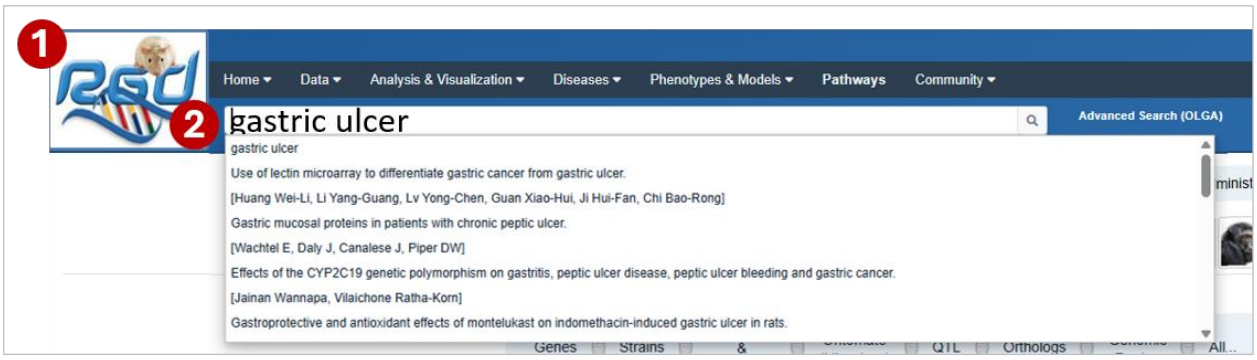
1) Uncheck "Link views" to adjust each track individually (recommended, but not required)
2) Select the down arrow next to the zoom icons → "Horizontally flip" for rat and mouse (as shown) OR just for the human track (depending on which option you choose, just note the direction of the transcript arrows and chromosome coordinates)
3) Doing this makes the gene symbols overlap. You can avoid this by selecting the 3 vertical dots menu in the track name → "Display mode" → "compact"

Adcy3 gene symbol is highlighted in yellow and *Dnajc27* is starred for reference.

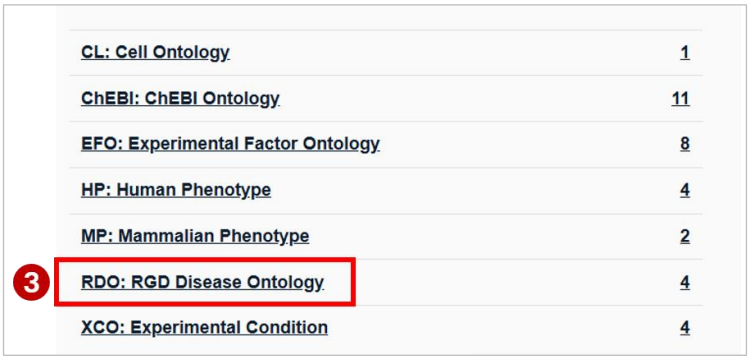
Import and view curated expression data from the RGD Downloads page

Video tutorial: https://rgd.mcw.edu/wg/viewing-expression-coverage-profiles-from-the-rgd-downloads-page/

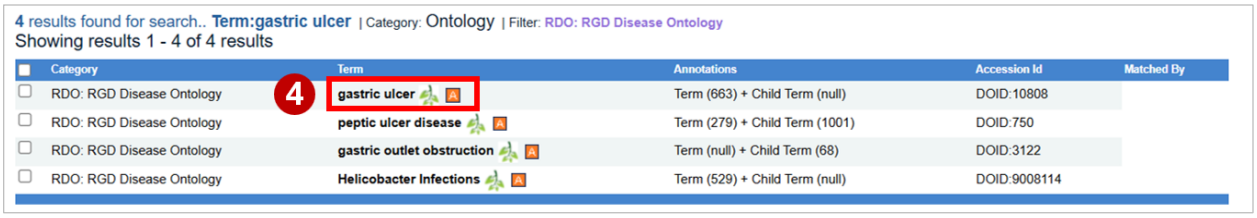
1. RGD homepage (**rgd.mcw.edu**)



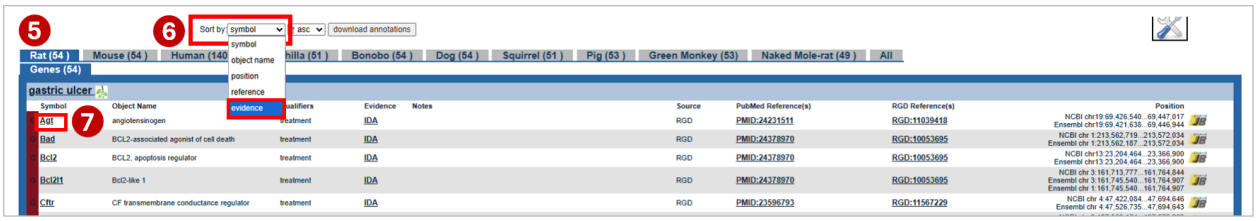
2. Search for disease of interest (e.g., gastric ulcer)
    - You could also search for a gene of interest, but for this example we will use a disease



3. Select **RDO: Disease Ontology**

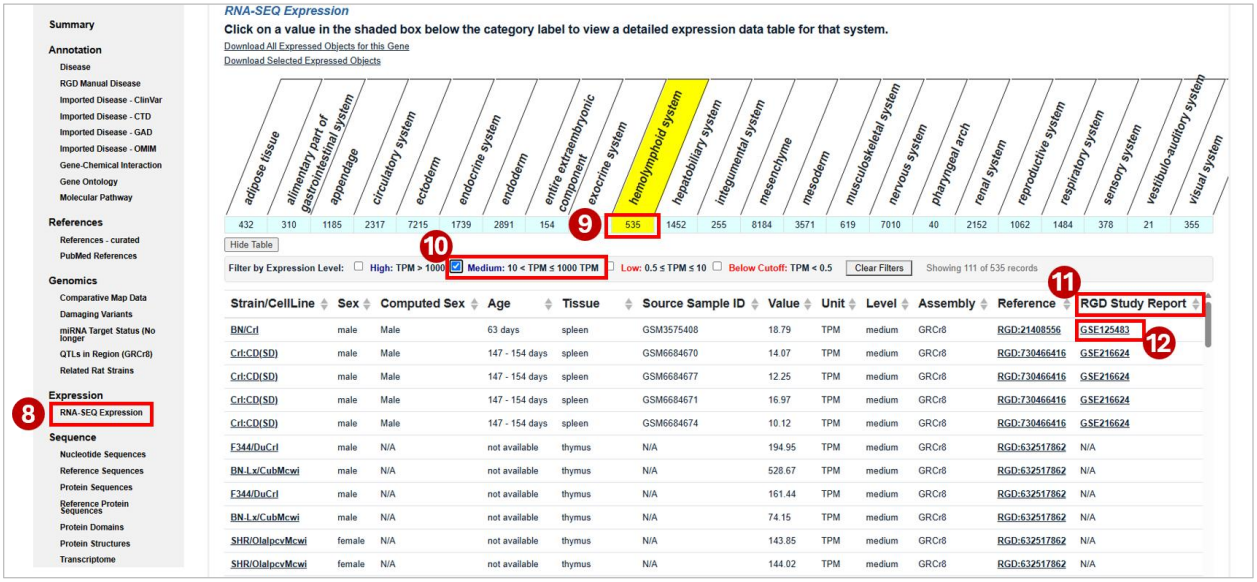


4. Select **gastric ulcer**



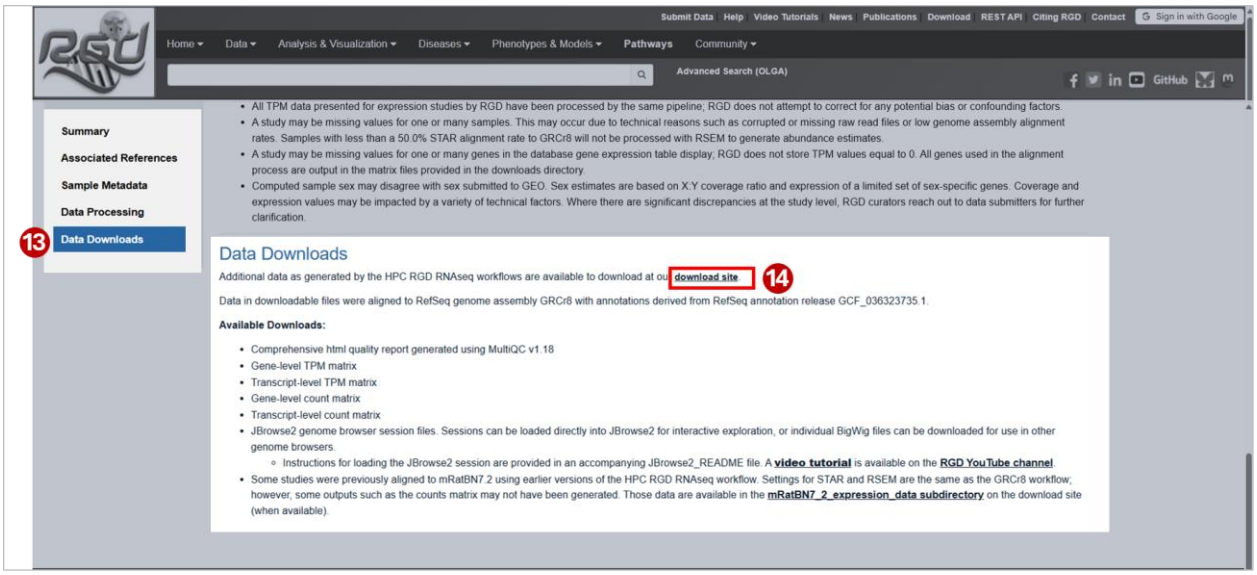
https://rgd.mcw.edu/rgdweb/ontology/annot.html?acc_id=DOID:10808&sort_by=evidence#annot

5. Scroll down on the RGD Disease Ontology annotations page for gastric ulcer
6. Sort table by **evidence**
7. Select **gene** (e.g., *Agt*)



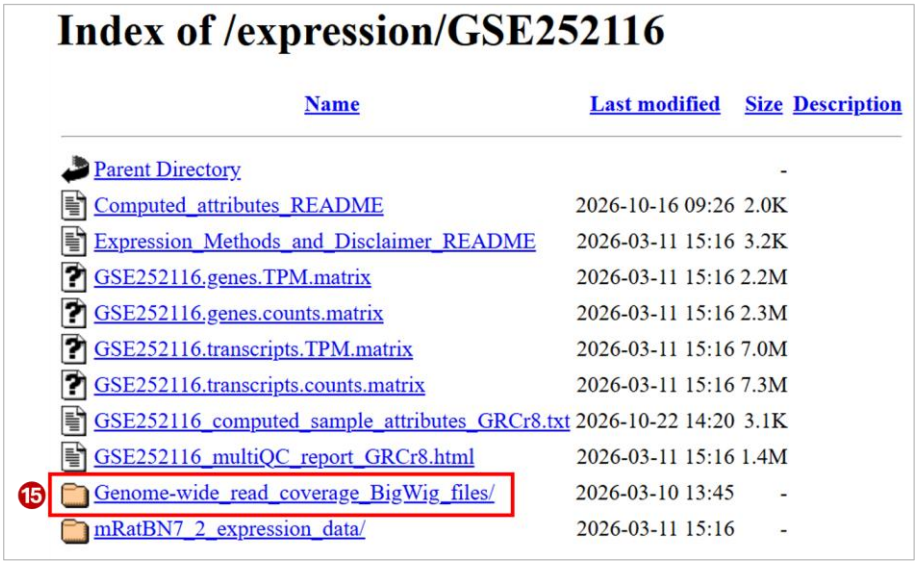
https://rgd.mcw.edu/rgdweb/report/gene/main.html?id=2069#rnaSeqExpression

8. Gene Report Page → **RNA-SEQ Expression** section
9. Select **category** (e.g., hemolymphoid system)
10. Table can be **filtered** by expression level (e.g., Medium)
11. Table can be **sorted** by column (e.g., RGD Study Report)
12. Select **GEO Study** link



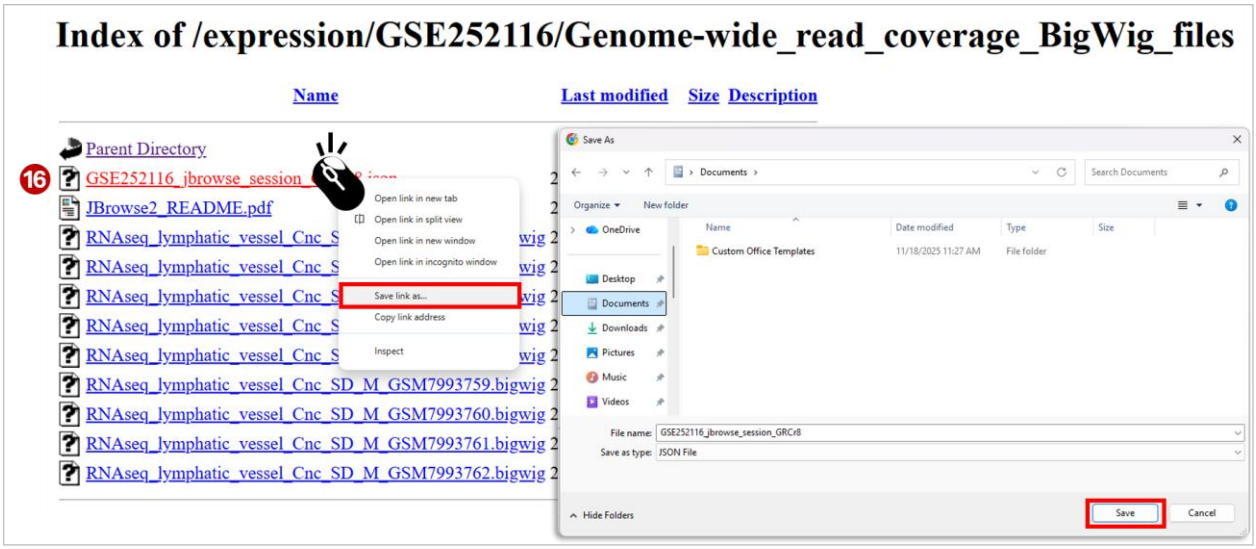
https://rgd.mcw.edu/rgdweb/report/expressionStudy/main.html?geoAcc=GSE252116#dataDownloads

13. On the Expression Study report page, navigate to **Data downloads**
14. Select **download site** link



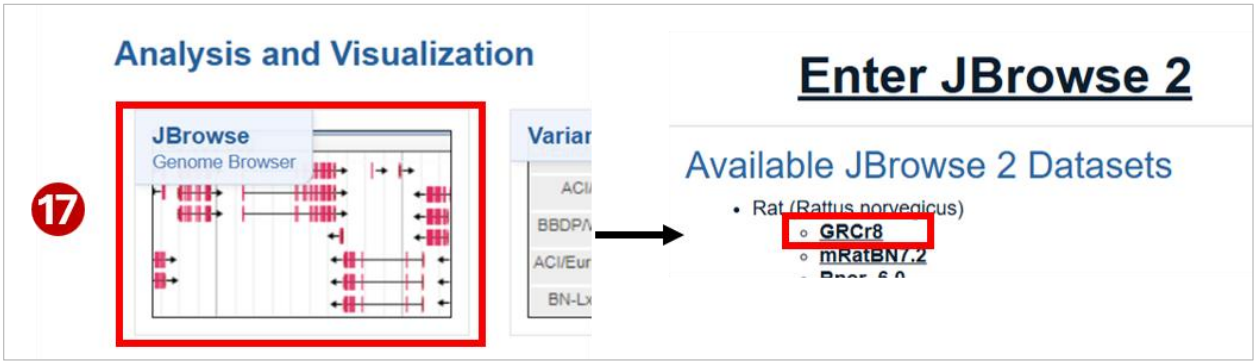
https://tinyurl.com/2b8t6jmn

15. Select **Genome-wide_read_coverage_BigWig_files/** link



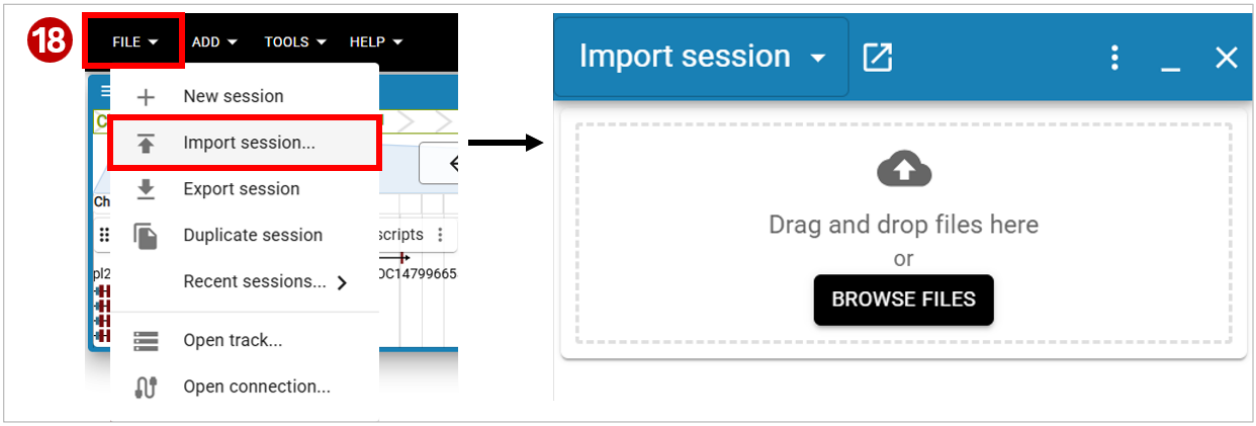
https://download.rgd.mcw.edu/expression/GSE252116/Genome-wide_read_coverage_BigWig_files/

16. To download the json file to your computer, right-click on the **GSE252116_jbrowse_session_GRCr8.json** file link and select "Save link as..." → Save

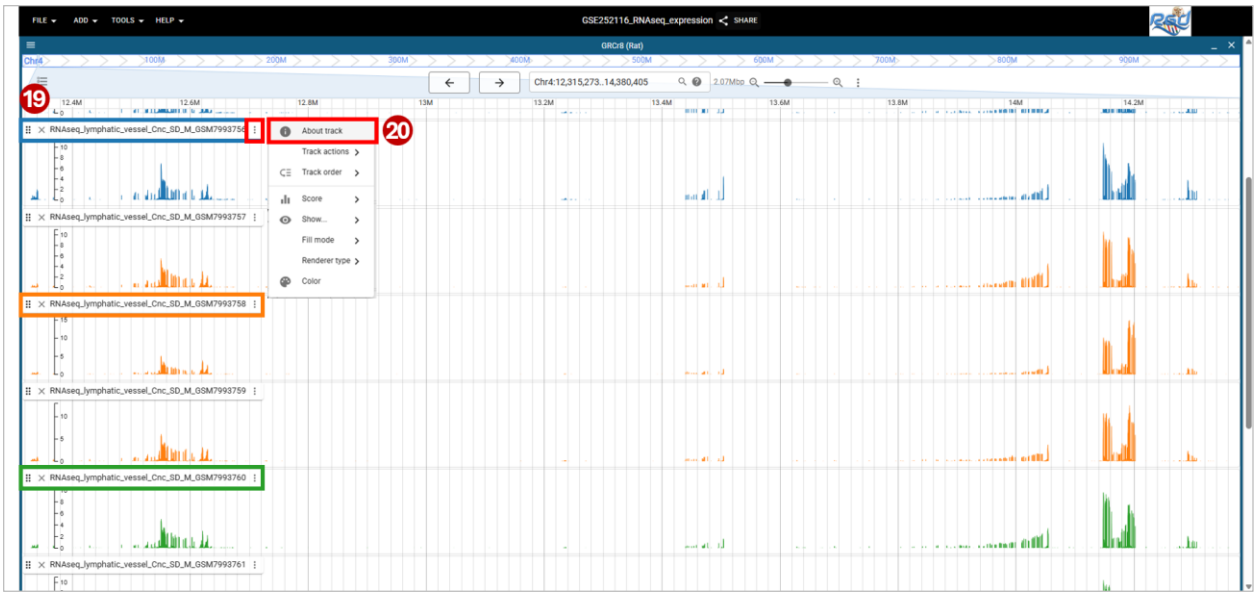


https://rgd.mcw.edu/rgdweb/jbrowse2/listing.jsp

17. Navigate to RGD **JBrowse 2** → GRCr8

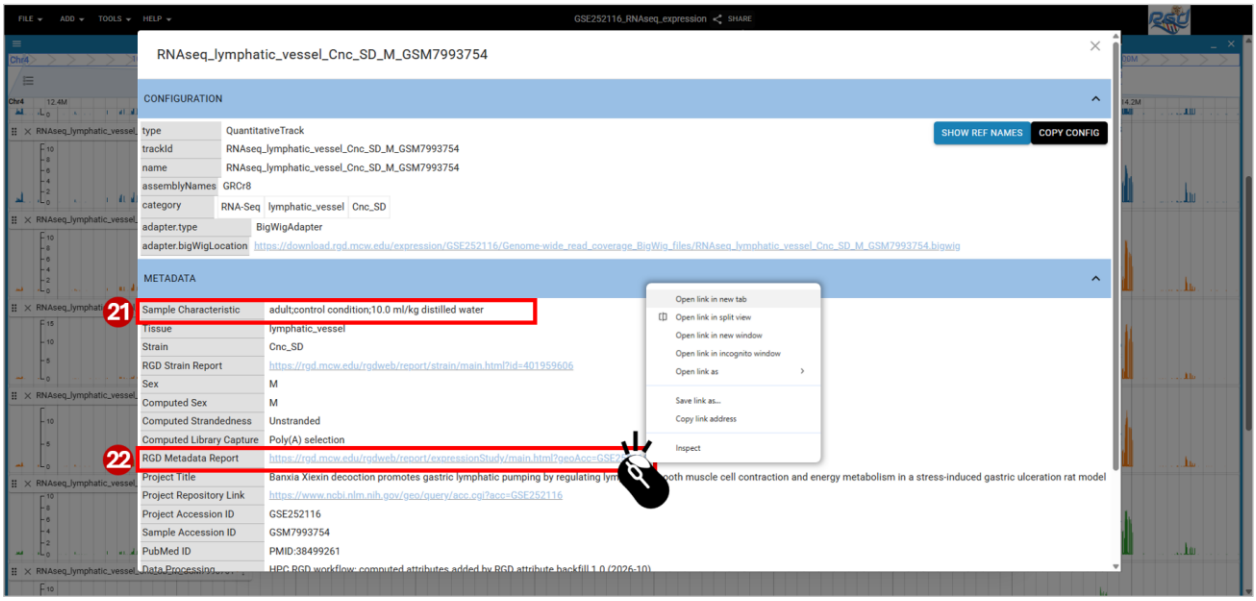


18. **Import** json file: "FILE" → "Import session" → upload json file in right sidebar

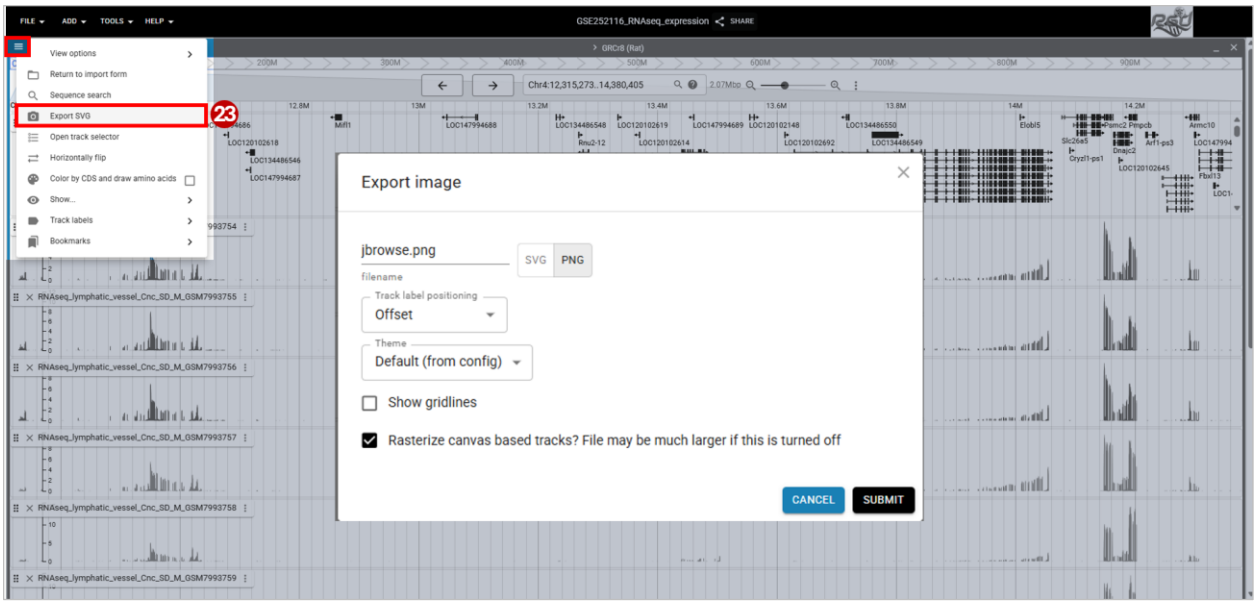


https://rgd.mcw.edu/jbrowse2/?session=share-g2fruvtads&password=zmvpw

19. **Tracks** are color-coded by matching strain, tissue, sex, and experimental conditions
    - Track names include tissue_strain_sex_GEO accession ID
20. To view experimental conditions, open the menu by clicking the 3 dots located after the track name → "About track" → Metadata display



21. Experimental conditions can be found in the "Sample Characteristic" field
22. Fully curated metadata values are available on the expression report page in the RGD user interface (linked in the "RGD Metadata Report" field)
    - Links in the metadata display will navigate away from the JBrowse 2 session, but do not open in a new tab. To avoid having to reload the session, right-click on the link and select "Open link in new tab"



23. Export JBrowse session as SVG or PNG

- At the top of the panel in the blue header, select the 3 horizontal lines menu
- Select “Export SVG”
- Select settings (e.g., switch to PNG)
- Click “SUBMIT”
- The image will download to your computer

Example PNG:

